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(54) **PERCUSSIVE THERAPY DEVICE WITH
ELECTRICALLY CONNECTED
ATTACHMENT**

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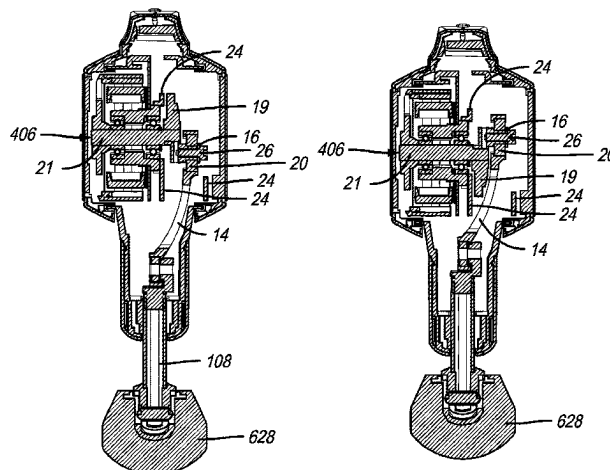
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(57)

ABSTRACT

A percussive therapy system includes a percussive therapy
device that includes a housing, an electrical source, a motor
positioned in the housing, a switch for activating the motor,
a push rod assembly operatively connected to the motor and
configured to reciprocate in response to activation of the
motor, and an attachment configured to be operatively
connected to a distal end of the push rod assembly of the
percussive massage device and to provide at least one
therapeutic effect to a user. The attachment may include at
least one of an actuator configured to provide the at least one
therapeutic effect to the user and a sensor configured to

(Continued)



obtain at least one of biometric data of the user and information regarding operation of the percussive therapy device.

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continuation of application No. 17/739,630, filed on May 9, 2022, now Pat. No. 11,890,253, which is a continuation-in-part of application No. 17/705,300, filed on Mar. 26, 2022, now Pat. No. 11,730,668, and a continuation-in-part of application No. 17/244,239, filed on Apr. 29, 2021, now Pat. No. 11,452,670, which is a continuation of application No. 17/244,278, filed on Apr. 29, 2021, now Pat. No. 11,564,860, which is a continuation-in-part of application No. 17/018,099, filed on Sep. 11, 2020, now Pat. No. 11,357,697, which is a continuation-in-part of application No. 16/869,402, filed on May 7, 2020, now Pat. No. 10,857,064, which is a continuation-in-part of application No. 16/796,143, filed on Feb. 20, 2020, now Pat. No. 10,940,081, and a continuation-in-part of application No. 16/675,772, filed on Nov. 6, 2019, now Pat. No. 10,702,448.

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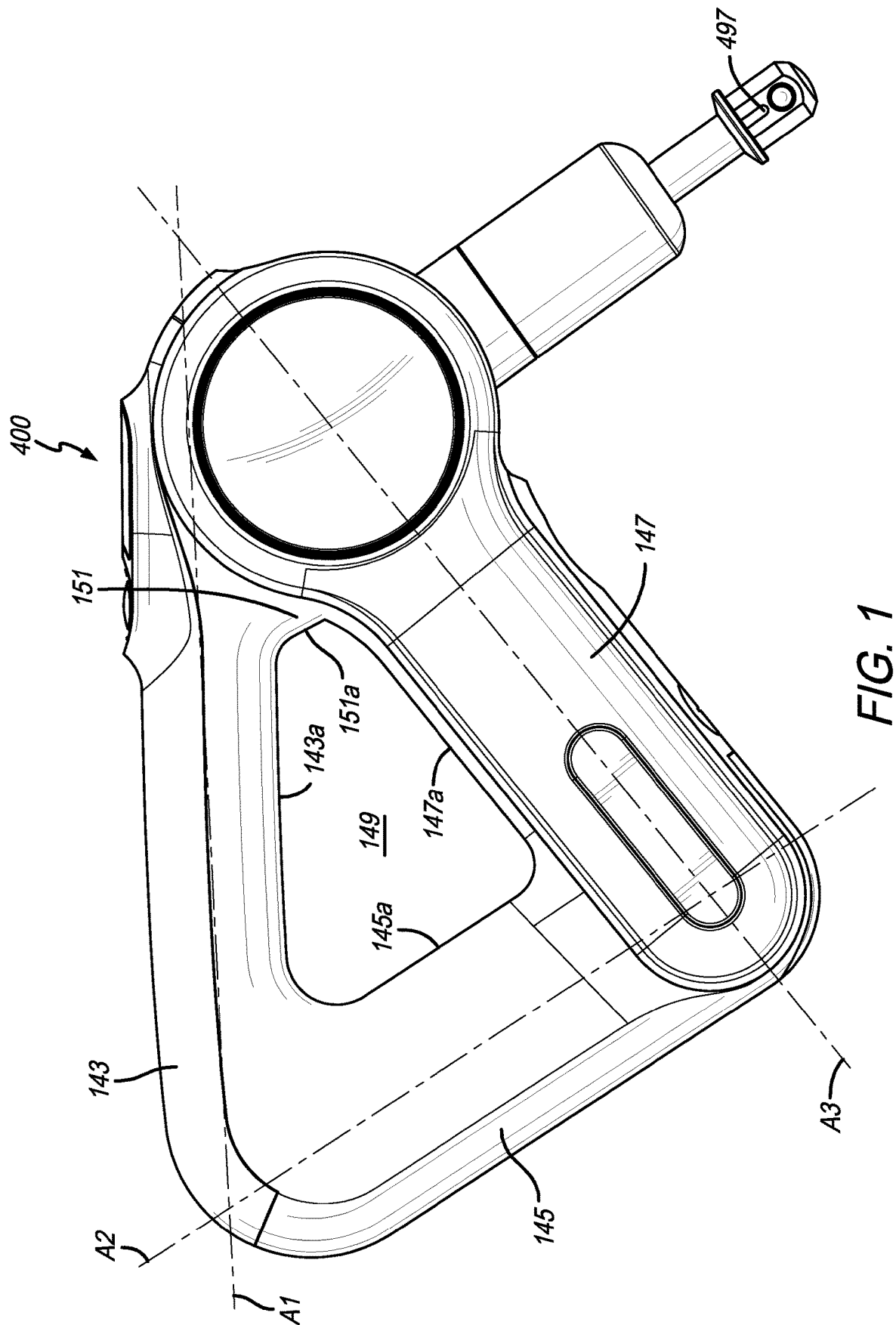
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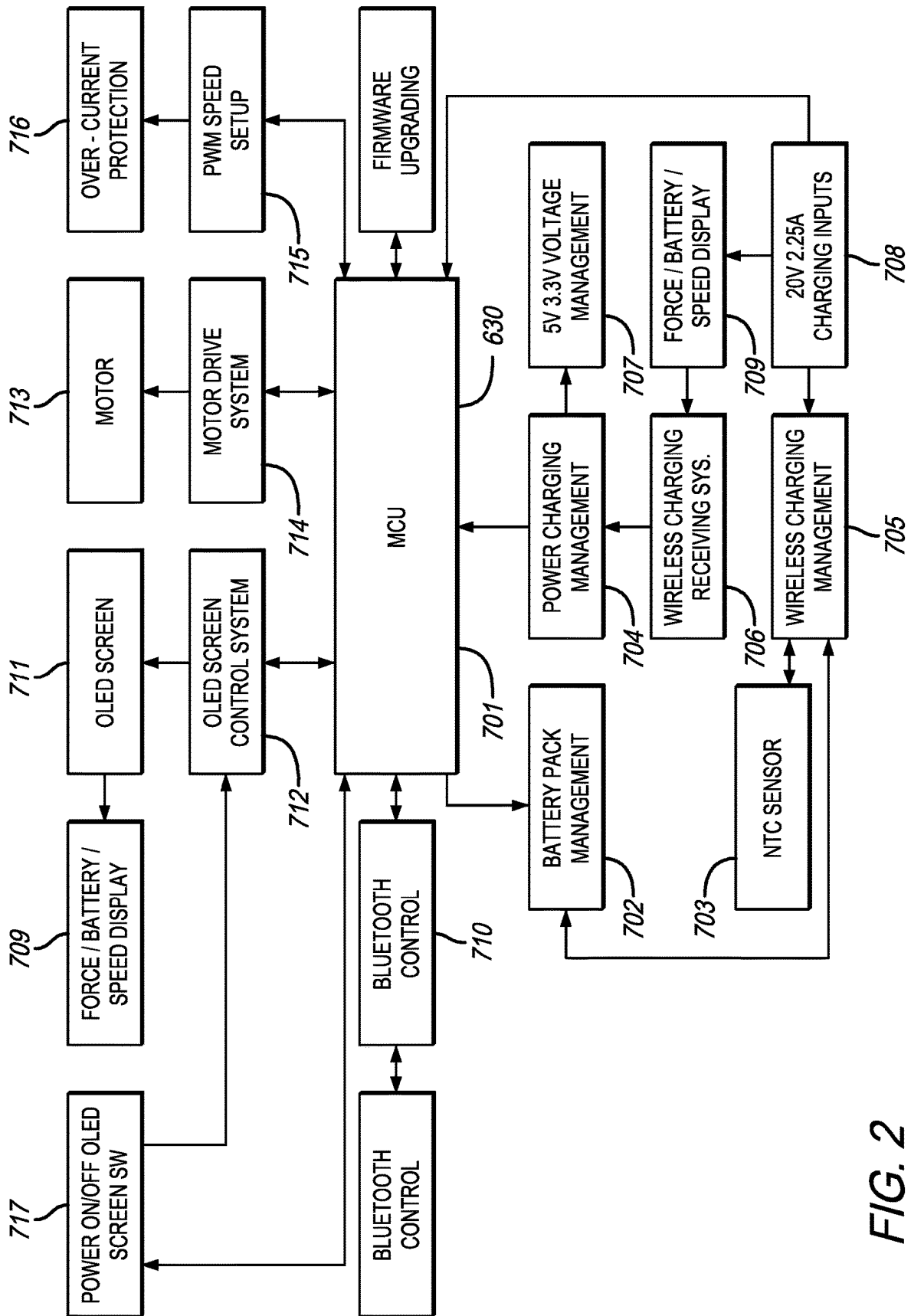


FIG. 2

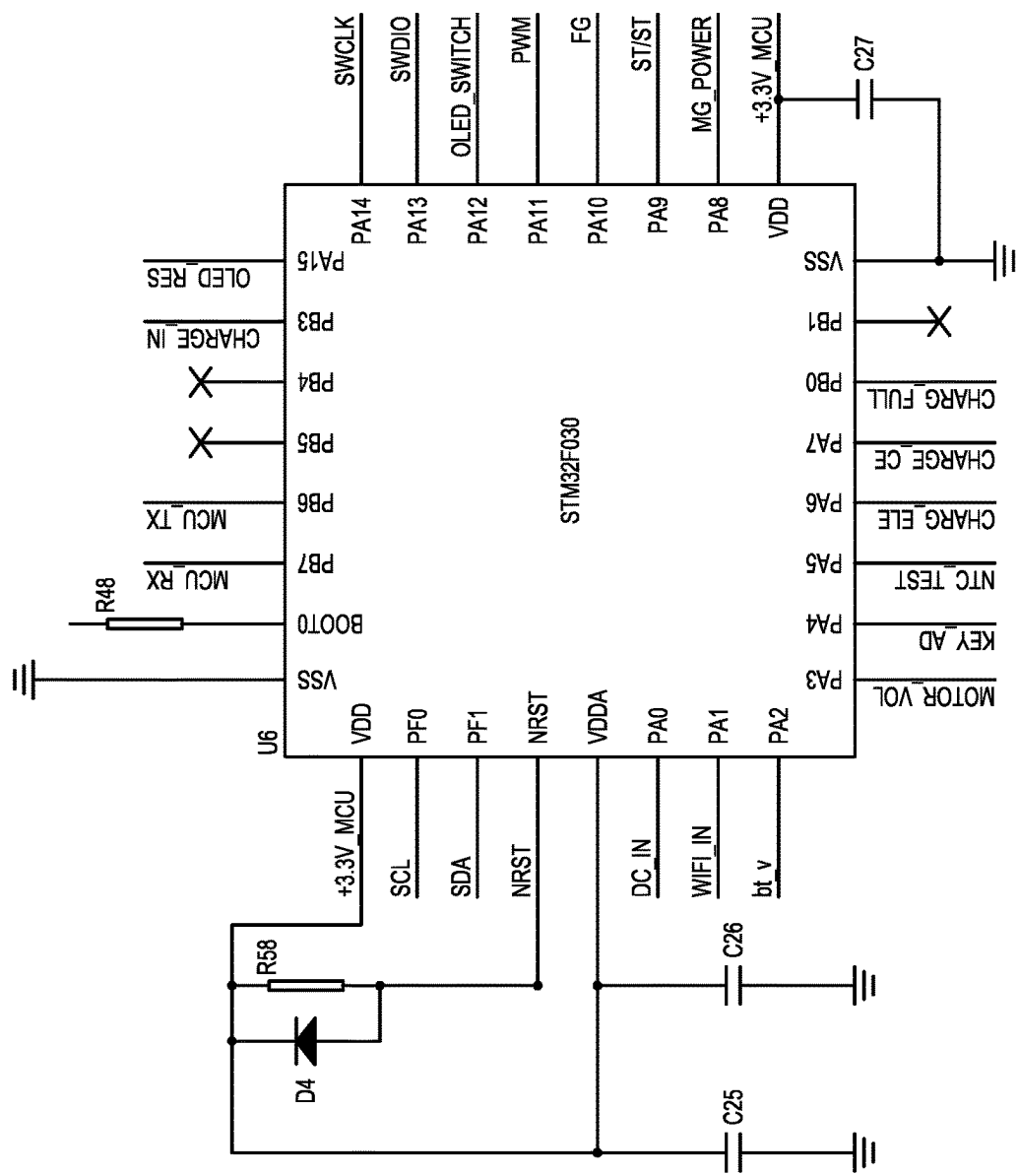


FIG. 3

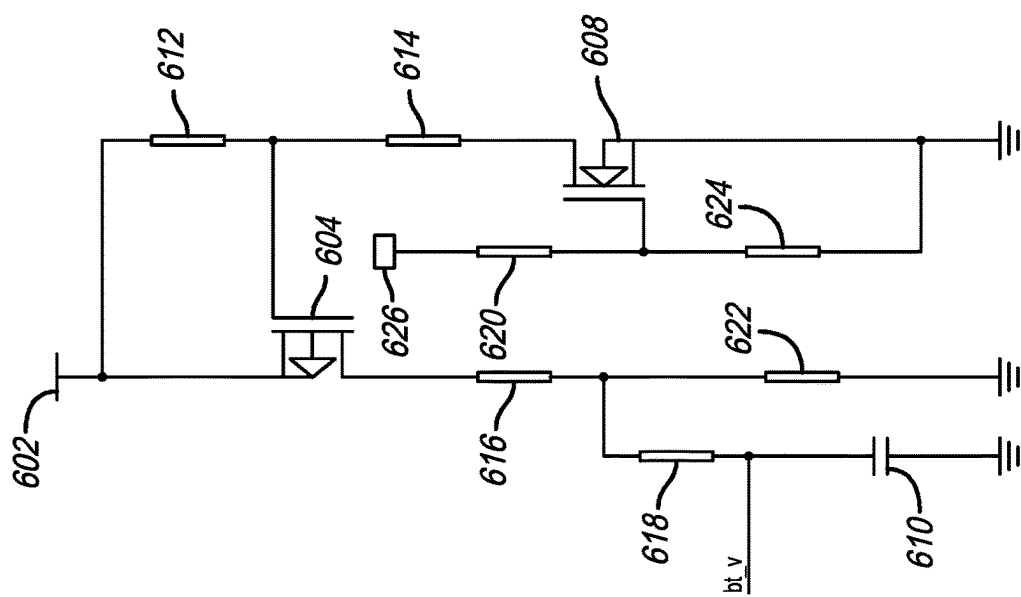


FIG. 4

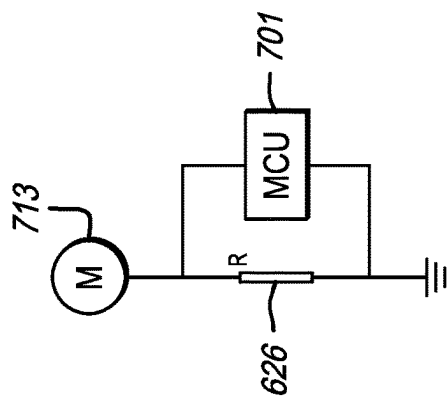
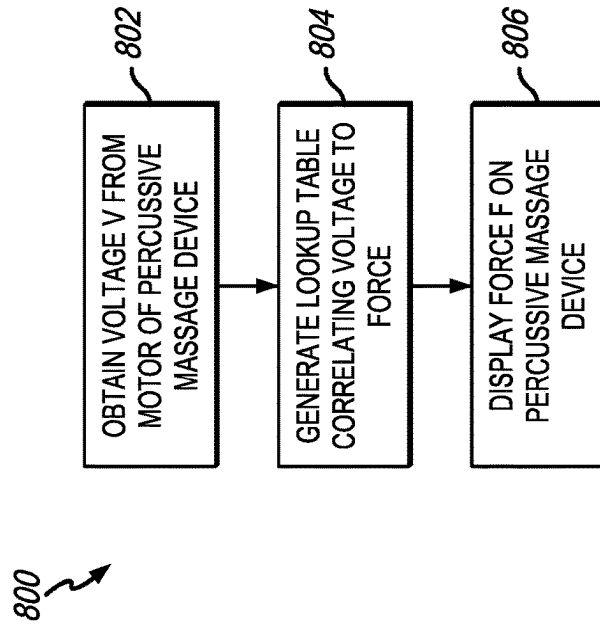
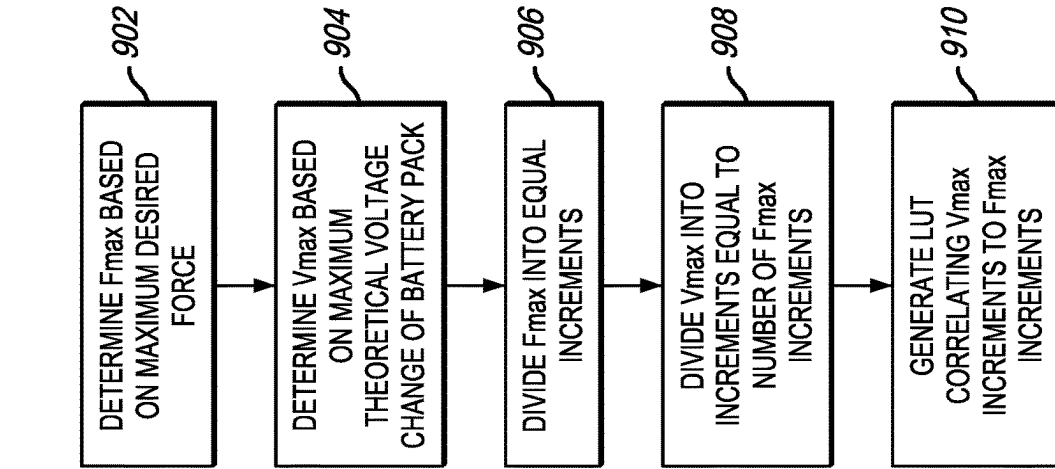


FIG. 5



POWER (W)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49
FORCE (LB)	0.03	0.06	0.09	0.12	0.15	0.18	0.21	0.24	0.27	0.3	0.33	0.36	0.39	0.42	0.45	0.48	0.51	0.54	0.57	0.6	0.63	0.66	0.69	0.72	0.75	0.78	0.81	0.84	0.87	0.9	0.93	0.96	0.99	1.02	1.05	1.08	1.11	1.14	1.17	1.2	1.23	1.26	1.29	1.32	1.35	1.38	1.41	1.44	1.47

POWER (W)	50	51	52	53	54	55	56	57	58	59	60
FORCE (LB)	1.5	1.53	1.56	1.59	1.62	1.65	1.68	1.71	1.74	1.77	1.8

LOOKUP OF FORCE METER

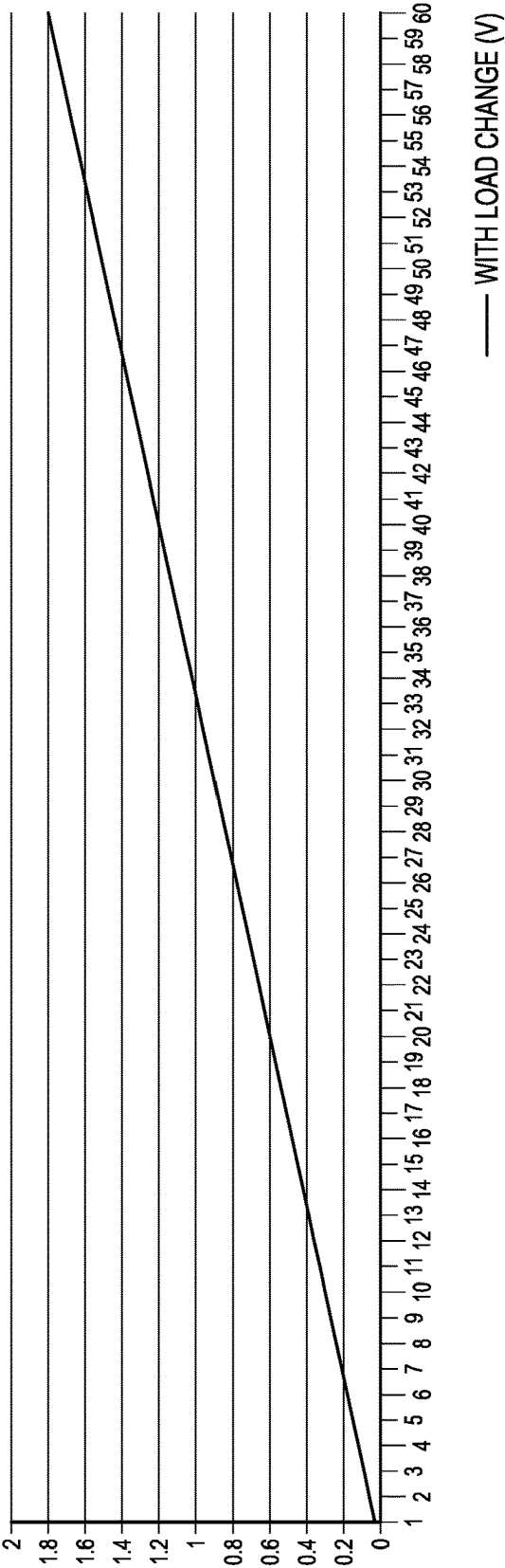
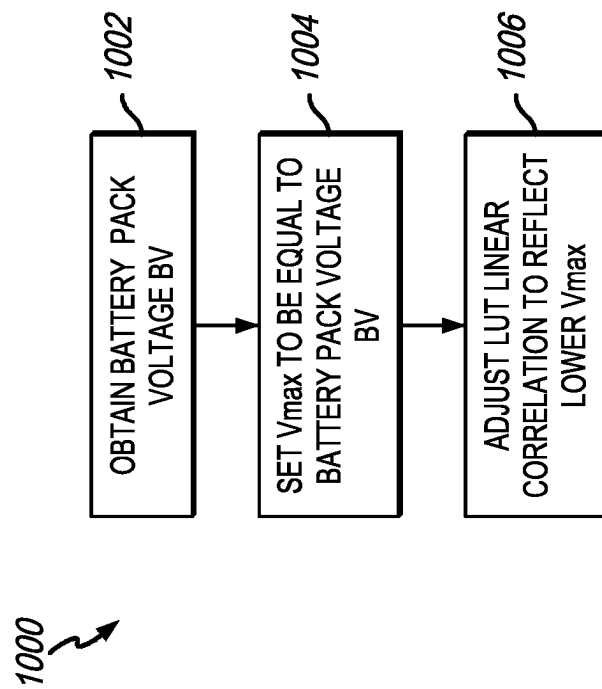


FIG. 8

*FIG. 9*

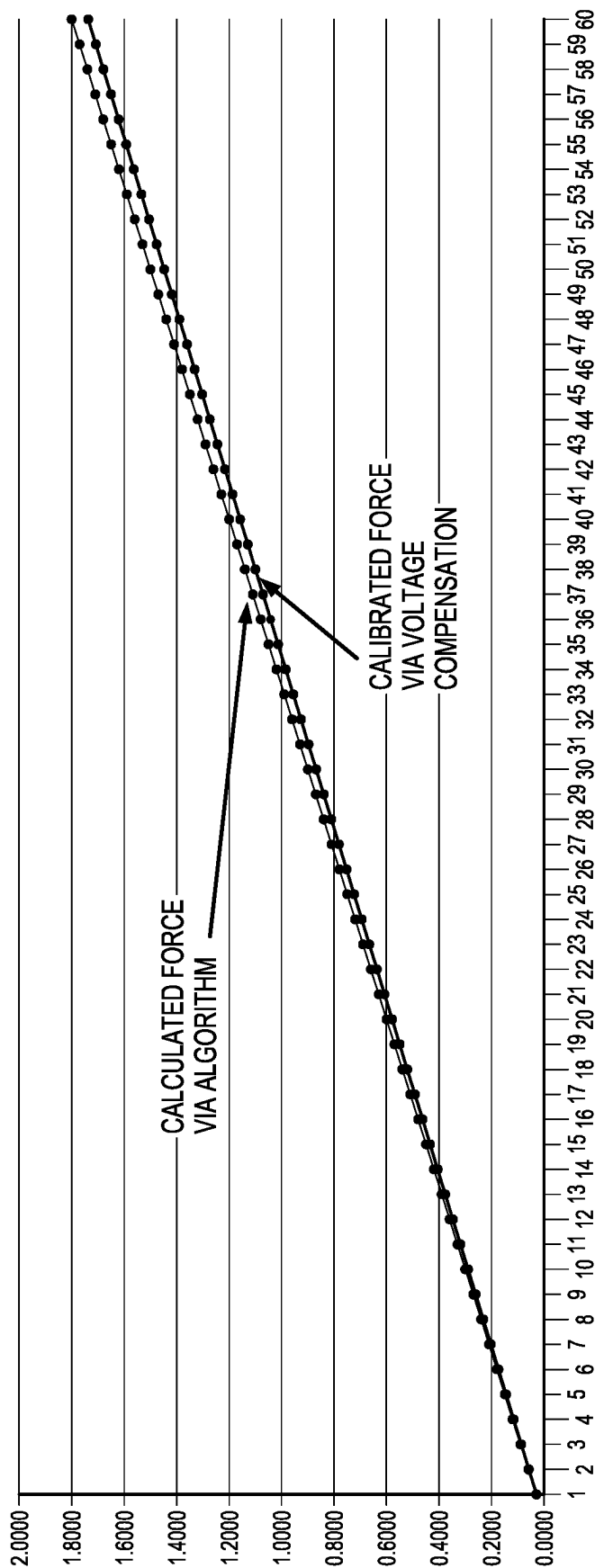
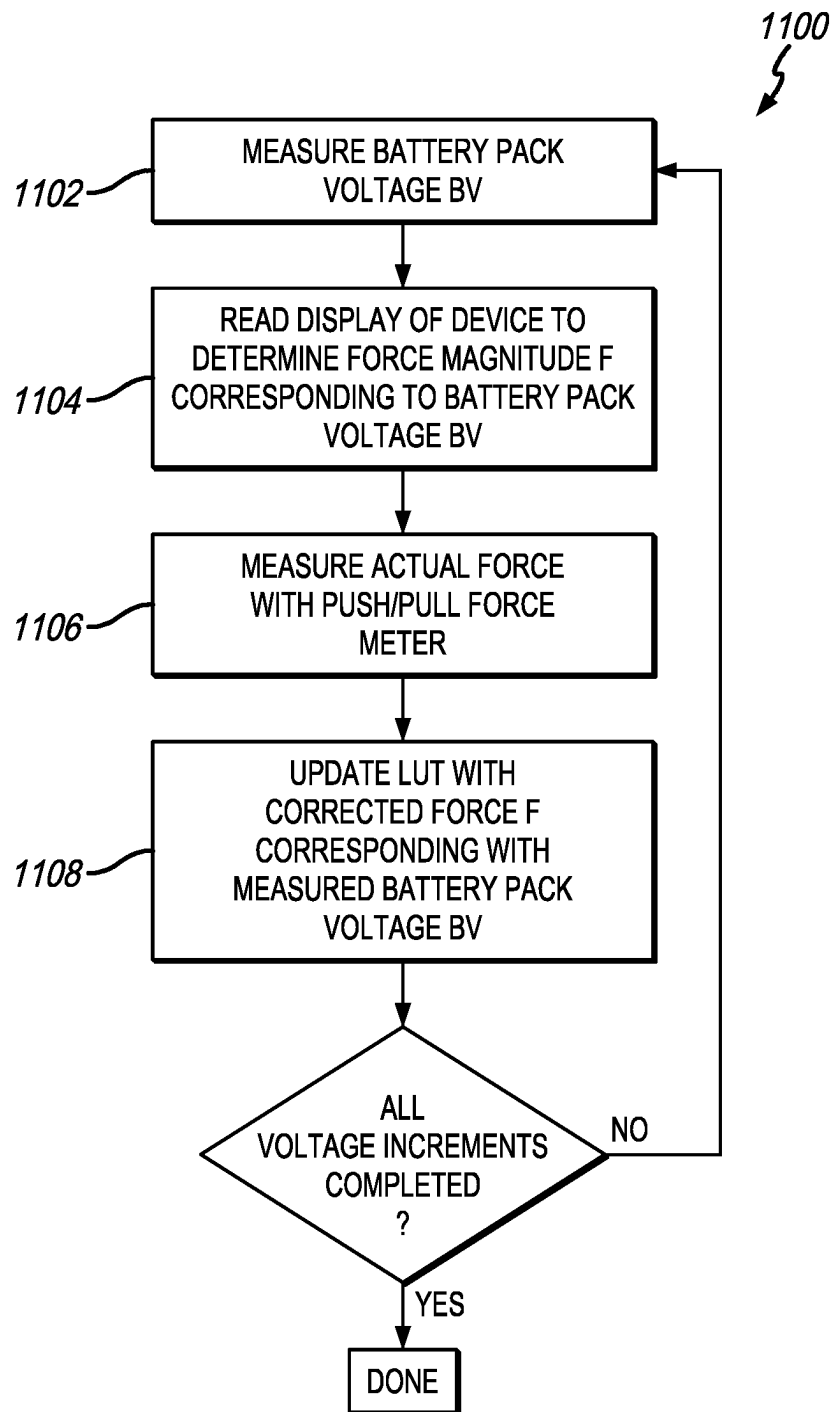


FIG. 10

**FIG. 11**

FINAL CALIBRATED FORCE METER

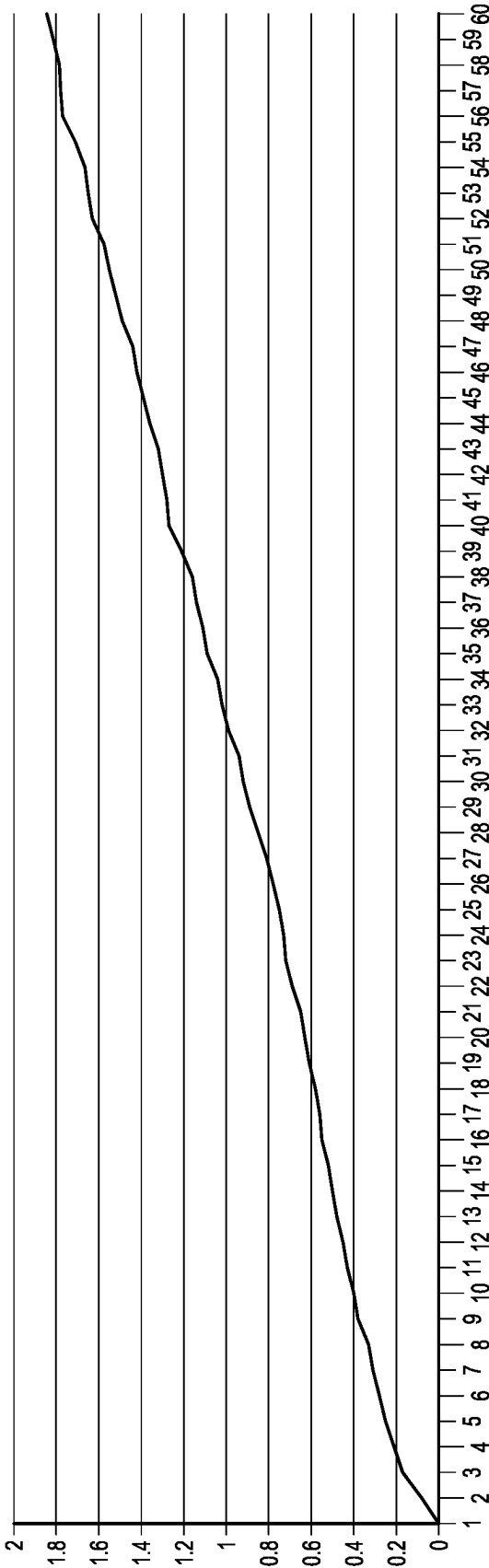


FIG. 12

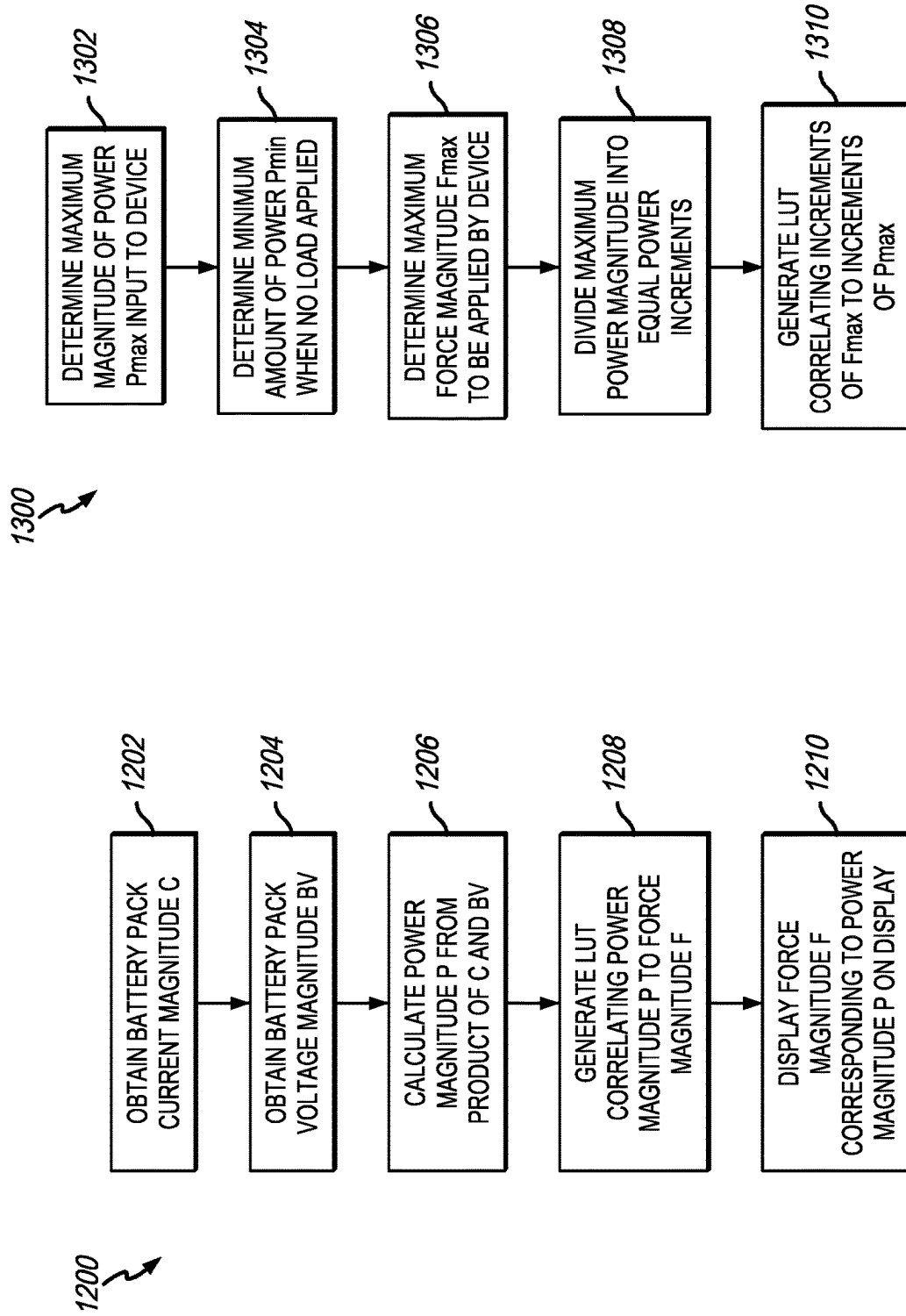


FIG. 14

FIG. 13

POWER (W)	3	6	9	12	15	18	21	24	27	30	33	36	39	42	45	48	51	54	57	60	63	66	69	72	75	78	81	84	87	90	93	96	99	102	105	108	111	114	117	120	123	126	129	132	135	138	141	144	147
FORCE (LB)	0	0	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46

POWER (W)	150	153	156	159	162	165	168	171	174	177	180
FORCE (LB)	47	48	49	50	51	52	53	54	55	56	57

FORCE METER - POWER MODE

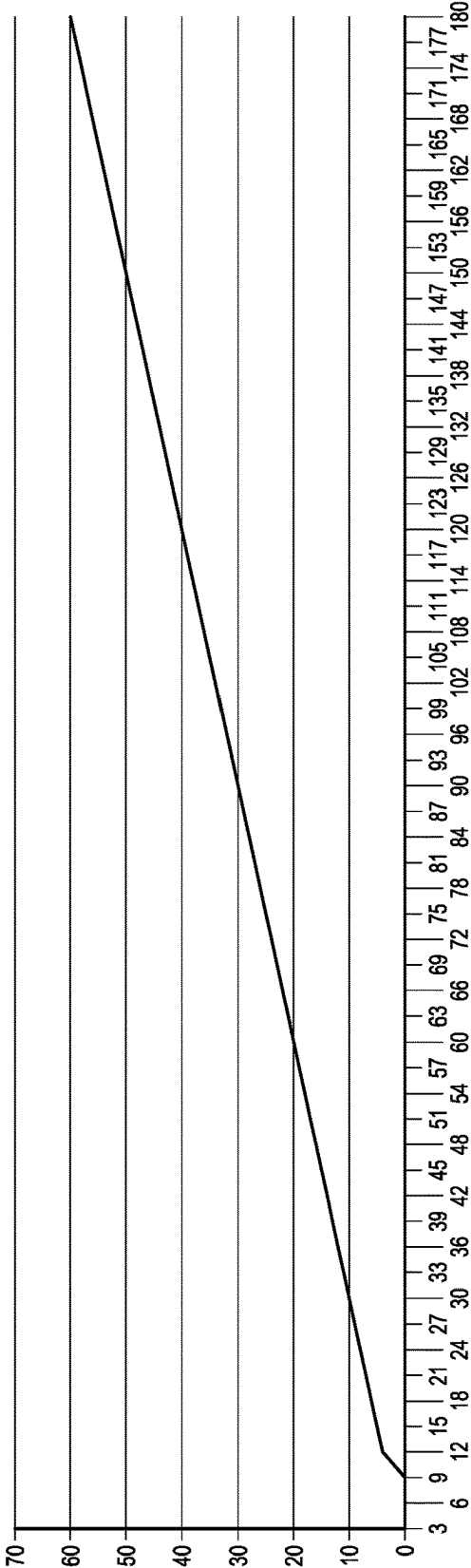


FIG. 15

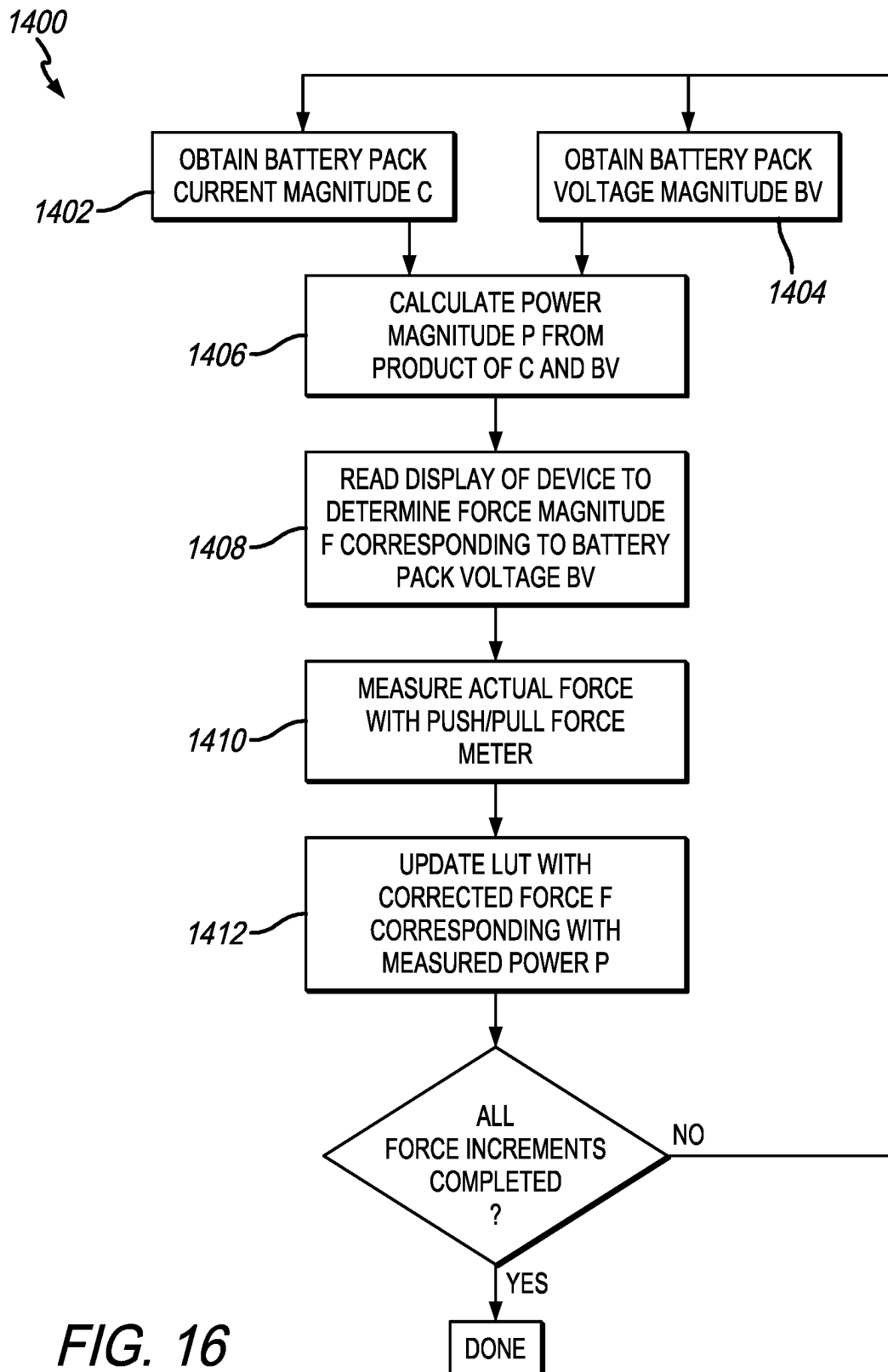


FIG. 16

FINAL CALIBRATED FORCE METER - POWER MODE

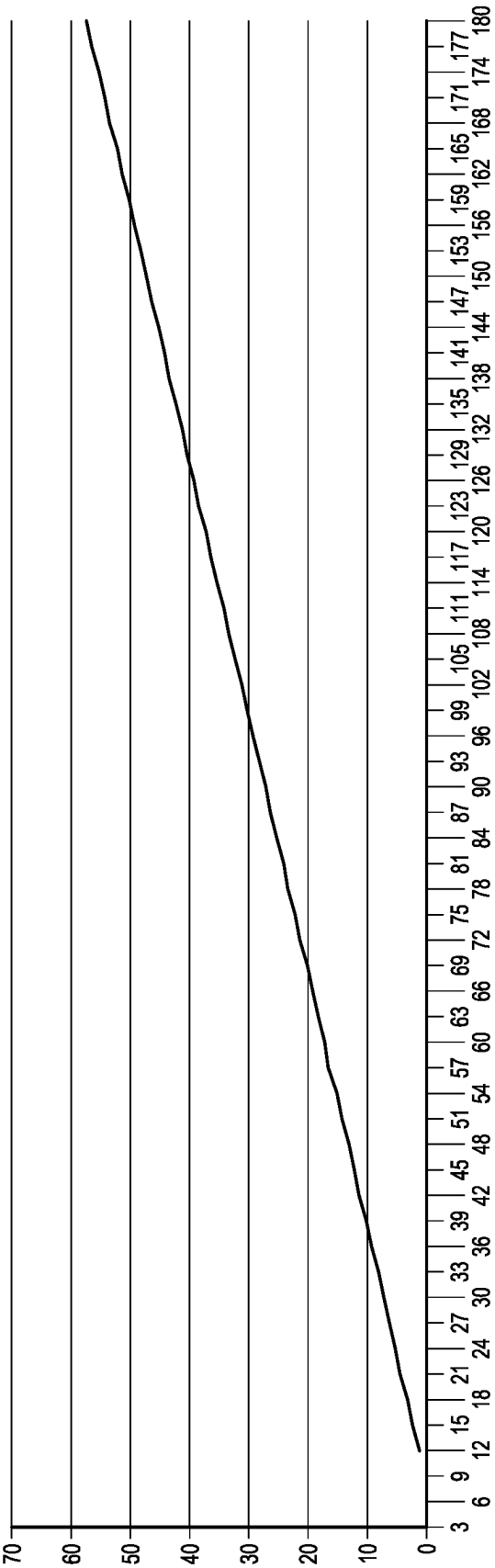
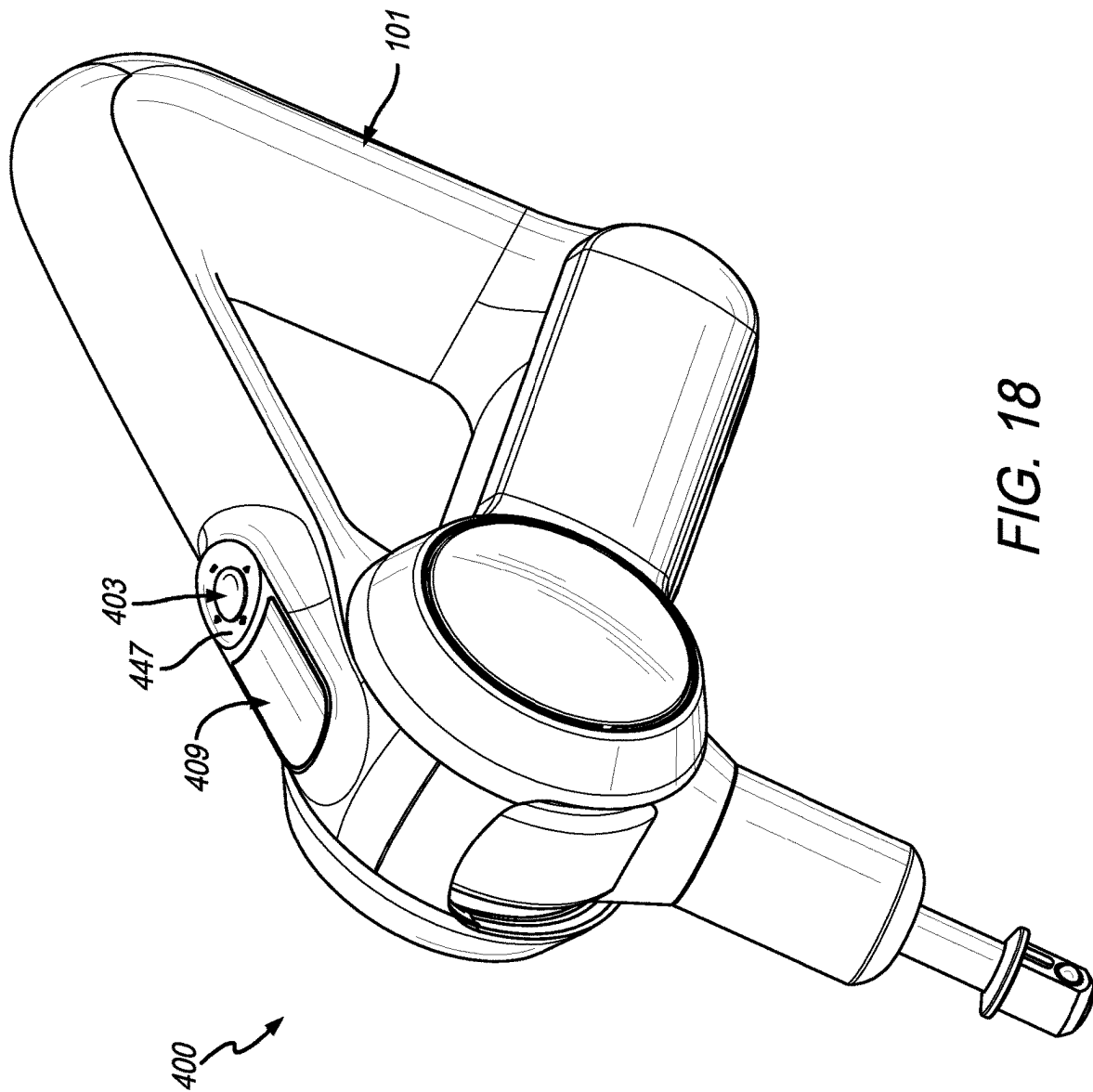


FIG. 17



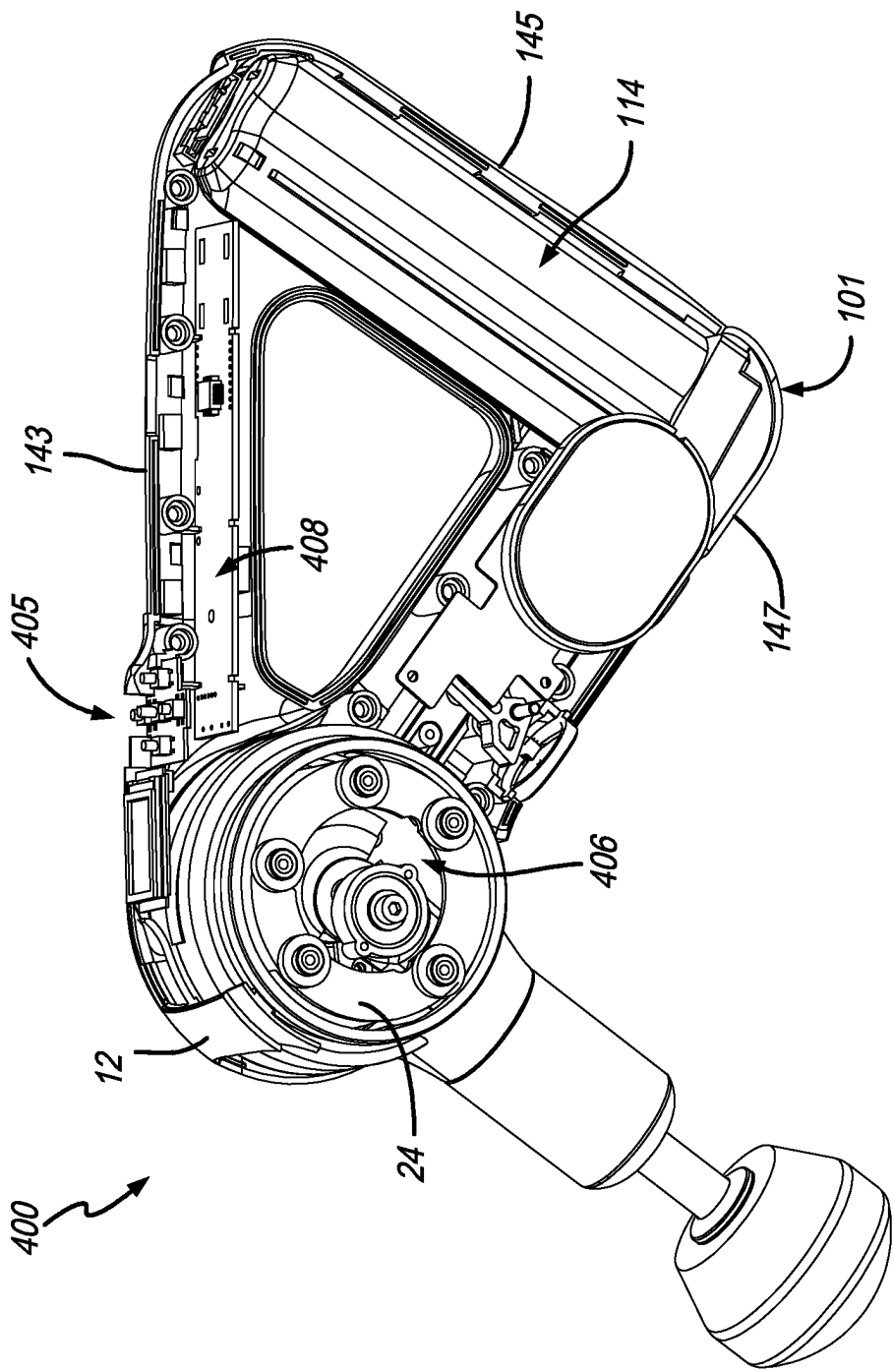
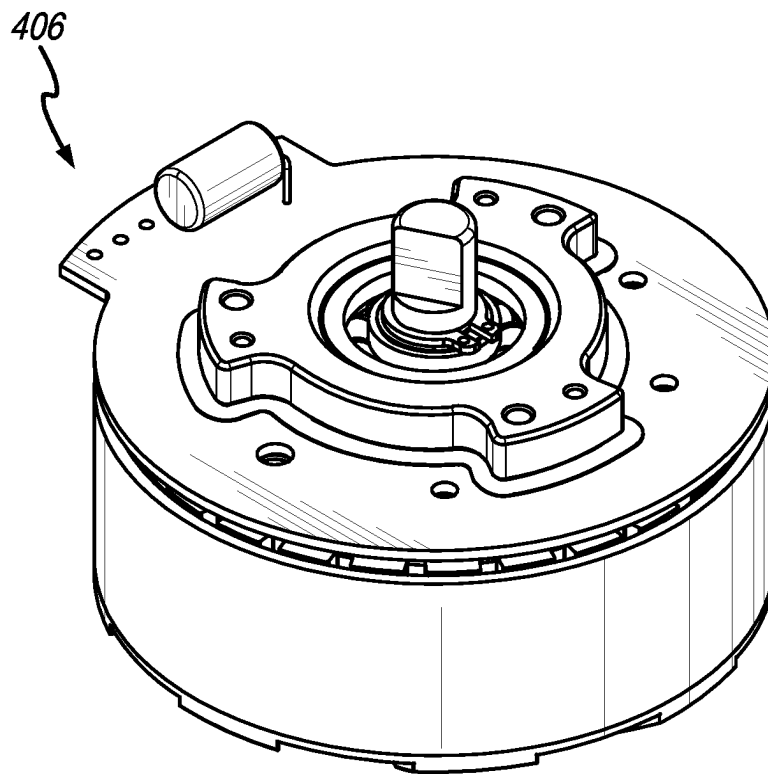


FIG. 19

*FIG. 20*

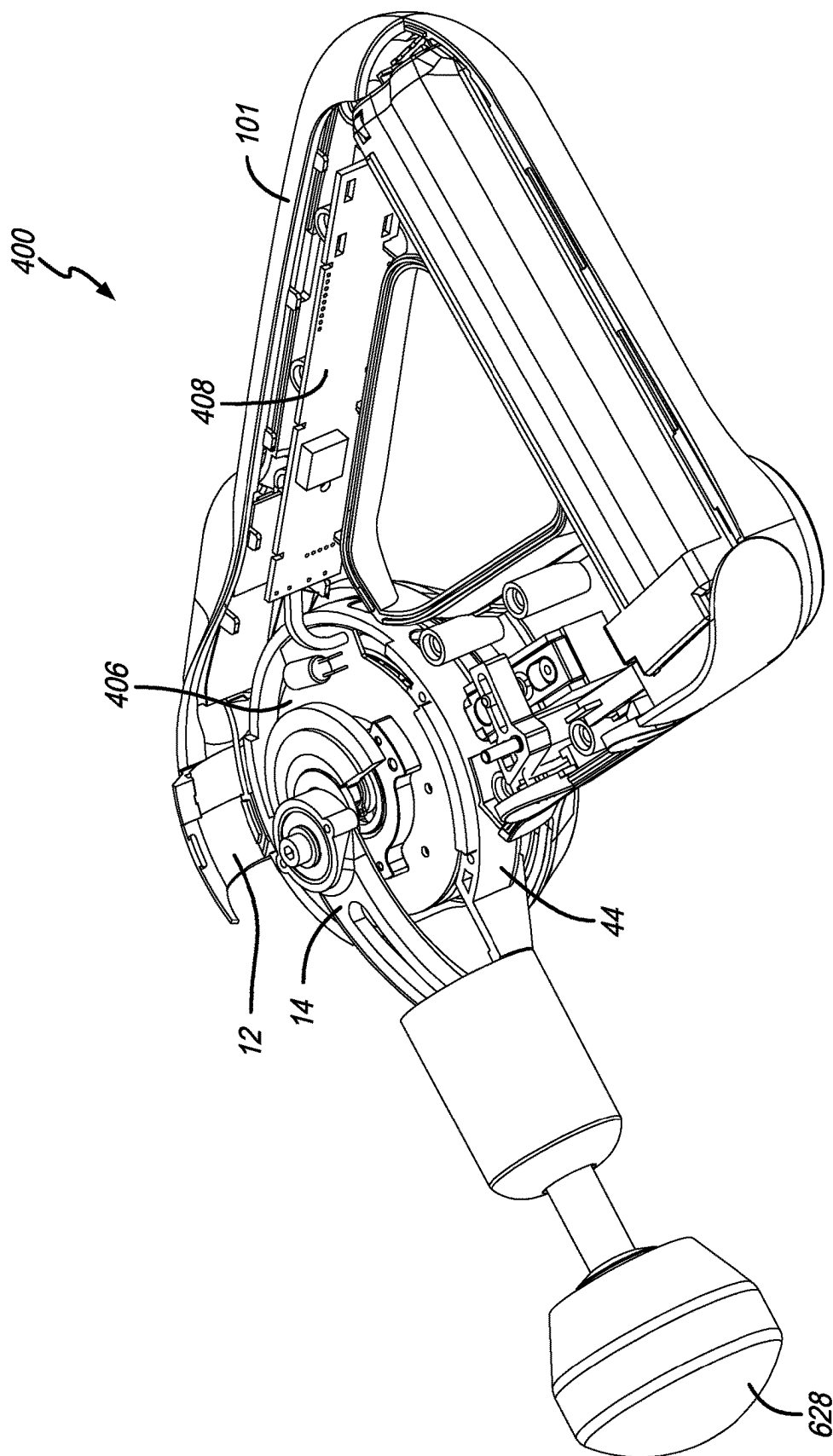


FIG. 21

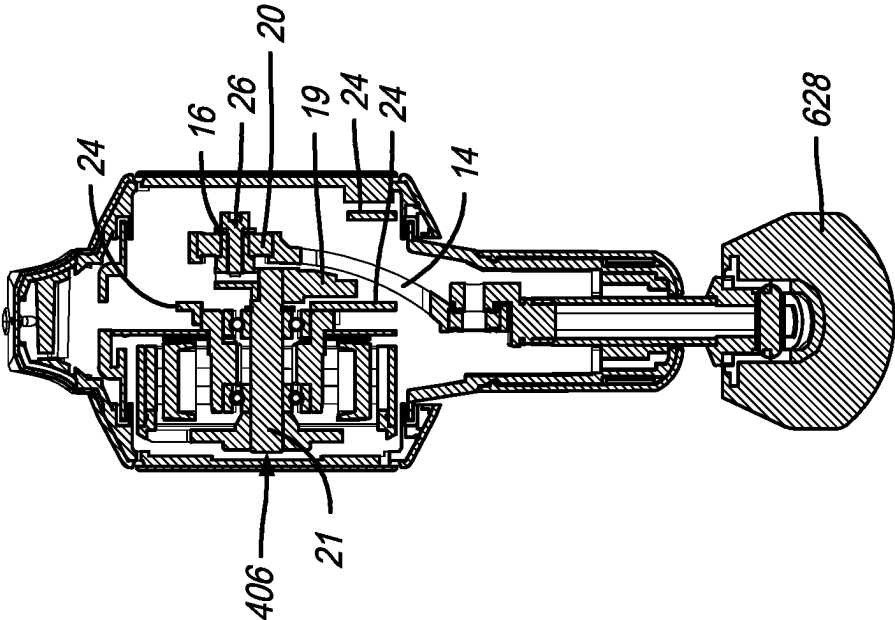


FIG. 22B

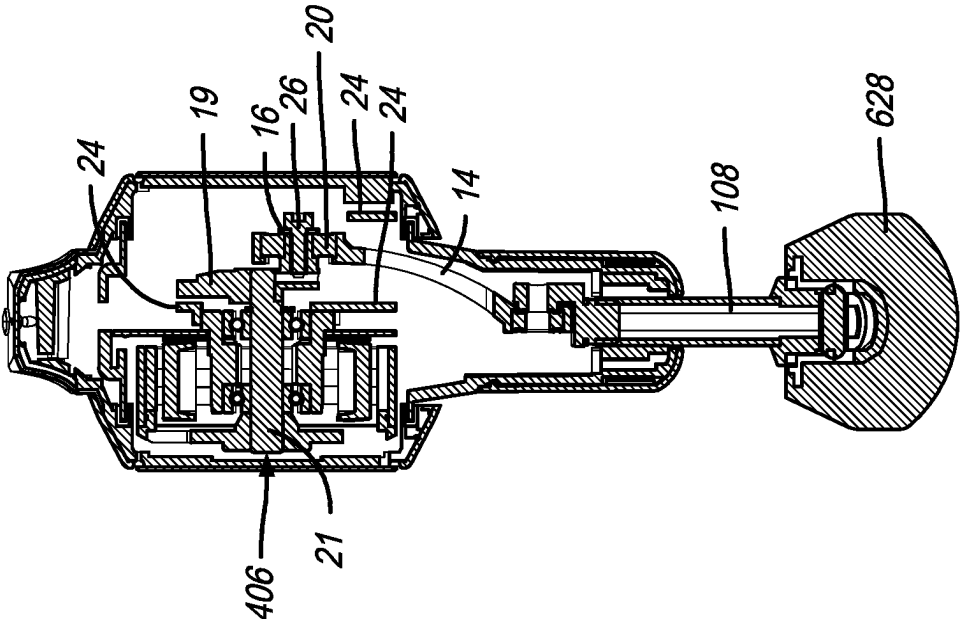


FIG. 22A

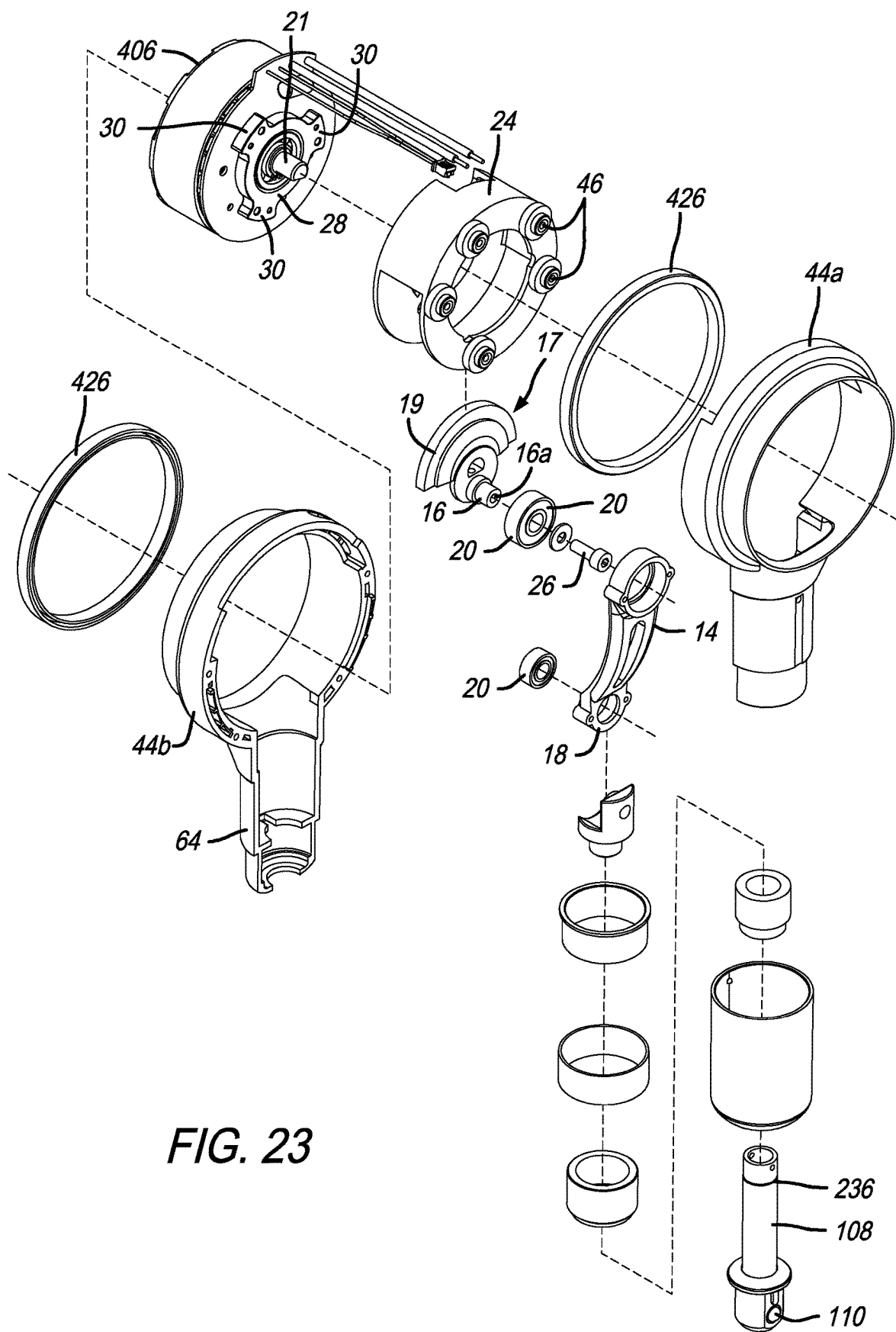


FIG. 23

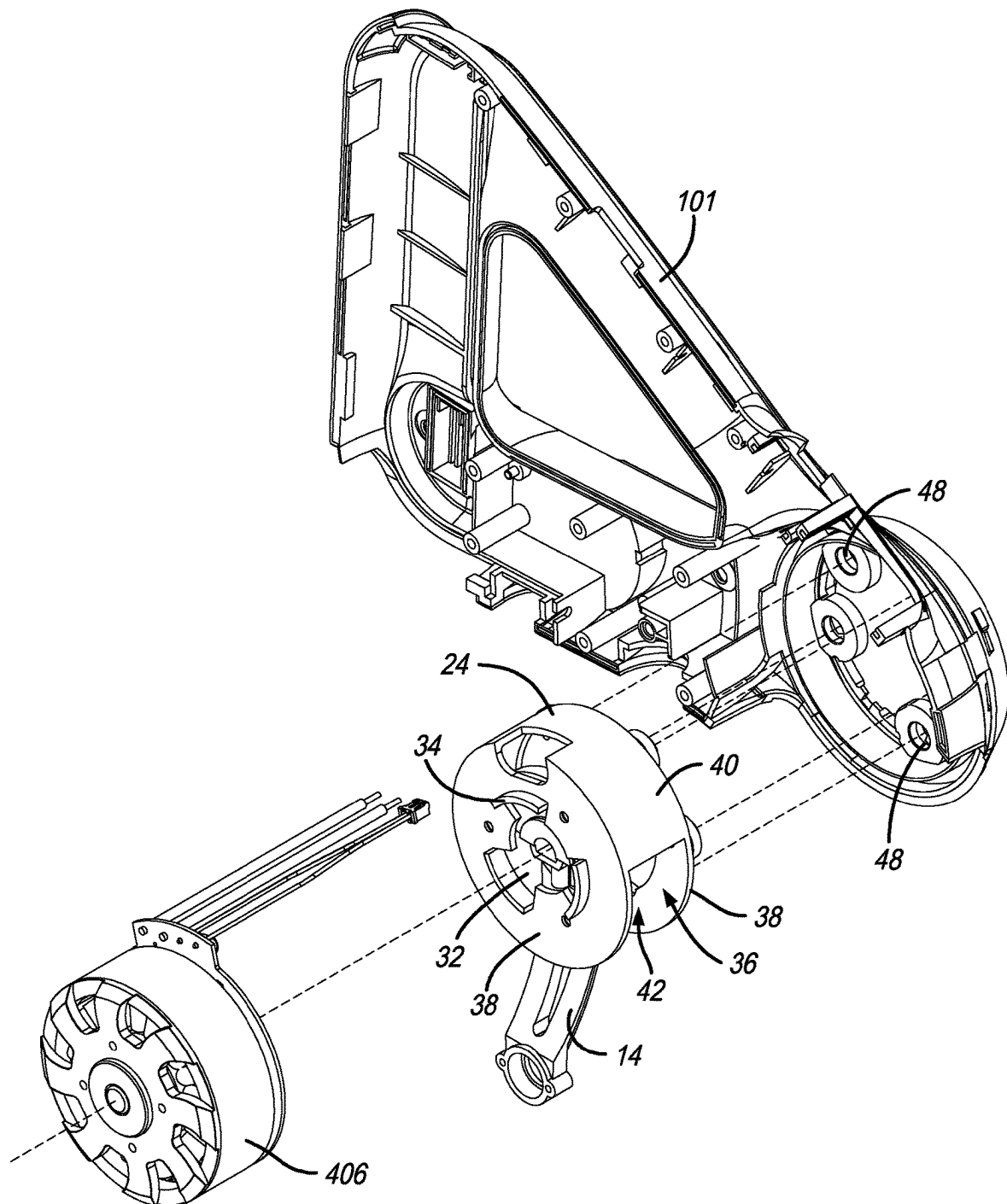


FIG. 23A

PROTOCOL 1

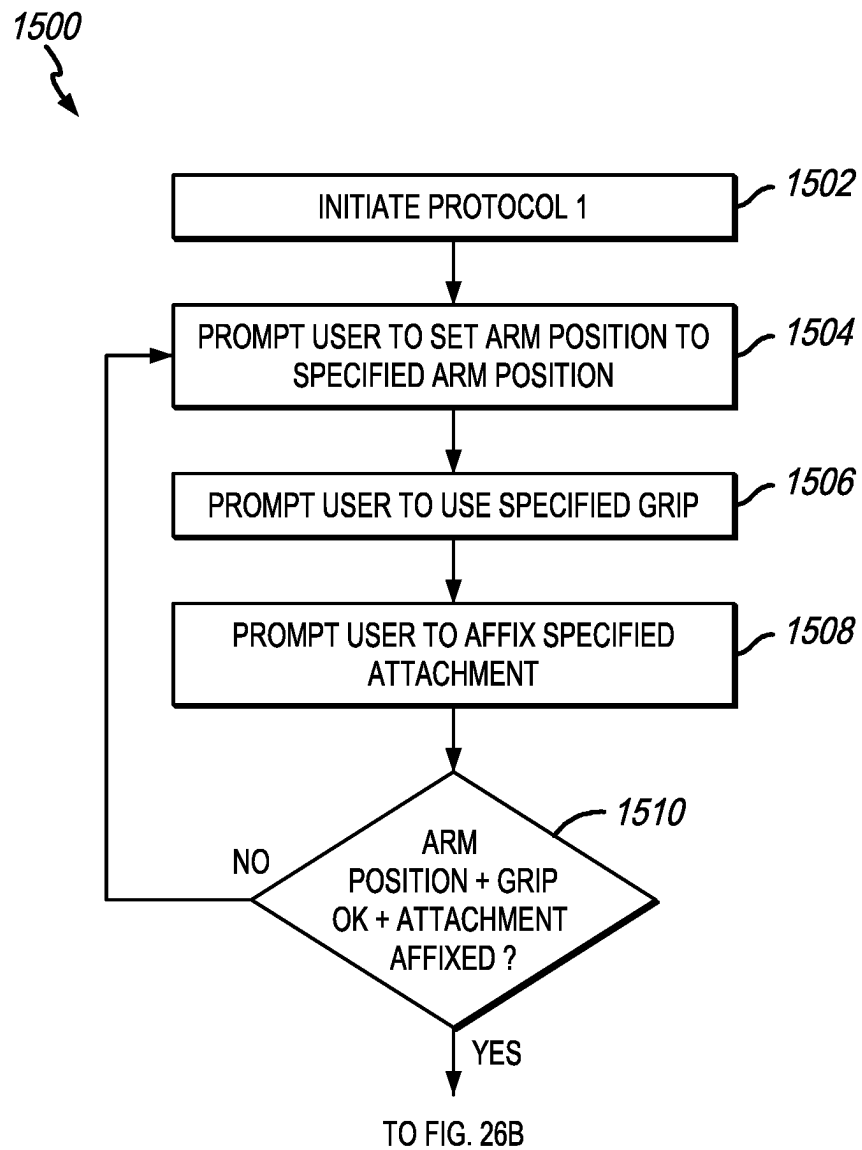
STEP	1	2	3	4
TIME(M)	0:30	0:15	0:30	0:45
SPEED (RPM)	1550	2100	2200	2400
AMPLITUDE	2	3	1	4
ATTACHMENT	DAMPENER	SMALL BALL	DAMPENER	LARGE BALL
FORCE	1	3	3	2
TEMPERATURE (°C)	21	26	29	32
GRIP	1	1	1	1

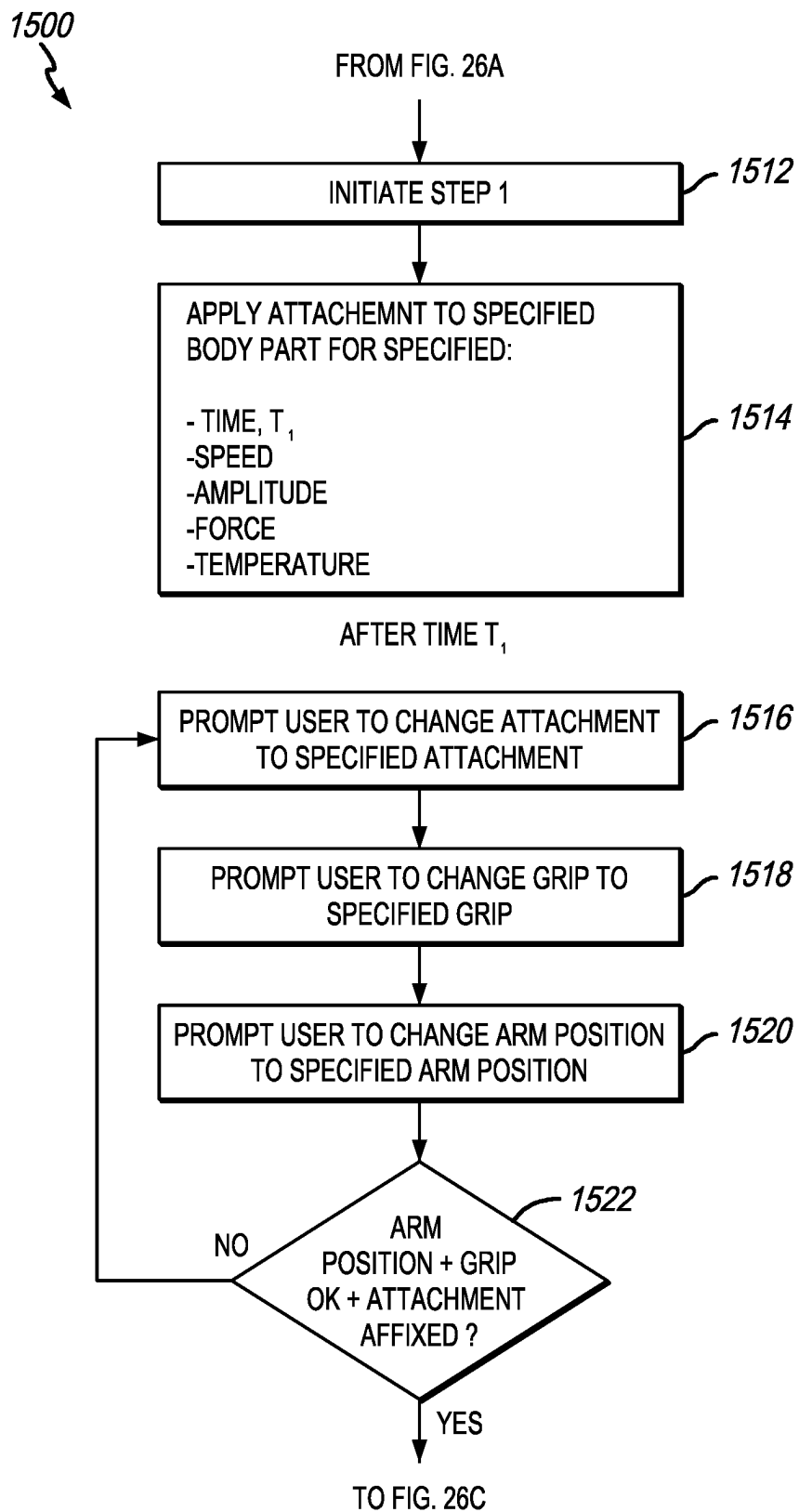
FIG. 24

PROTOCOL: SHIN SPLINTS

STEP	1	2	3	4
TIME(M)	1:00	1:00	1:00	1:00
SPEED (RPM)	1500	1500	2000	2000
AMPLITUDE	1	1	3	3
ATTACHMENT	DAMPENER	DAMPENER	DAMPENER	DAMPENER
FORCE	2	2	3	3
TEMPERATURE (°C)	21	21	24	24
GRIP	REVERSE	REVERSE	BASE	BASE
ARM POSITION	1	1	1	1
BODY PART	R. SHIN	L. SHIN	R. CALF	L. CALF

FIG. 25

**FIG. 26A**

**FIG. 26B**

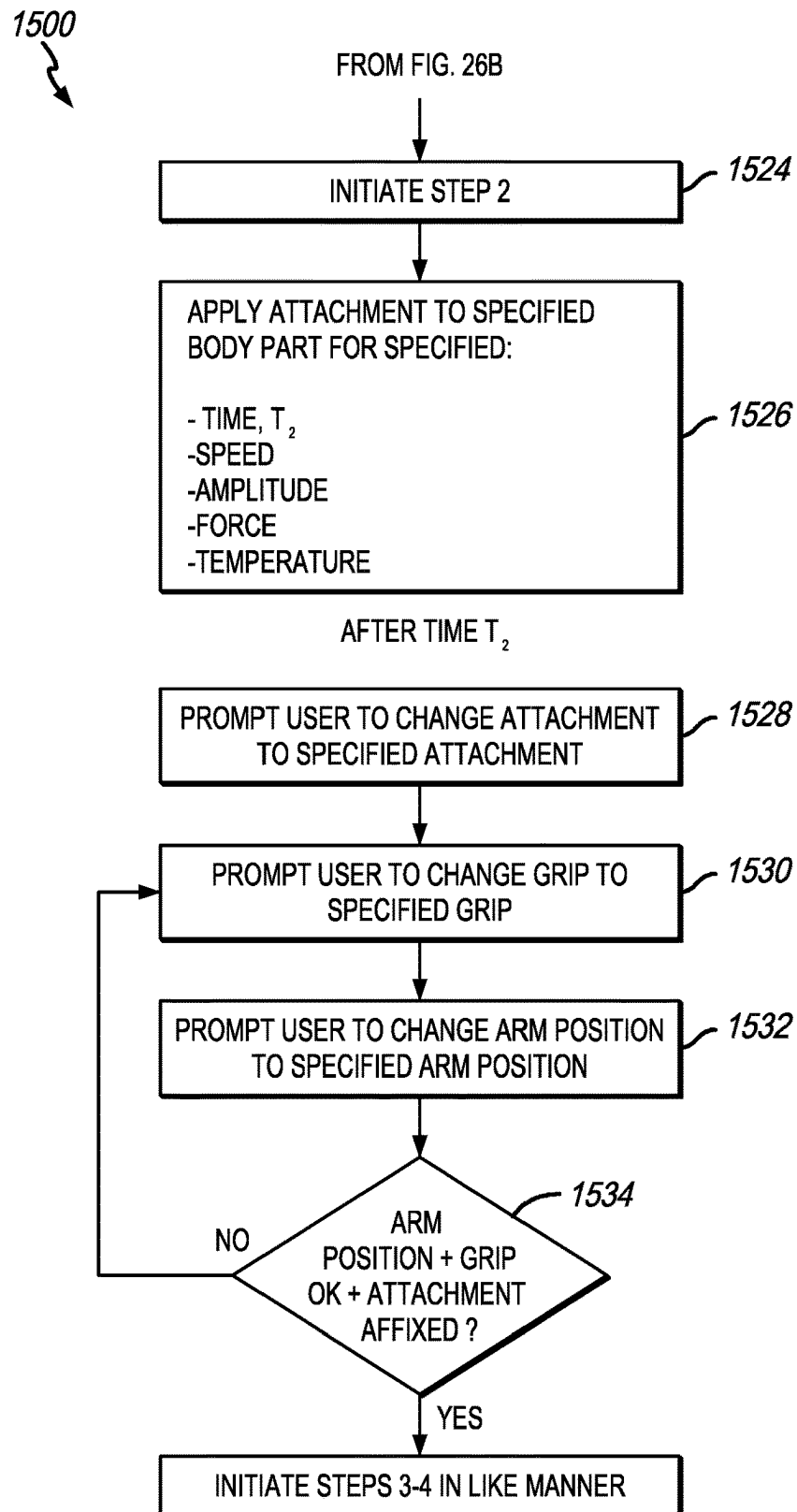
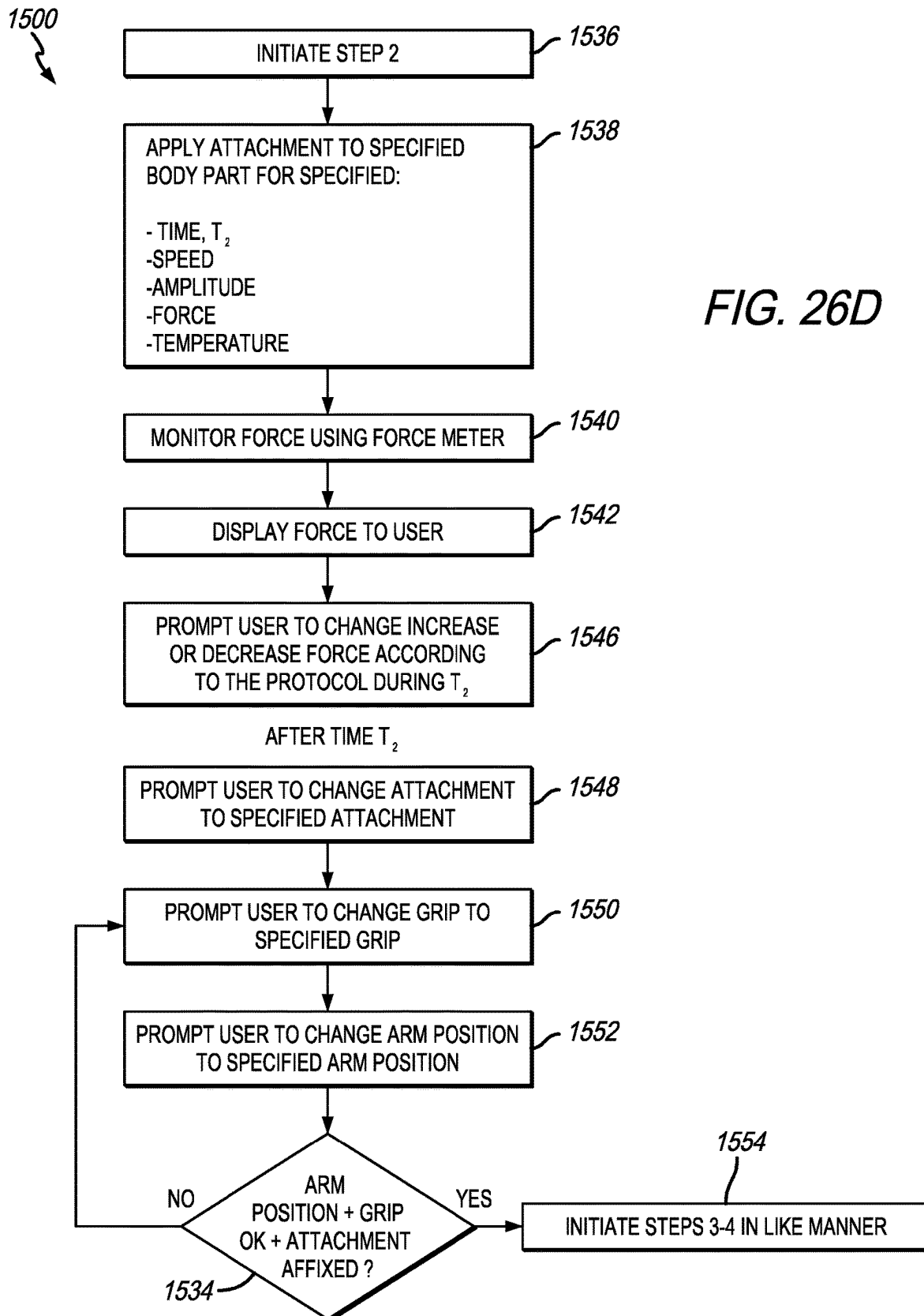


FIG. 26C



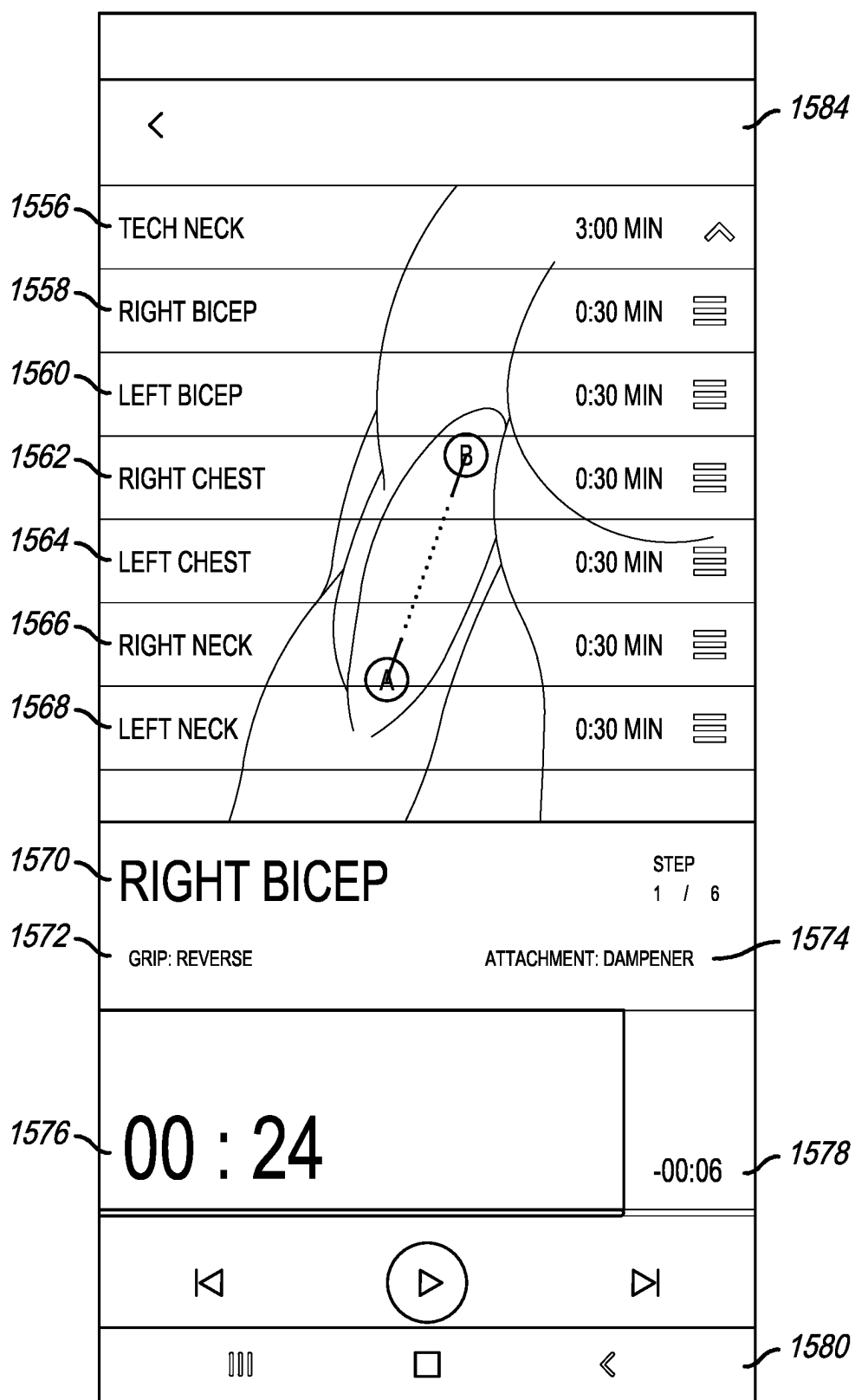


FIG. 27

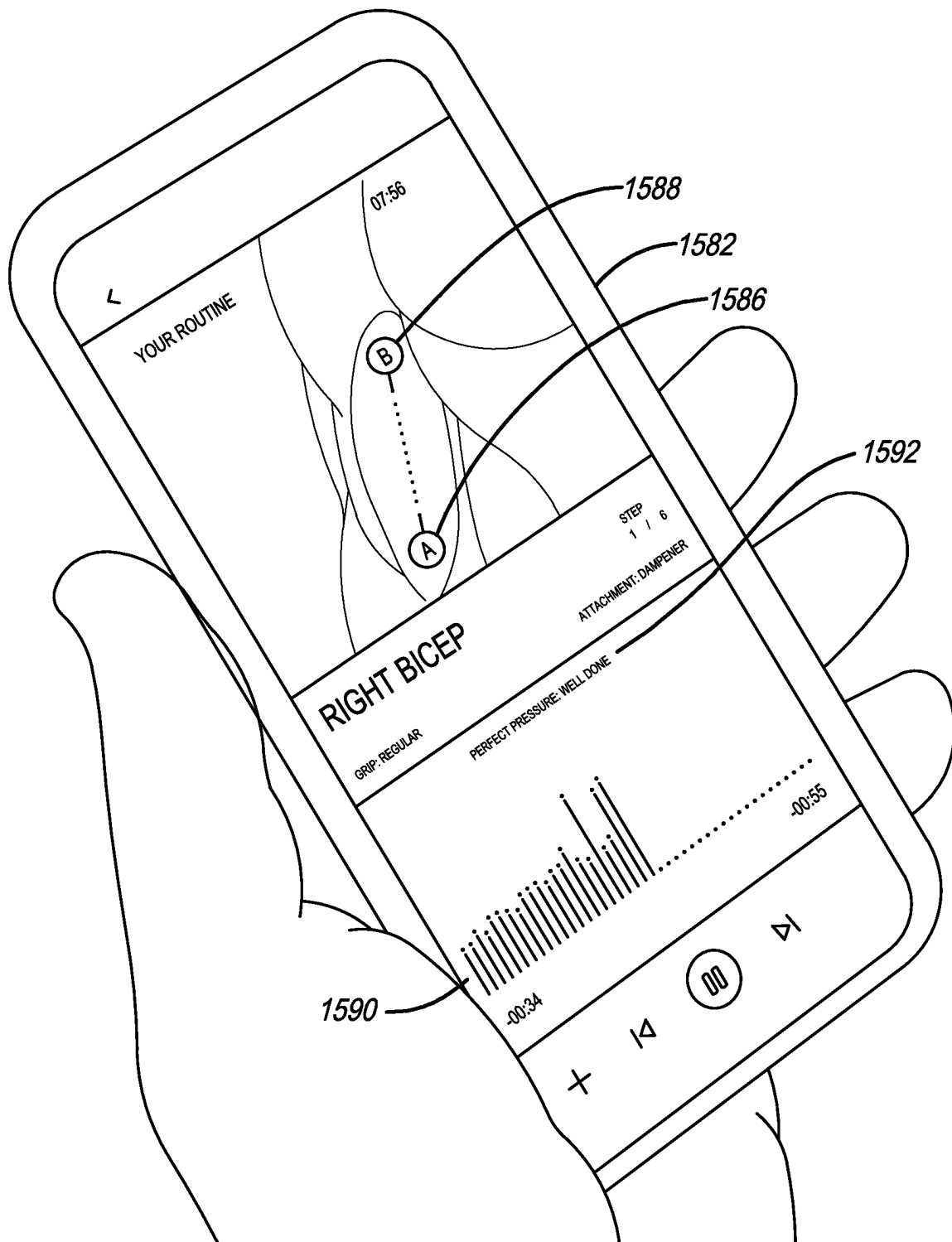
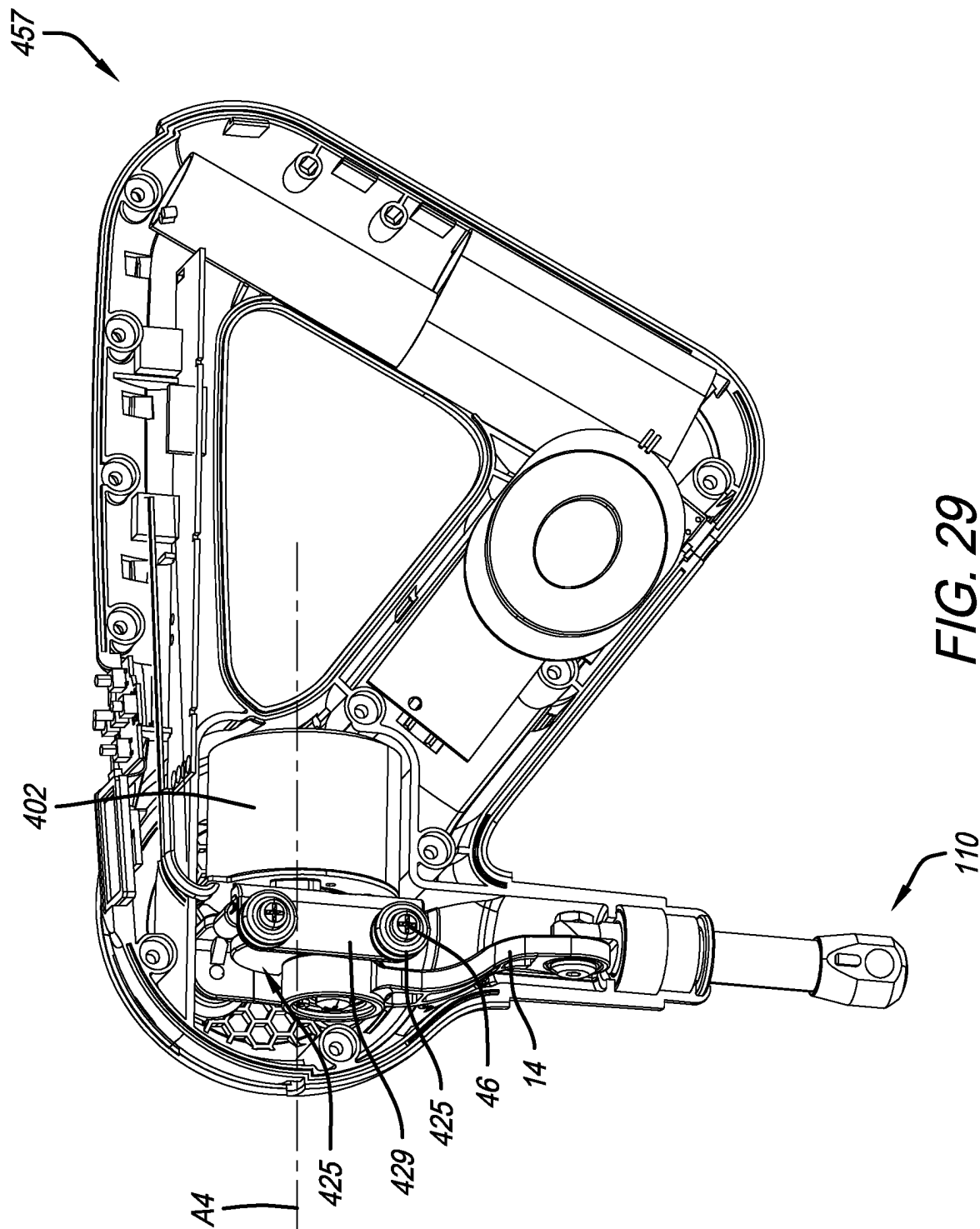


FIG. 28



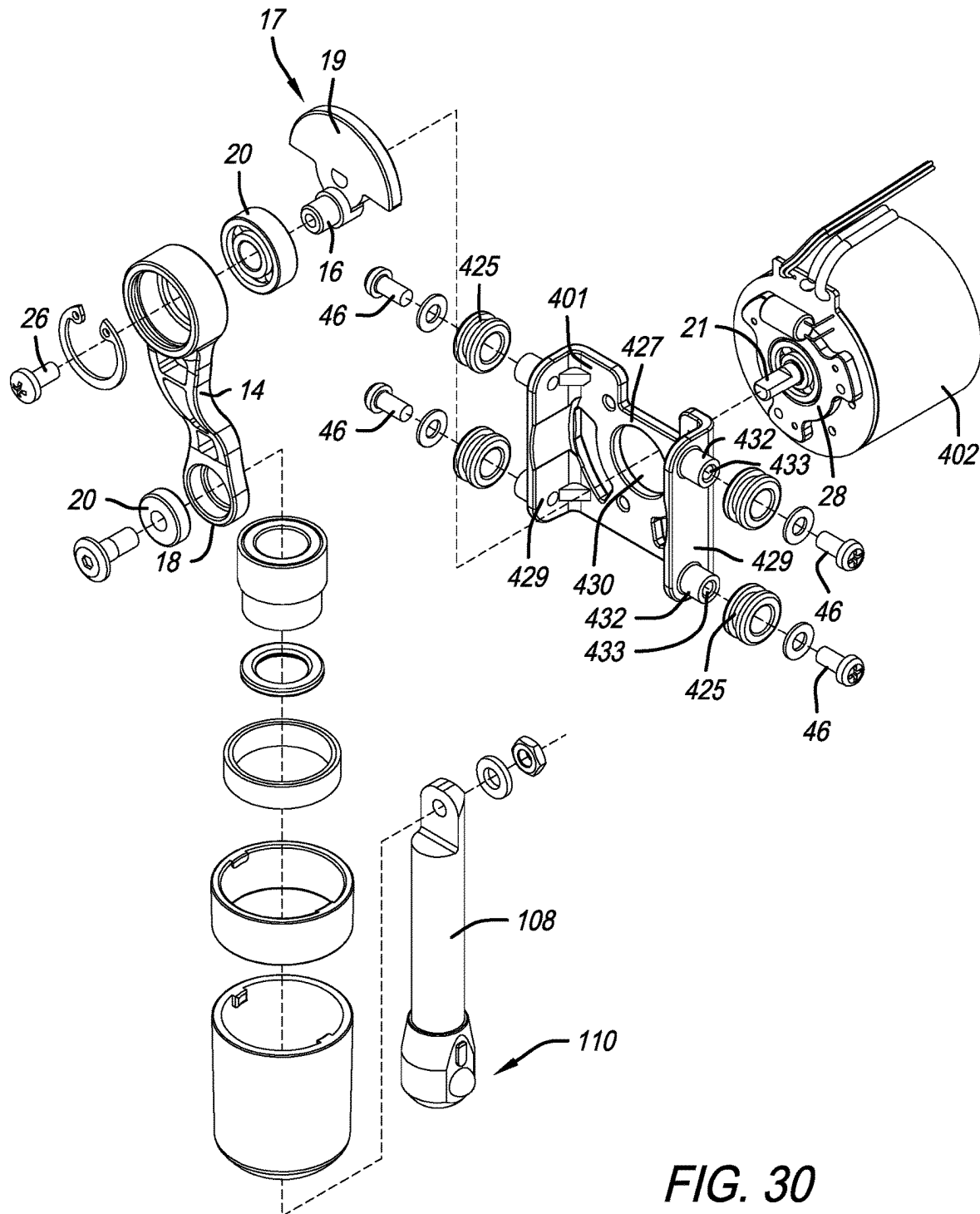


FIG. 30

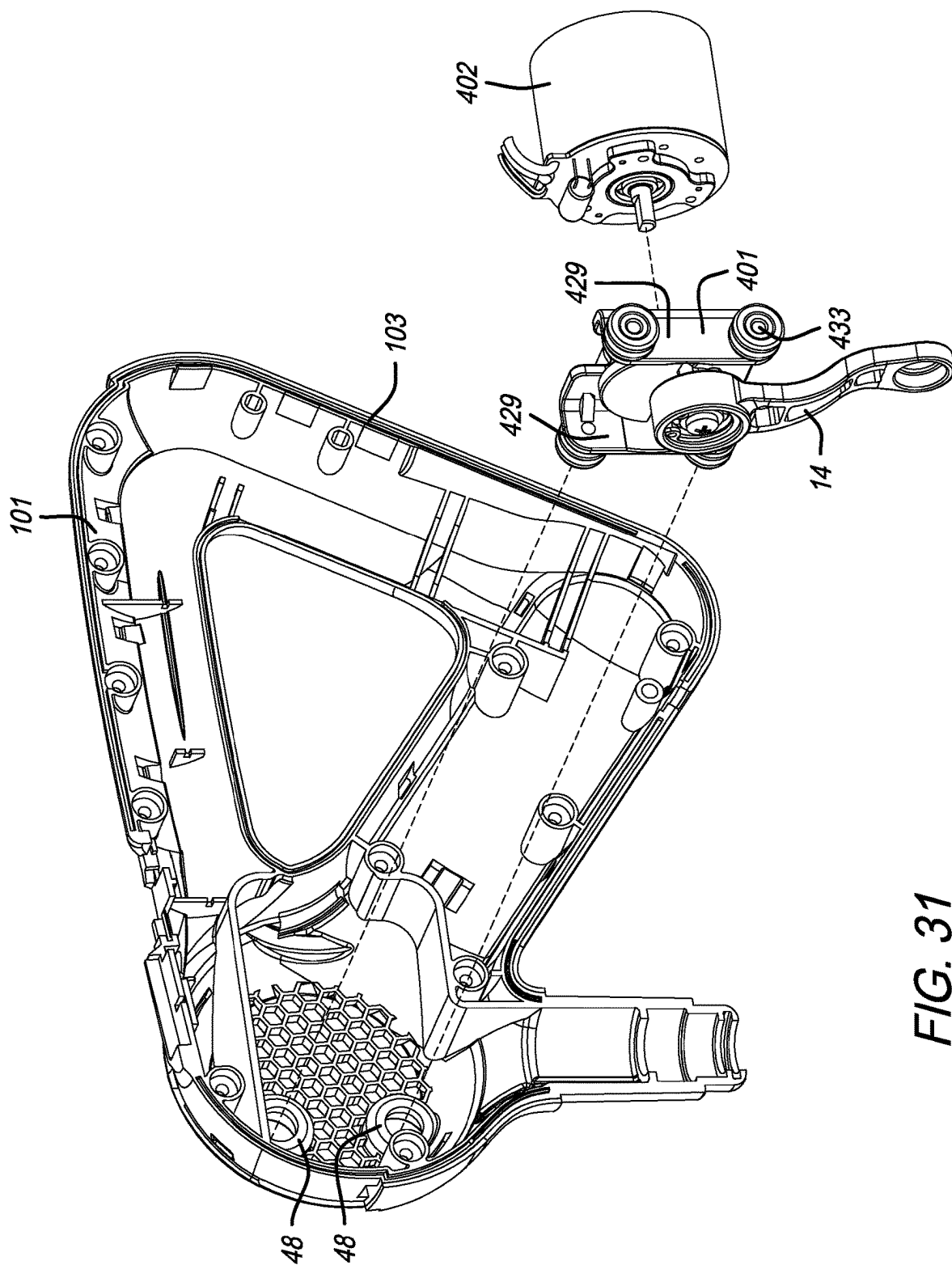


FIG. 31

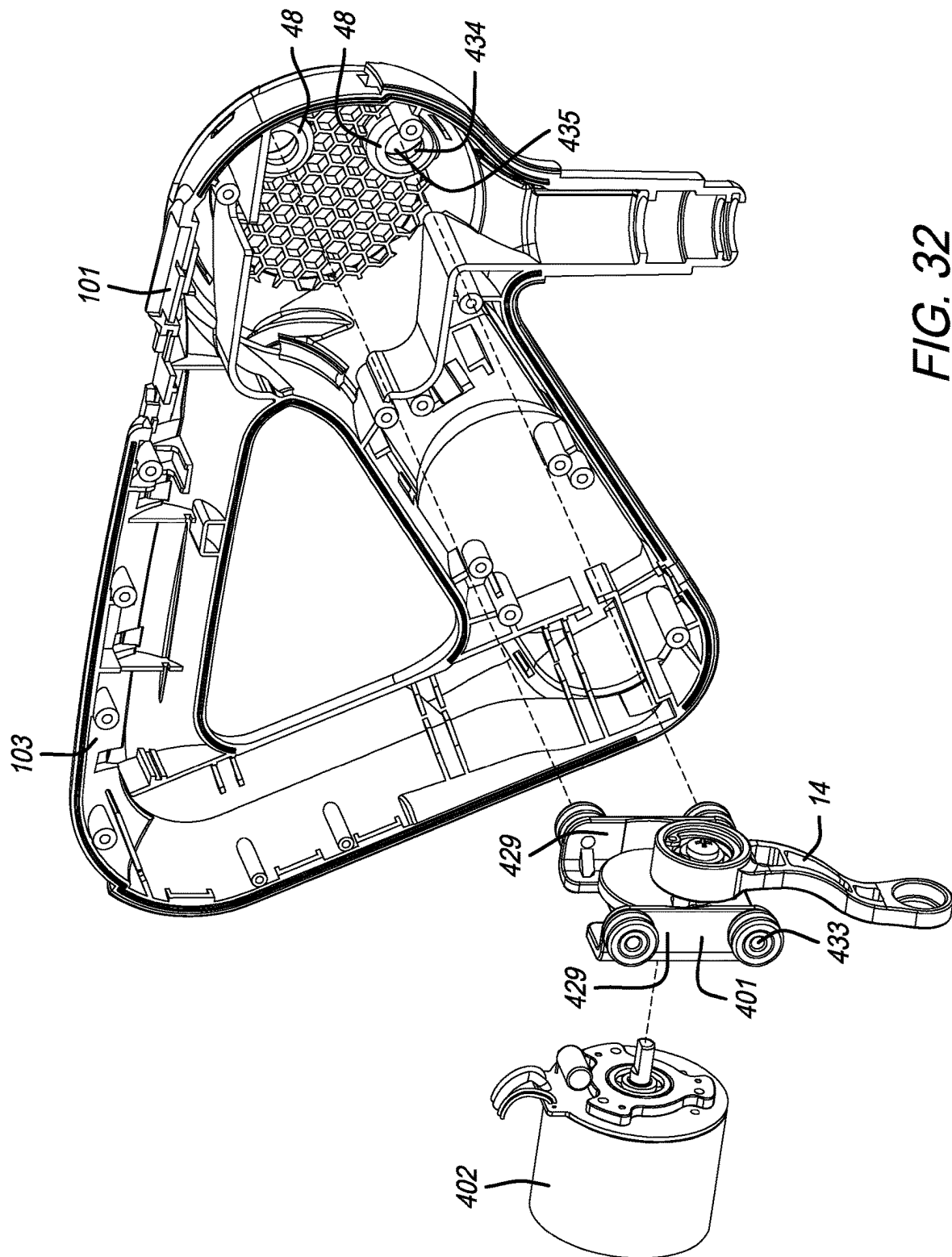


FIG. 32

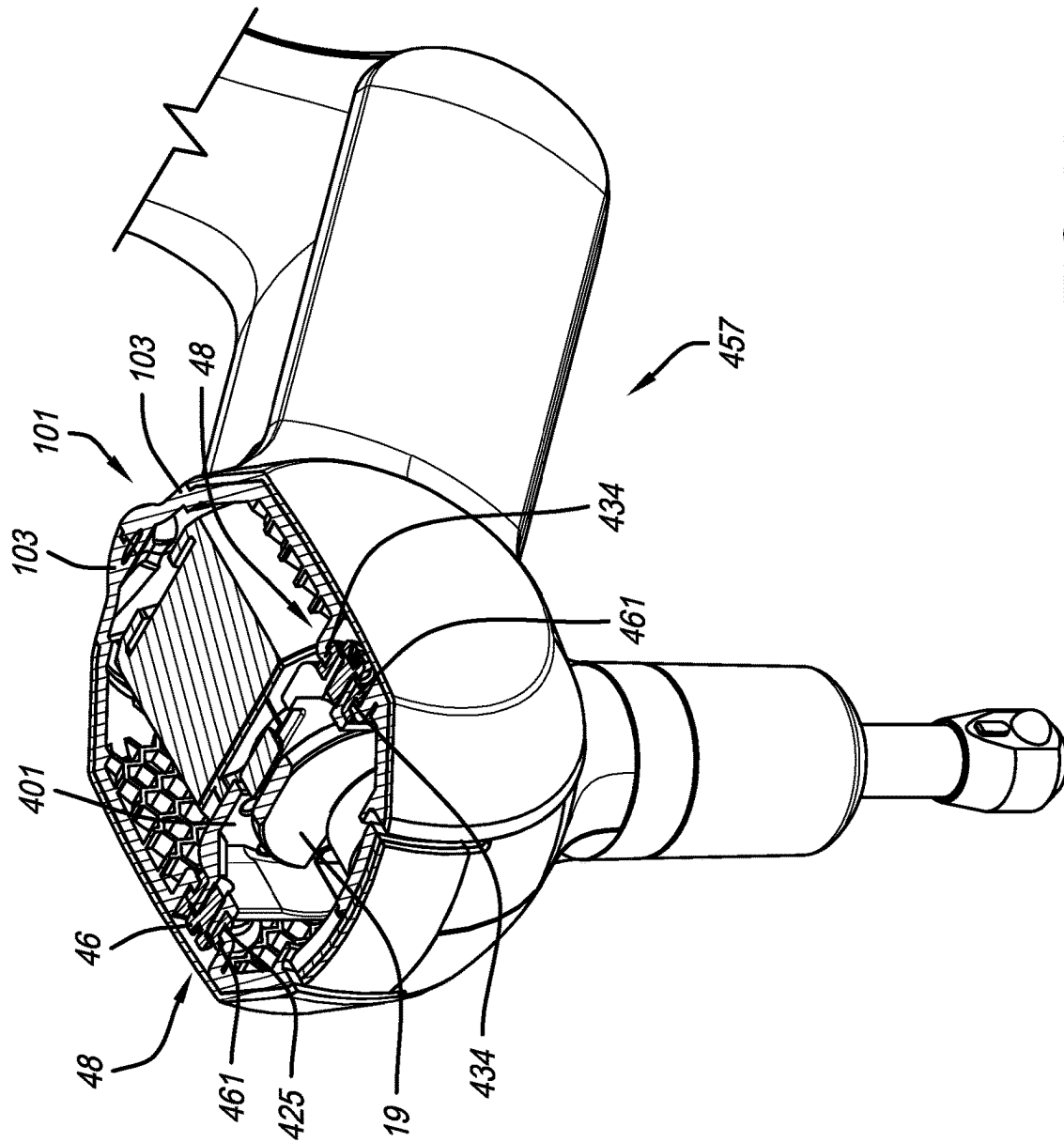


FIG. 33

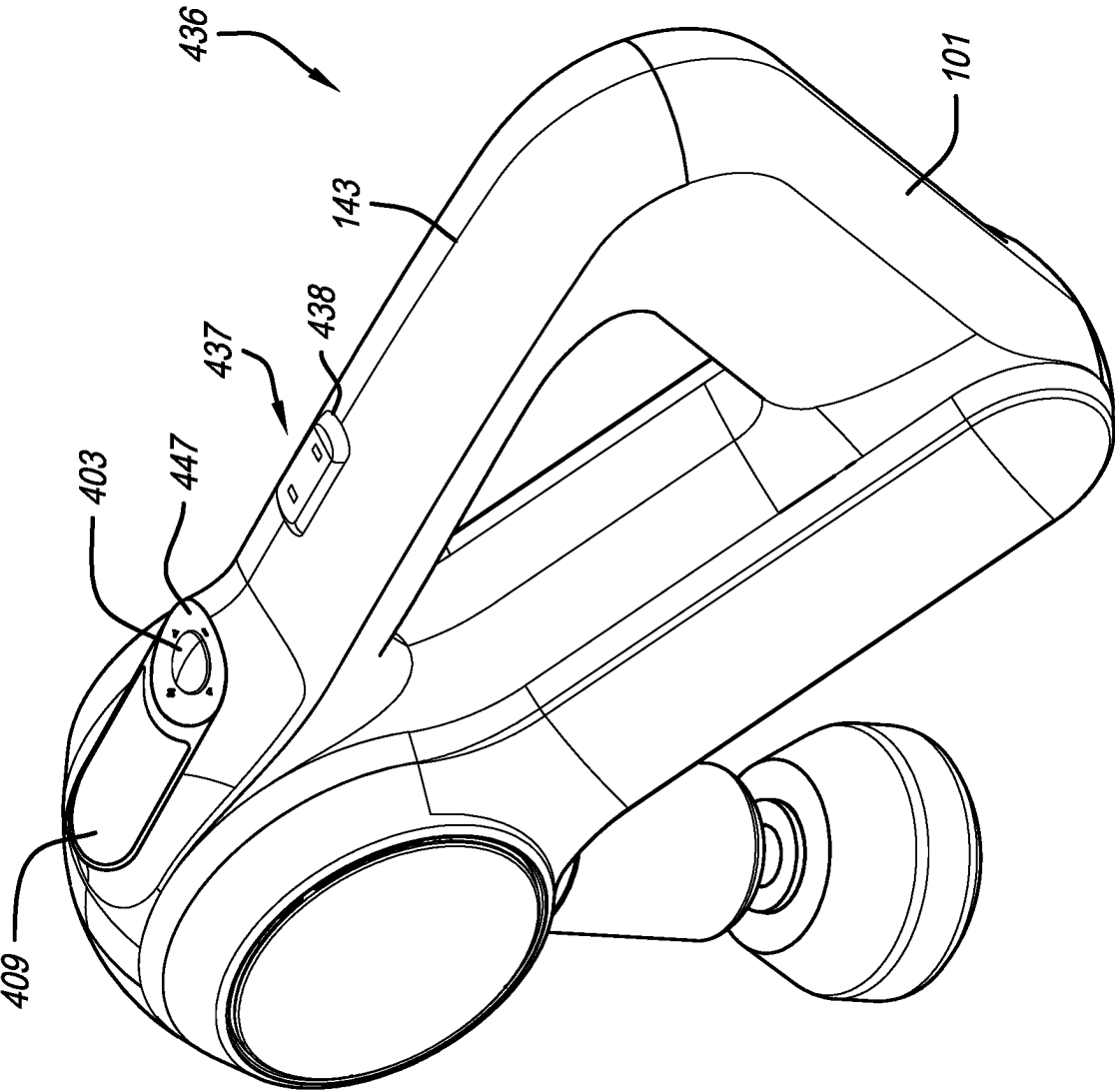


FIG. 34

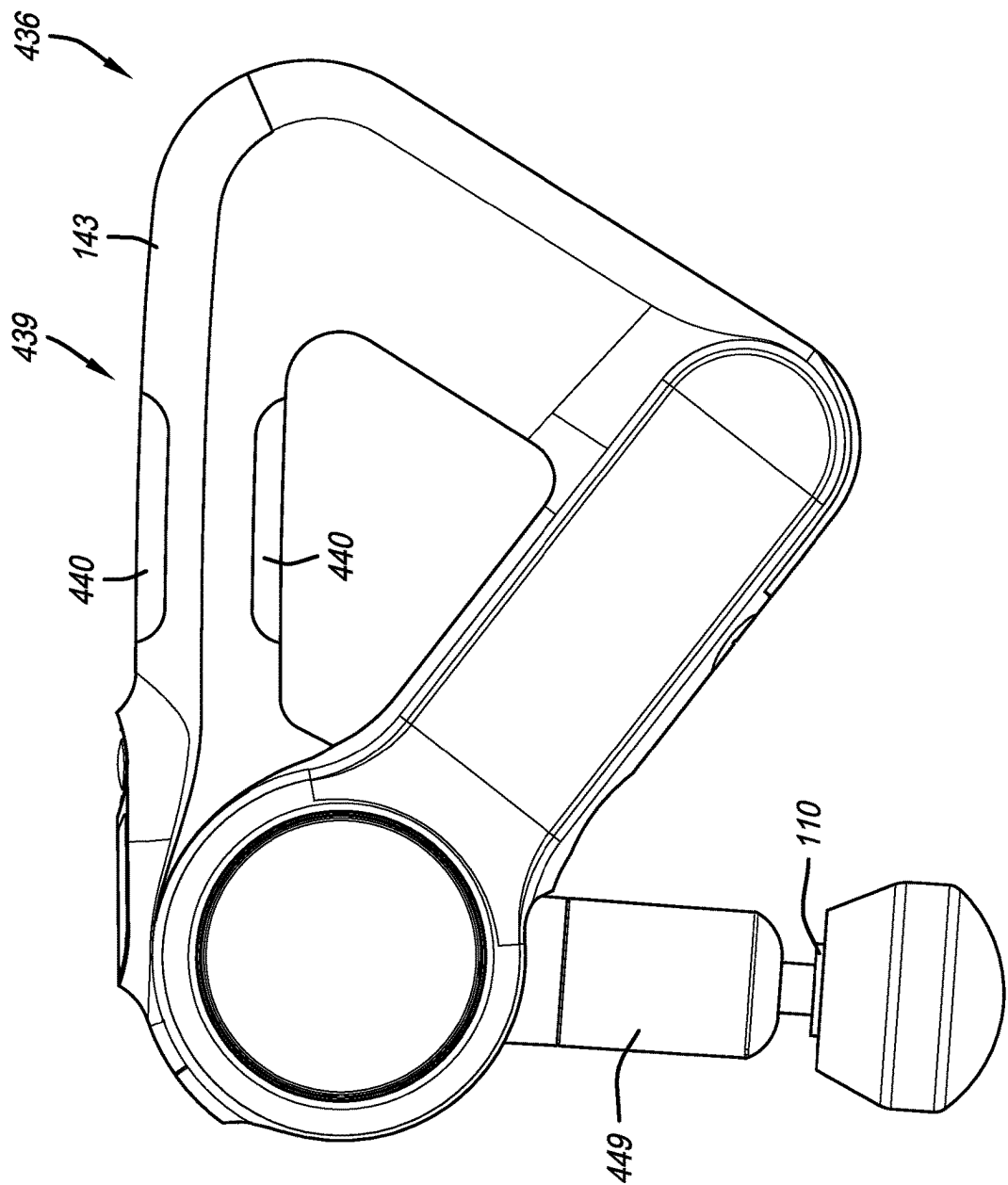


FIG. 35

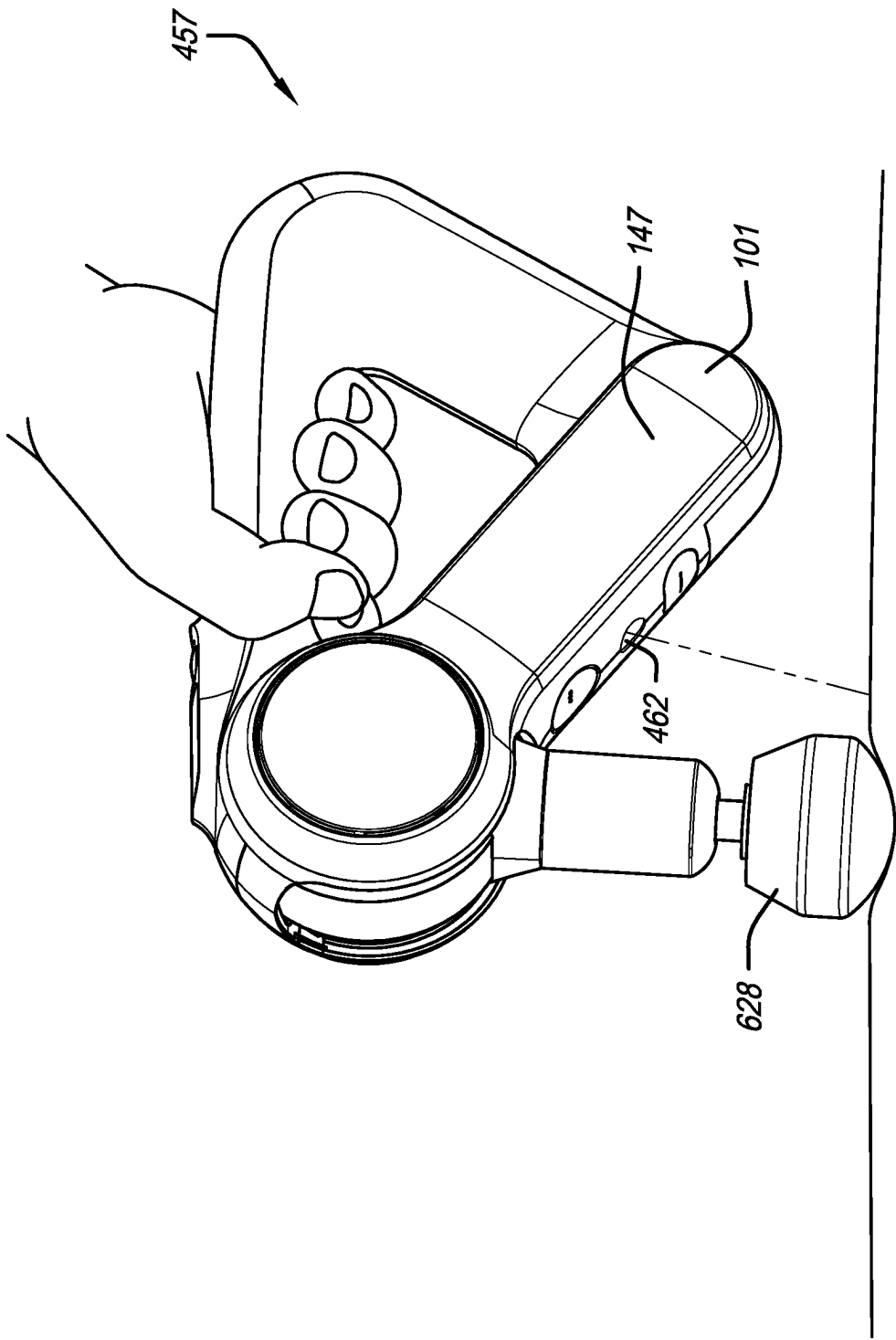


FIG. 36

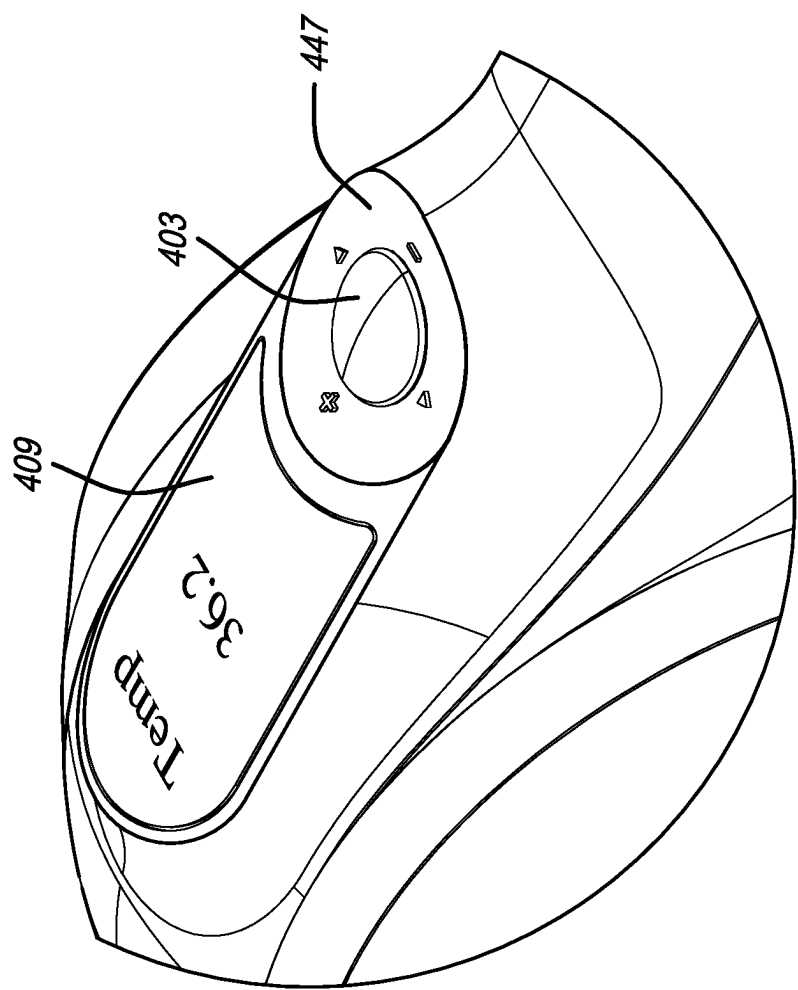


FIG. 36A

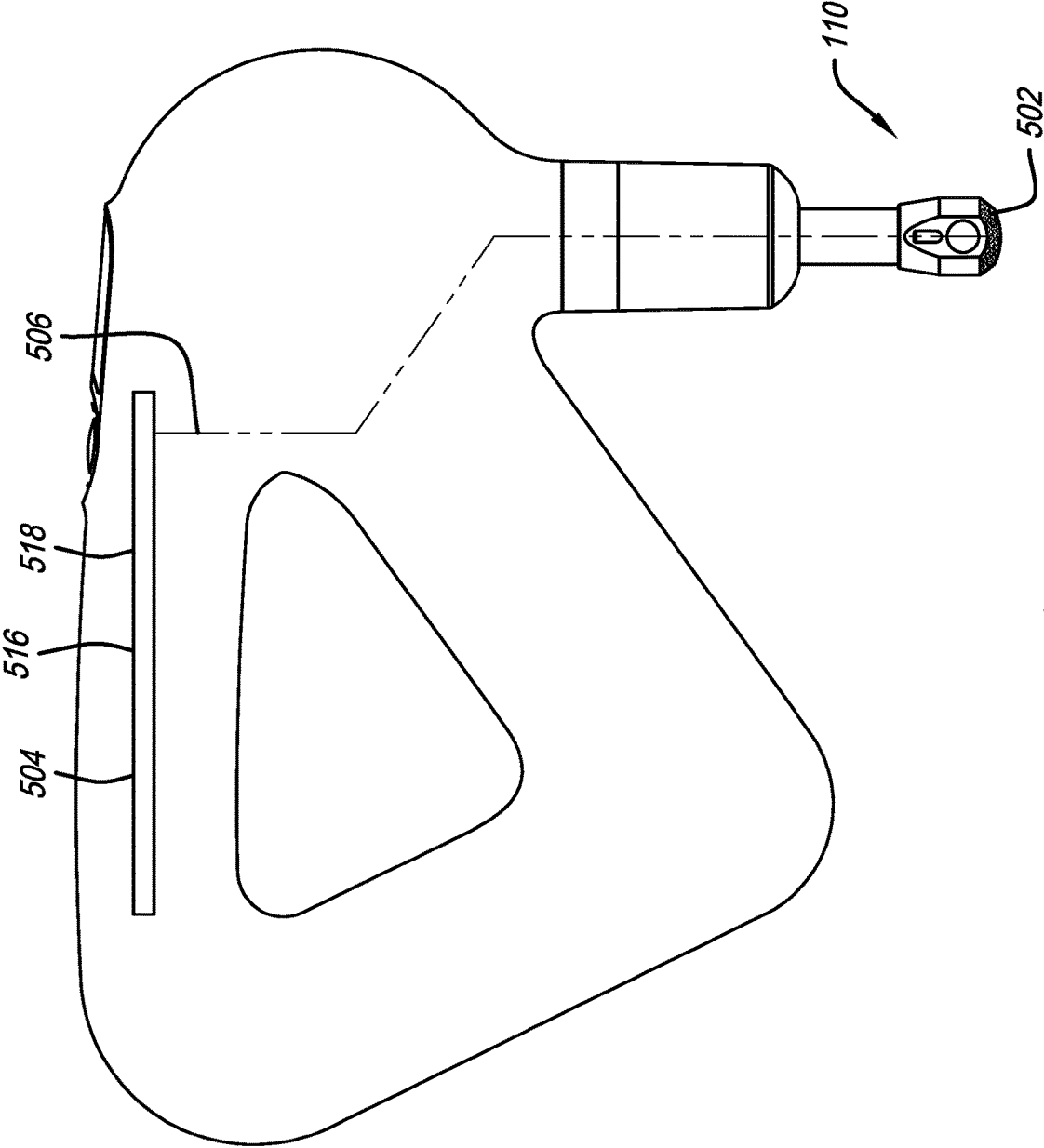


FIG. 37

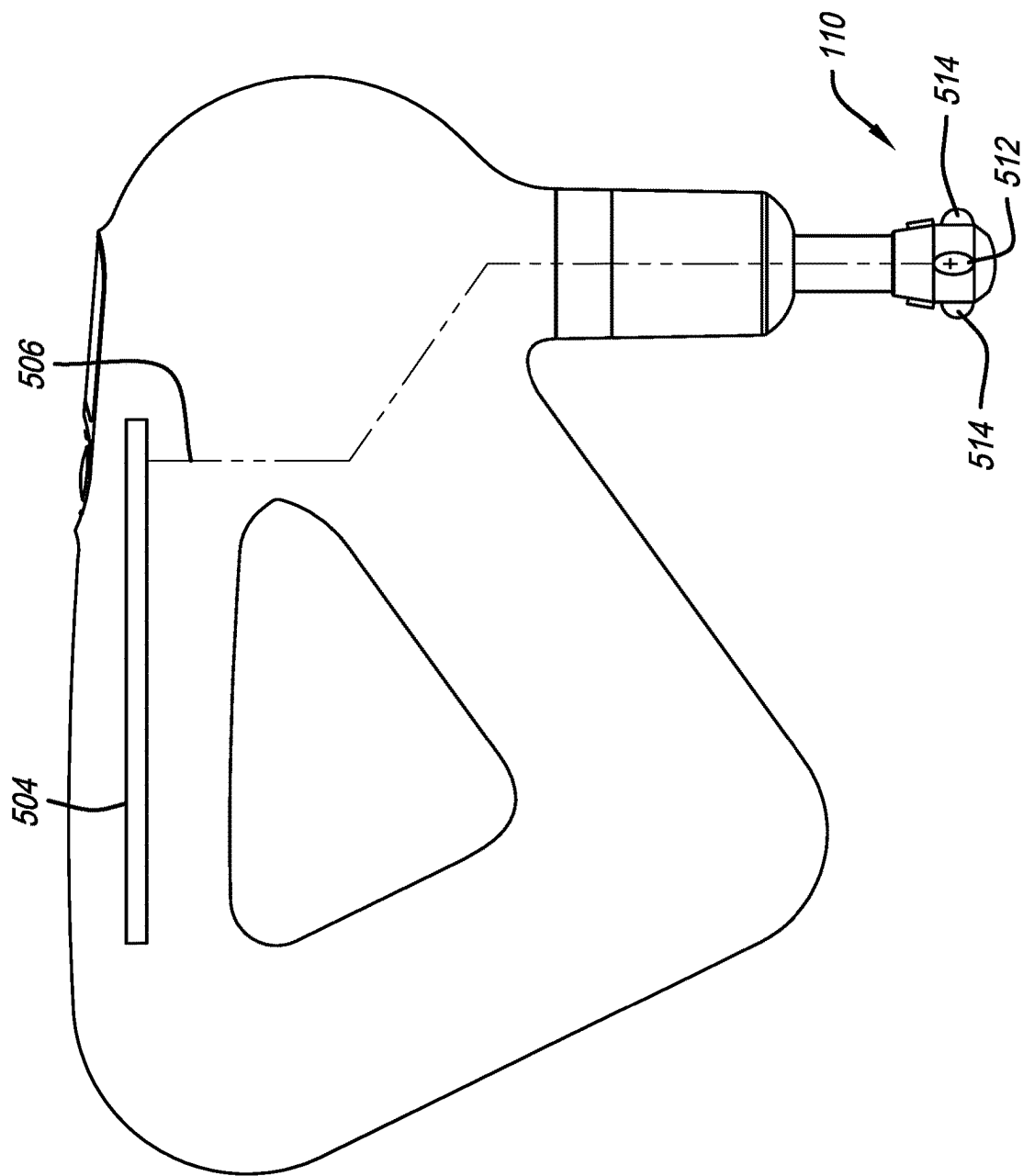


FIG. 38

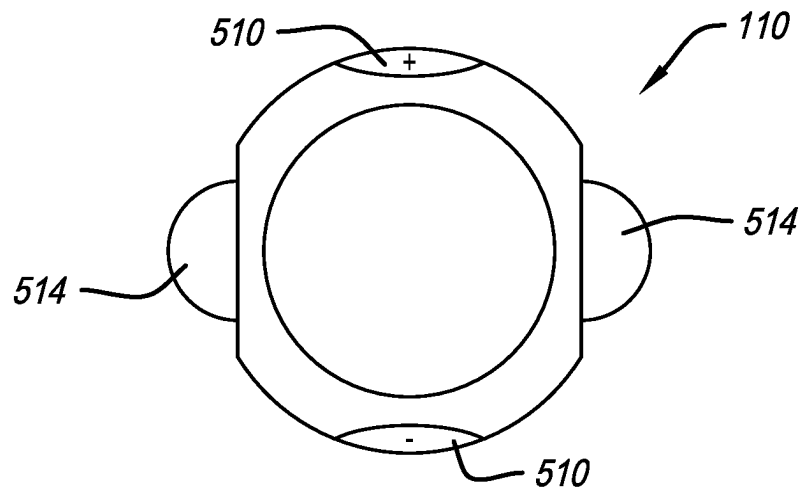


FIG. 39

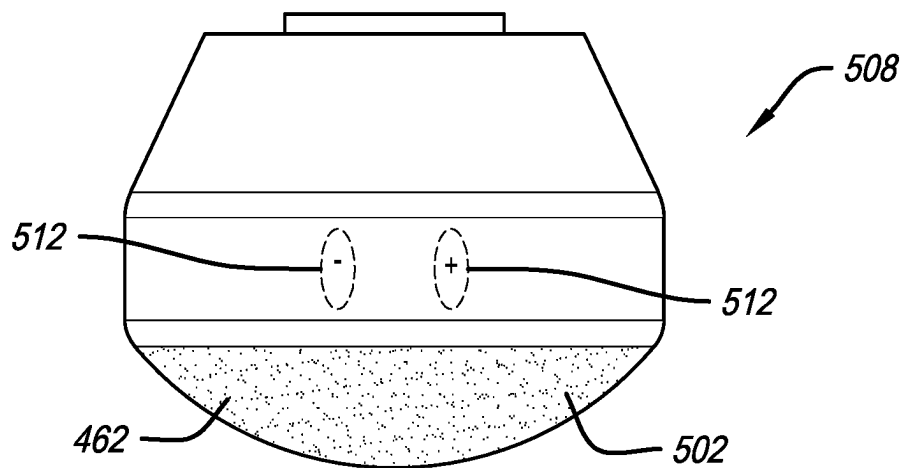
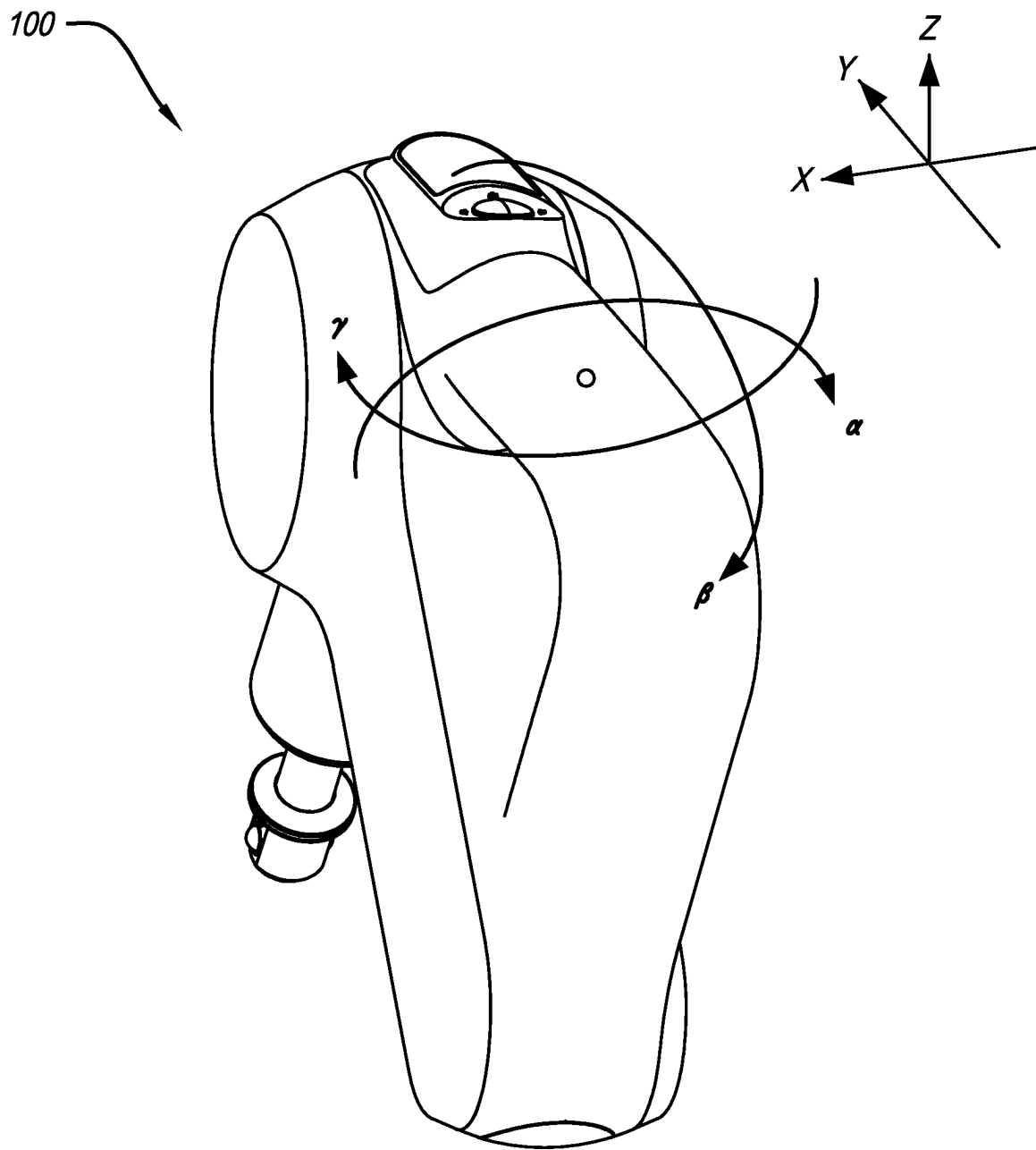


FIG. 40

**FIG. 41**

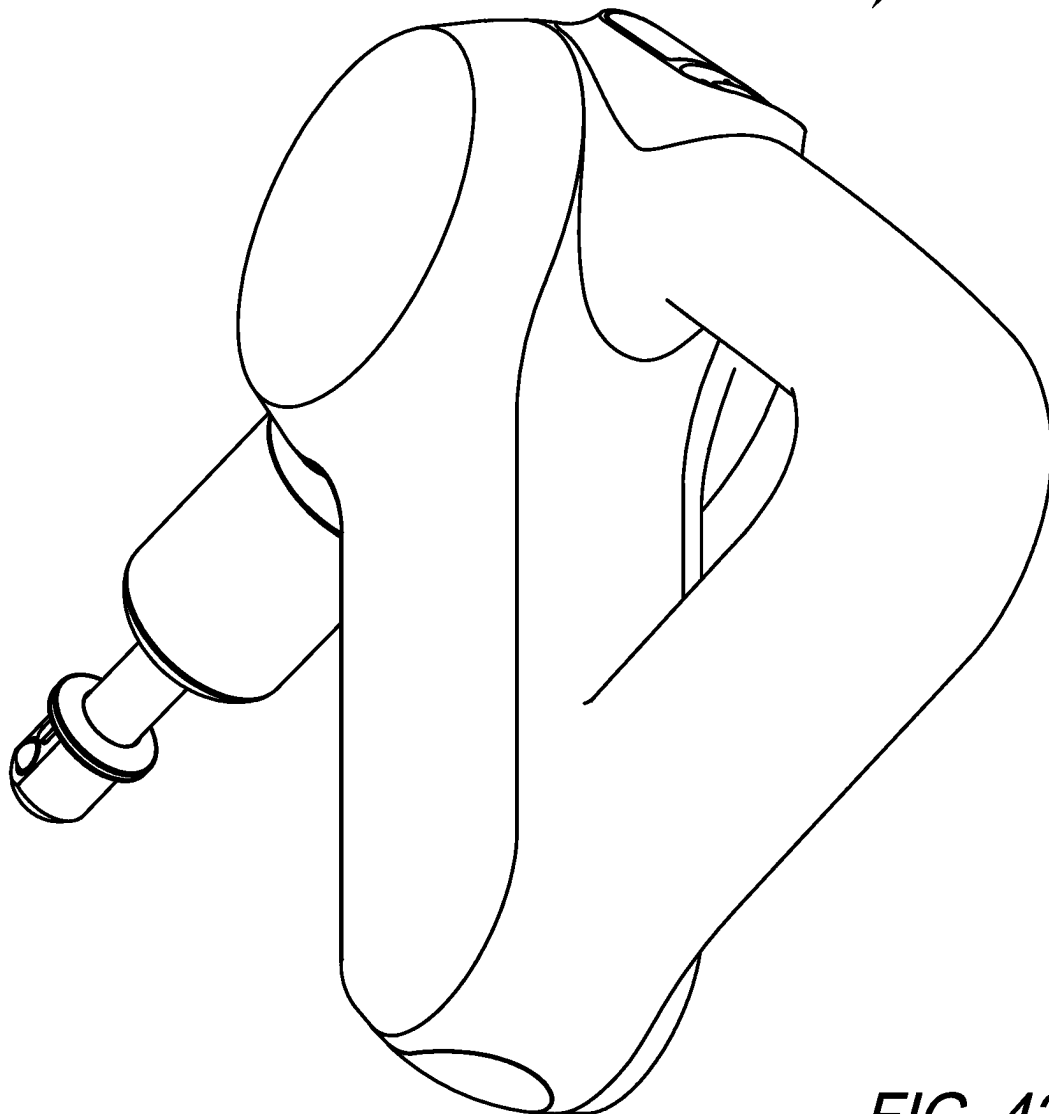
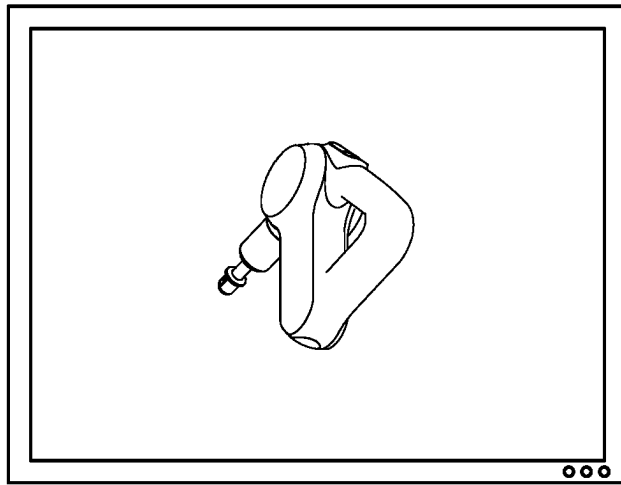


FIG. 42A

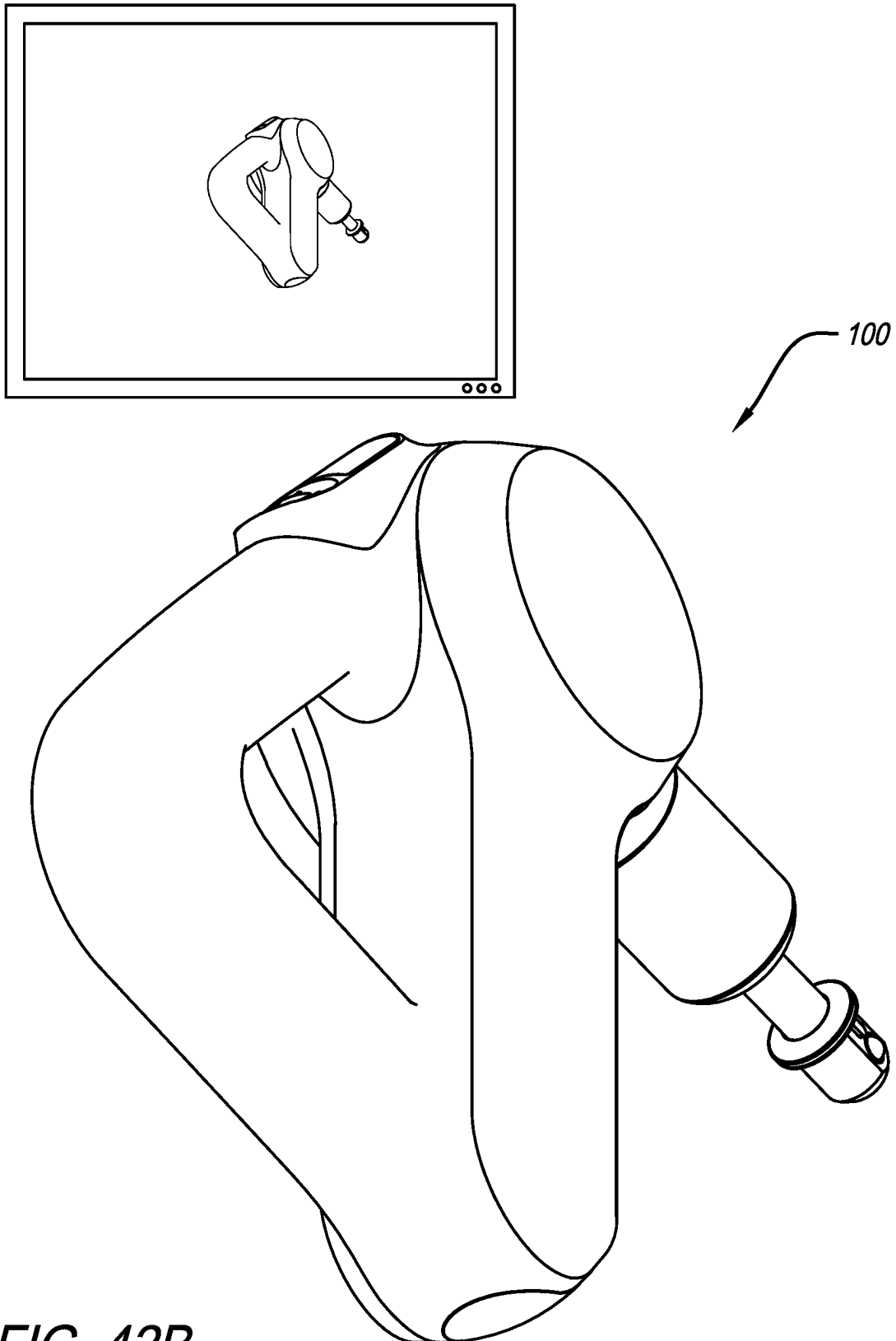
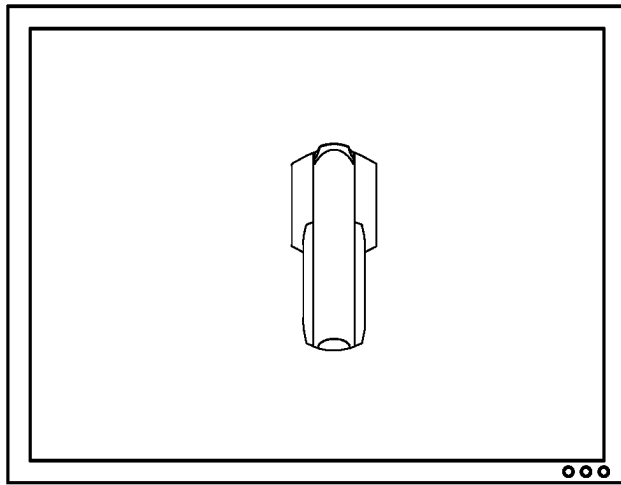


FIG. 42B



100

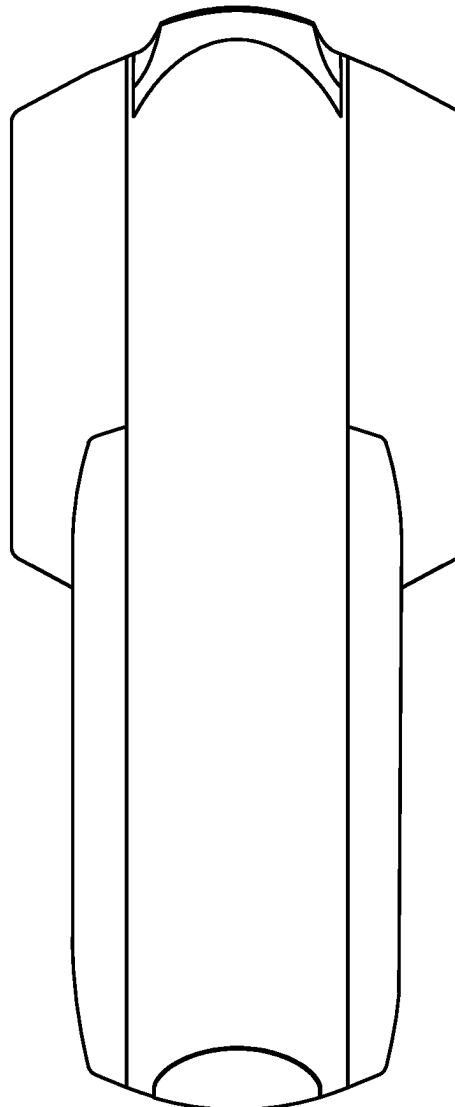


FIG. 42C

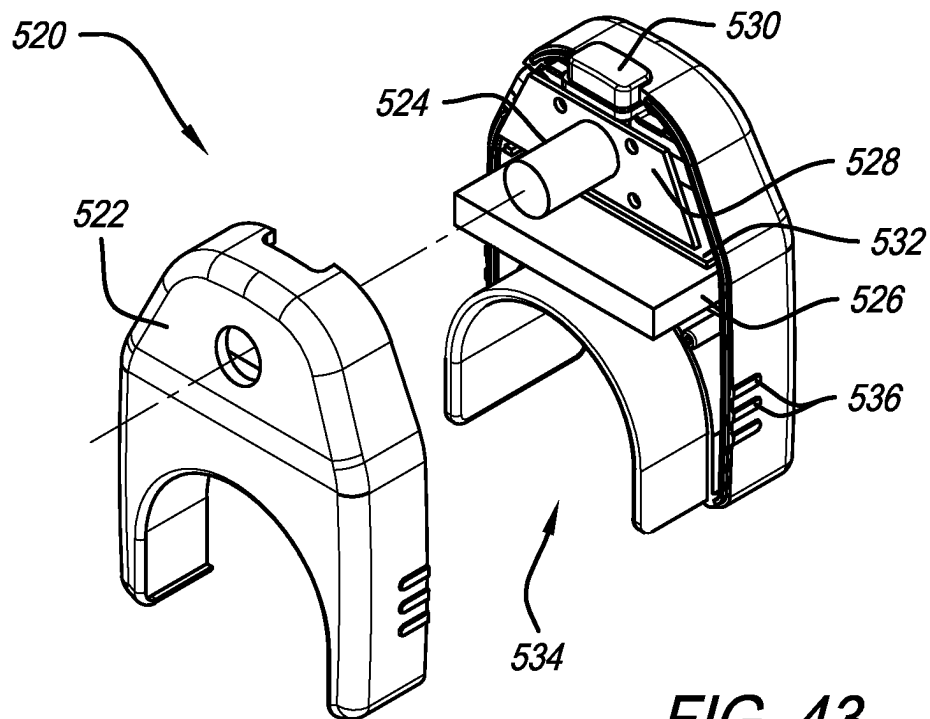


FIG. 43

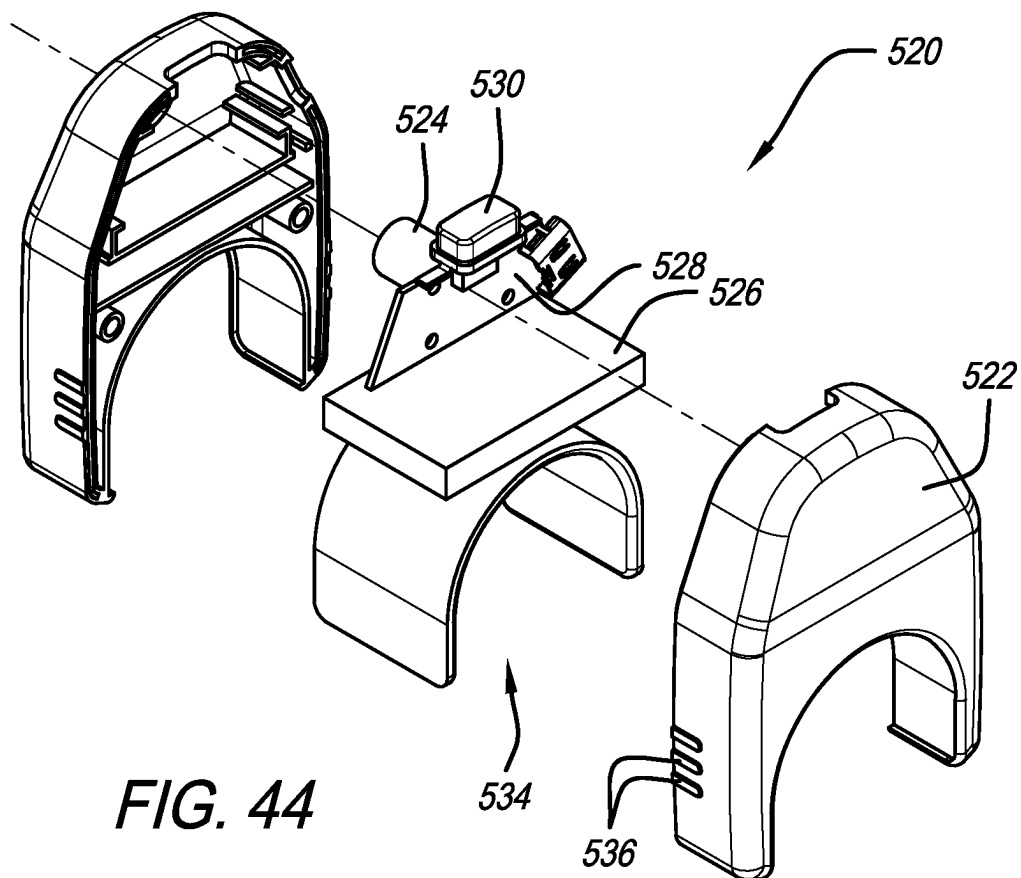
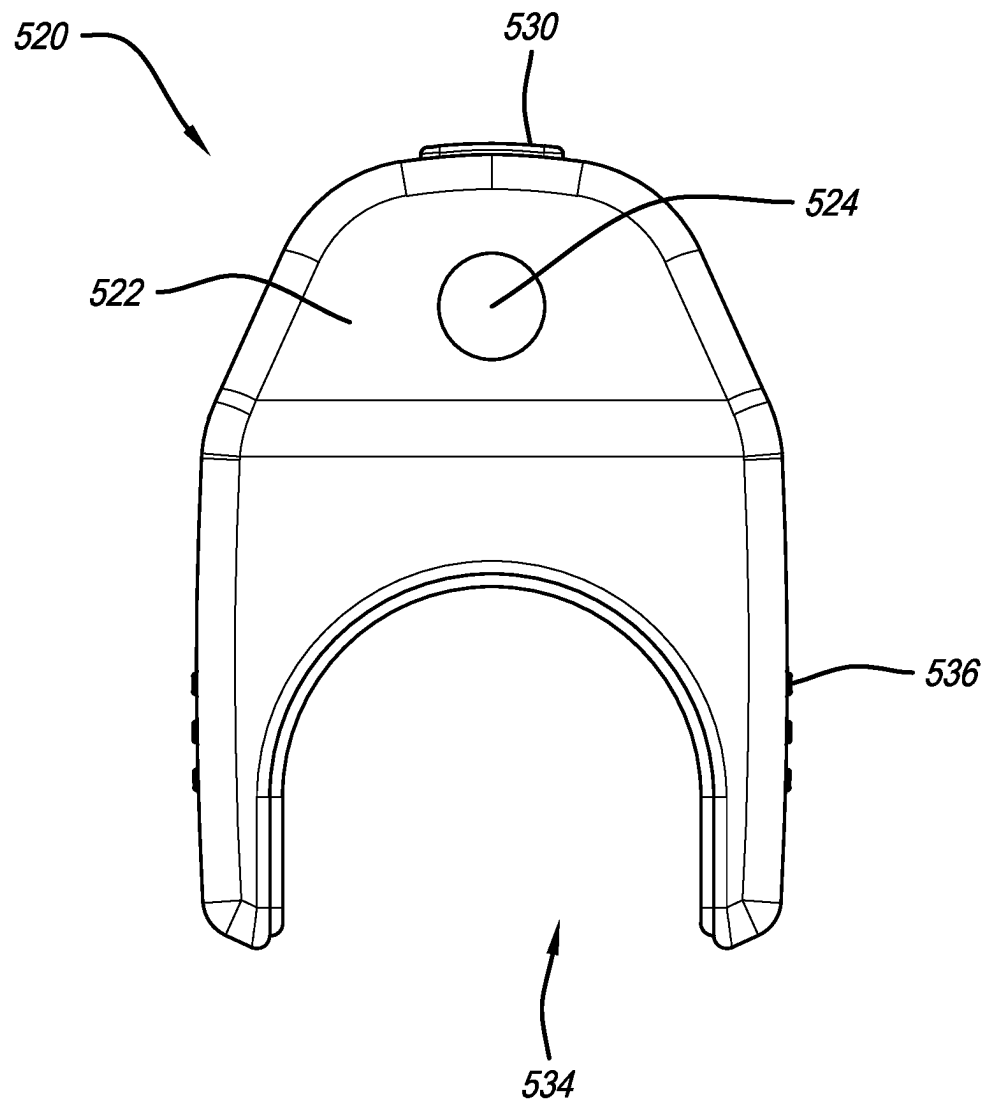


FIG. 44

**FIG. 45**

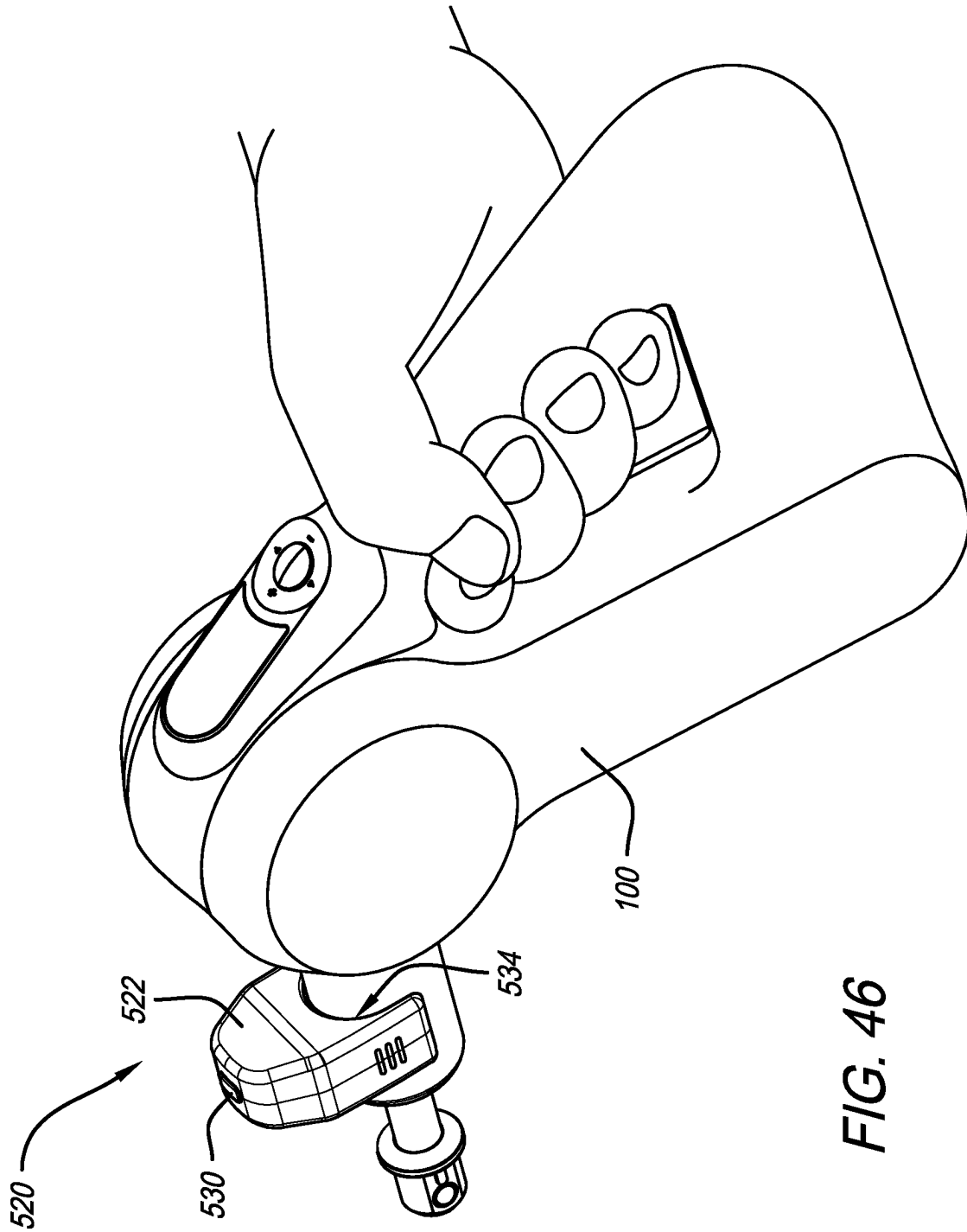


FIG. 46

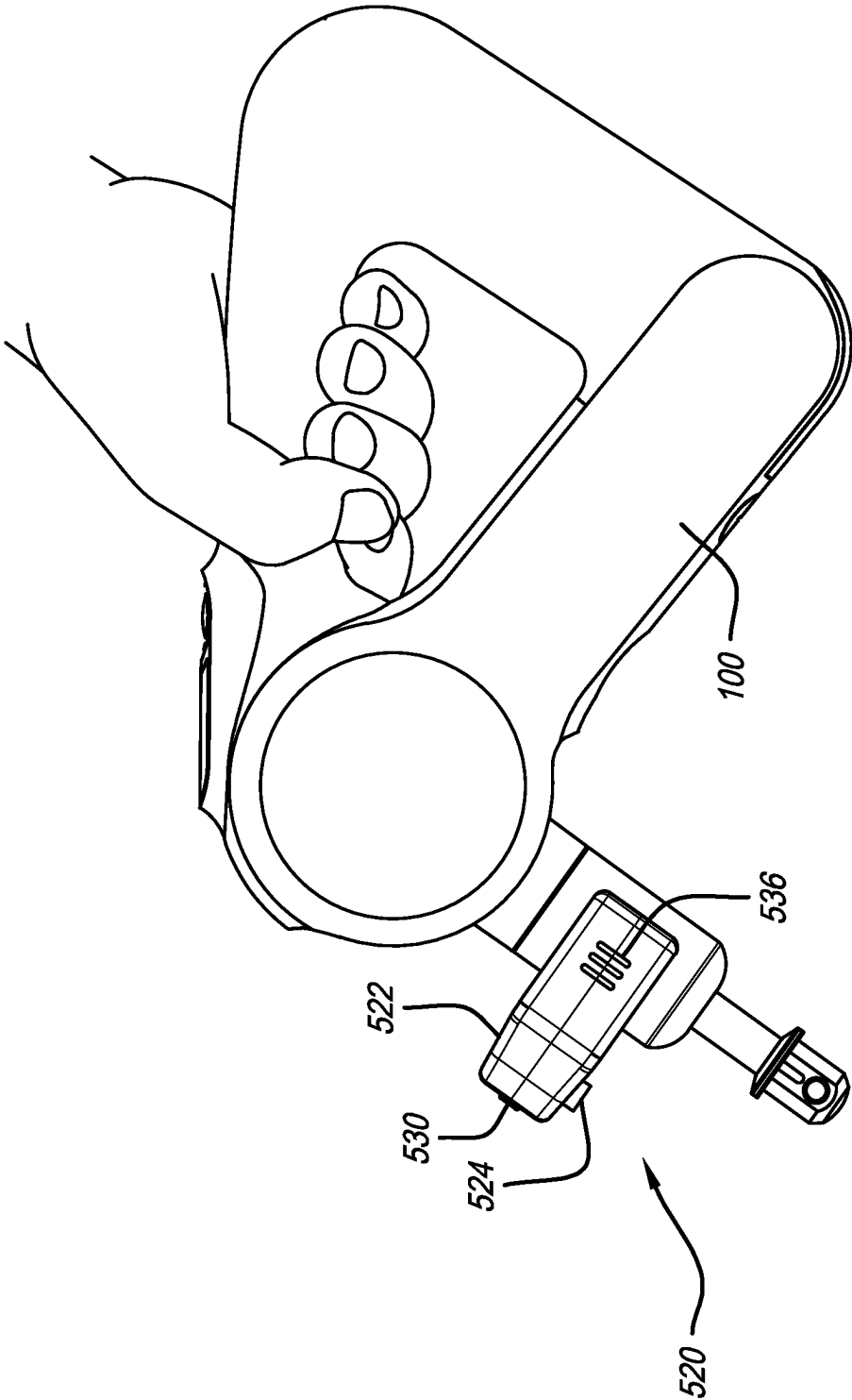


FIG. 47

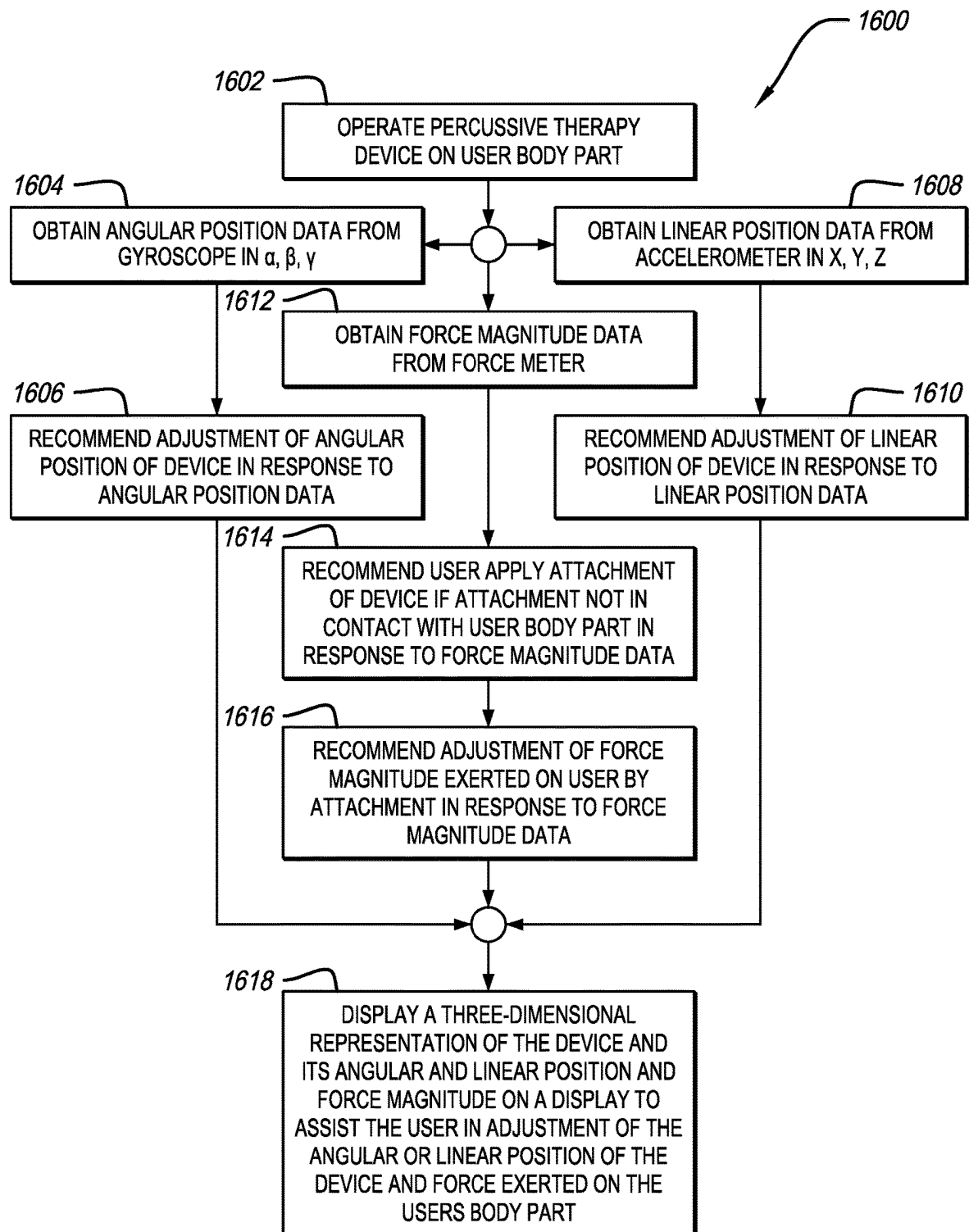
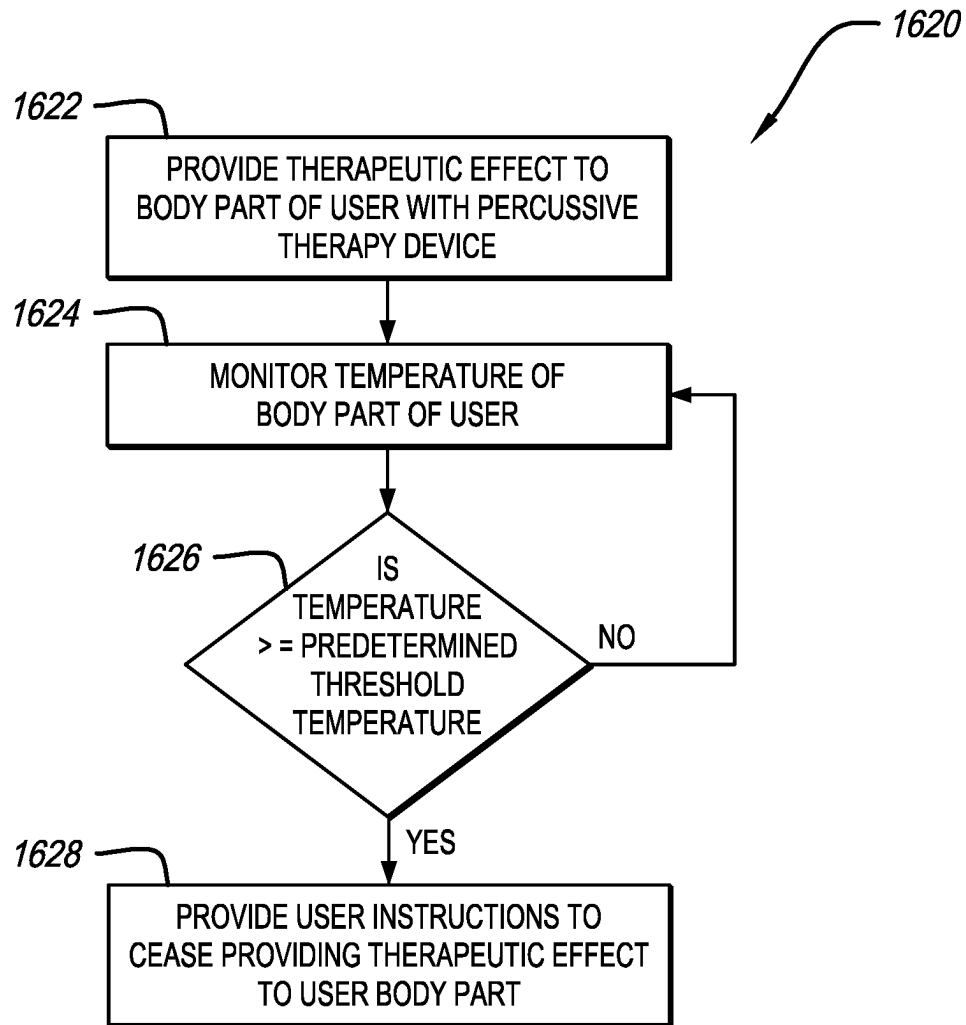


FIG. 48

**FIG. 49**

PERCUSSIVE THERAPY DEVICE WITH ELECTRICALLY CONNECTED ATTACHMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/162,499, filed Jan. 31, 2023, which is a continuation of U.S. patent application Ser. No. 17/244,278 filed Apr. 29, 2021, now U.S. Pat. No. 11,564,860. U.S. patent application Ser. No. 18/162,499 is a continuation of Ser. No. 17/739,630, filed May 9, 2022, now U.S. Pat. No. 11,890,253. U.S. patent application Ser. No. 17/244,278 is a continuation-in-part of U.S. patent application Ser. No. 17/018,099, filed Sep. 11, 2020, now U.S. Pat. No. 11,357,697, which is a continuation-in-part of U.S. patent application Ser. No. 16/869,402, filed May 7, 2020, now U.S. Pat. No. 10,857,064, which is a continuation-in-part of U.S. patent application Ser. No. 16/796,143, filed Feb. 20, 2020, now U.S. Pat. No. 10,940,081, which claims the benefit of U.S. Provisional Application No. 62/844,424, filed May 7, 2019, U.S. Provisional Application No. 62/899,098, filed Sep. 11, 2019, and U.S. Provisional Application No. 62/912,392, filed Oct. 8, 2019. U.S. patent application Ser. No. 16/869,402 is also a continuation-in-part of U.S. patent application Ser. No. 16/675,772, filed Nov. 6, 2019, now U.S. Pat. No. 10,702,448, which claims the benefit of U.S. Provisional Application No. 62/785,151, filed Dec. 26, 2018. U.S. patent application Ser. No. 17/244,278 also claims the benefit of U.S. Provisional Application No. 63/133,951, filed Jan. 5, 2021 and U.S. Provisional Application No. 63/017,472, filed Apr. 29, 2020. U.S. patent application Ser. No. 17/739,630 is also a continuation-in-part of U.S. patent application Ser. No. 17/244,239, filed Apr. 29, 2021, now U.S. Pat. No. 11,452,670, which is a continuation-in-part of U.S. patent application Ser. No. 17/018,099, filed Sep. 11, 2020, now U.S. Pat. No. 11,357,697. U.S. patent application Ser. No. 17/244,239 also claims the benefit of U.S. Provisional Application No. 63/133,951, filed Jan. 4, 2021 and U.S. Provisional Application No. 63/017,472, filed Apr. 29, 2020. U.S. patent application Ser. No. 17/739,630 is also a continuation-in-part of U.S. patent application Ser. No. 17/705,300, filed Mar. 26, 2022, now U.S. Pat. No. 11,730,668. All applications listed above are incorporated by reference herein in their entireties.

FIELD OF THE INVENTION

The present invention relates generally to massage devices and more particularly to a percussive therapy device that includes an electrically connected or smart attachment.

BACKGROUND OF THE INVENTION

Massage devices often provide ineffective massages that are superficial and do not provide any real benefit. Accordingly, there is a need for an improved massage device. Furthermore, percussive massage devices are often used in an ineffective manner. Accordingly, there is a need for a percussive therapy device to be automated to provide effective massage or recovery.

SUMMARY OF THE PREFERRED EMBODIMENTS

In accordance with a first aspect of the present invention there is provided a percussive therapy system that includes

a percussive therapy device that includes a housing, an electrical source, a motor positioned in the housing, a switch for activating the motor, a push rod assembly operatively connected to the motor and configured to reciprocate in response to activation of the motor, and an attachment configured to be operatively connected to a distal end of the push rod assembly of the percussive massage device and to provide at least one therapeutic effect to a user.

In a preferred embodiment, the attachment comprises at least one of an actuator configured to provide the at least one therapeutic effect to the user and a sensor configured to obtain at least one of biometric data of the user and information regarding operation of the percussive therapy device. The actuator may include at least one of a vibration actuator, a heating actuator, a cooling actuator, and an exfoliating actuator. The sensor may include at least one of a thermal sensor, an oxygen sensor, a blood flow sensor, a force meter, a gyroscope, and an accelerometer.

In a preferred embodiment, the system also includes a routine controller that is configured to initiate a protocol configured to provide user instructions to apply the attachment to a first body part until a thermal sensor senses that the first body part has reached a predetermined temperature.

In a preferred embodiment, the system is configured to determine at least one characteristic of the attachment. The at least one characteristic of the attachment may include a type of the attachment, a sensor of the attachment, and an actuator of the attachment. Preferably, the system also includes a wireless communication module configured to transmit the at least one characteristic to at least one of the percussive therapy device and a remote device.

In a preferred embodiment, the attachment includes a first set of electrical contacts. In a preferred embodiment, the distal end of the push rod assembly includes an attachment member that includes first and second balls biased outwardly therefrom. The first and second balls may be the first set of electrical contacts.

In accordance with another aspect of the present invention there is provided a method of providing at least one therapeutic effect to a user that includes obtaining a percussive therapy device that includes a housing, an electrical source, a motor positioned in the housing, a switch for activating the motor, a push rod assembly operatively connected to the motor and configured to reciprocate in response to activation of the motor, obtaining an attachment configured to be operatively connected to the percussive massage device and configured to provide at least one therapeutic effect to a user, and operating the percussive therapy device using the attachment. The at least one therapeutic effect may include vibration, percussion, heating, cooling, and exfoliation.

In a preferred embodiment, the attachment is further configured to obtain at least one of thermal data, blood-oxygen content data, blood flow data, angular position data, linear position data, and force magnitude data.

The method can also include the step of providing a recommendation to the user. In a preferred embodiment, the recommendation is generated from at least one of the thermal data, the angular position data, the linear position data, and the force magnitude data to assist in providing the at least one therapeutic effect to the user.

The method can also include the steps of providing the at least one therapeutic effect to a first body part of the user, monitoring a temperature of the first body part of the user, determining that the first body part of the user has reached a predetermined temperature, and providing user instructions to the user to cease providing the at least one therapeutic effect to the first body part when the first body part has

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reached the predetermined temperature. The at least one therapeutic effect may be provided in accordance with a protocol.

In a preferred embodiment, the method can further include the step of determining at least one characteristic of the attachment. The method can also include the step of providing a prompt communicating the at least one characteristic of the attachment to the user.

BRIEF DESCRIPTION OF THE DRAWINGS

The invention may be more readily understood by referring to the accompanying drawings in which:

FIG. 1 is a side elevational view of a percussive massage device in accordance with a preferred embodiment of the present invention;

FIG. 2 is a block diagram showing interconnected components of a percussive massage device with a force meter;

FIG. 3 is a circuit diagram of a microcontroller unit with pin outputs in accordance with one embodiment;

FIG. 4 is a circuit diagram used for battery voltage detection in accordance with one embodiment;

FIG. 5 is a circuit diagram for detection and measurement of voltage of the motor of the percussive massage device in accordance with one embodiment;

FIG. 6 is a flow diagram showing a method of detecting force applied by the percussive massage device in accordance with a preferred embodiment;

FIG. 7 is a flow diagram showing a method of generating a lookup table correlating voltage to force in accordance with a preferred embodiment;

FIG. 8 is a graph plotting a lookup table for use by a method of detecting force applied by the percussive massage device that was generated by correlating voltage to force in accordance with a preferred embodiment;

FIG. 9 is a flow diagram showing a method of calibrating a lookup table according to a preferred embodiment;

FIG. 10 is a graph plotting a lookup table generated by a method of detecting force applied by the percussive massage device against a lookup table calibrated by using a method of calibrating a lookup table according to a preferred embodiment;

FIG. 11 is a flow diagram showing a method of calibrating a lookup table;

FIG. 12 is a graph plotting a lookup table after being calibrated in accordance with a preferred embodiment;

FIG. 13 is a flow diagram showing a method of detecting force applied by a percussive massage device in accordance with a preferred embodiment;

FIG. 14 is a flow diagram showing a method of generating a lookup table correlating power to force in accordance with a preferred embodiment;

FIG. 15 is a graph plotting a lookup table for use by a method of detecting force of that was generated by correlating power to force in accordance with a preferred embodiment;

FIG. 16 is a flow diagram showing a method of calibrating a lookup table in accordance with a preferred embodiment;

FIG. 17 is a graph plotting a lookup table after being calibrated in accordance with a preferred embodiment;

FIG. 18 is a perspective view of a percussive massage device in accordance with a preferred embodiment of the present invention;

FIG. 19 is a perspective view of the percussive massage device with a portion of the housing removed;

FIG. 20 is a perspective view of the motor,

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FIG. 21 is a perspective view of the percussive massage device of FIG. 18 with a portion of the housing removed;

FIGS. 22A and 22B are cross sectional views of the head portion and motor;

FIG. 23 is an exploded view of some of the internal components of percussive massage device of FIG. 18;

FIG. 23A is an exploded view of the motor and motor mount;

FIG. 24 is a chart showing steps of Protocol 1 in accordance with a method of performing a routine for a percussive massage device;

FIG. 25 is a chart showing steps of a "Shin Splints" protocol in accordance with a method of performing a routine for a percussive massage device;

FIGS. 26A, 26B, 26C, and 26D are methods of performing a routine for a percussive massage device;

FIG. 27 is a front view of a graphical user interface showing a "Right Bicep" protocol;

FIG. 28 is a front view of a graphical user interface showing a "Right Bicep" protocol;

FIG. 29 is perspective view of a percussive massage device with a portion of the housing removed and showing the motor mount orienting the motor shaft axis extending longitudinally;

FIG. 30 is an exploded perspective view of the motor mount, motor and other components from FIG. 29;

FIG. 31 is a perspective view showing the motor and motor mount exploded out of the housing;

FIG. 32 is a perspective view showing the motor and motor mount exploded out of the housing on the opposite side as FIG. 31;

FIG. 33 is a cross-sectional perspective view;

FIG. 34 is a perspective view of a percussive massage device that includes a heart rate monitor;

FIG. 35 is a perspective view of a percussive massage device that includes a heart rate monitor with first and second pulse contacts;

FIG. 36 is a perspective view of a percussive massage device that includes a temperature sensor or monitor;

FIG. 36A is a detailed view of the temperature reading on the screen taken from FIG. 34;

FIG. 37 is a side elevational schematic of a percussive therapy device with a heated male attachment member;

FIG. 38 is a side elevational schematic of a percussive therapy device with a male attachment member with first and second electrical contacts;

FIG. 39 is a bottom view of male attachment member with first and second electrical contacts;

FIG. 40 is a massage member with a heating element therein;

FIG. 41 is a perspective view of a percussive therapy device that includes a gyroscope and accelerometer;

FIGS. 42A-C are perspective views of a percussive therapy device and graphical representations thereof on a display;

FIG. 43 is a perspective view of an attachment configured to be operably connected with a percussive therapy device;

FIG. 44 is a perspective view of an attachment configured to be operably connected with a percussive therapy device;

FIG. 45 is a bottom view of an attachment configured to be operably connected with a percussive therapy device;

FIG. 46 is a perspective view of a percussive therapy system including a percussive therapy device and an attachment thereon;

FIG. 47 is a perspective view of a percussive therapy system including a percussive therapy device and an attachment thereon;

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FIG. 48 is a flow diagram of a method of providing at least one therapeutic effect to a user in accordance with an embodiment of the present invention

FIG. 49 is a flow diagram of a method of preparing a user's body part for exercise in accordance with an embodiment of the present invention.

Like numerals refer to like parts throughout the several views of the drawings.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The following description and drawings are illustrative and are not to be construed as limiting. Numerous specific details are described to provide a thorough understanding of the disclosure. However, in certain instances, well-known or conventional details are not described in order to avoid obscuring the description. References to one or another embodiment in the present disclosure can be, but not necessarily are, references to the same embodiment; and, such references mean at least one of the embodiments.

Reference in this specification to "one embodiment" or "an embodiment" means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Appearances of the phrase "in one embodiment" in various places in the specification do not necessarily refer to the same embodiment, nor are separate or alternative embodiments mutually exclusive of other embodiments. Moreover, various features are described which may be exhibited by some embodiments and not by others. Similarly, various requirements are described which may be requirements for some embodiments but not other embodiments.

The terms used in this specification generally have their ordinary meanings in the art, within the context of the disclosure, and in the specific context where each term is used. Certain terms that are used to describe the disclosure are discussed below, or elsewhere in the specification, to provide additional guidance to the practitioner regarding the description of the disclosure. For convenience, certain terms may be highlighted, for example using italics and/or quotation marks: The use of highlighting has no influence on the scope and meaning of a term; the scope and meaning of a term is the same, in the same context, whether or not it is highlighted. It will be appreciated that the same thing can be said in more than one way.

Consequently, alternative language and synonyms may be used for any one or more of the terms discussed herein. Nor is any special significance to be placed upon whether or not a term is elaborated or discussed herein. Synonyms for certain terms are provided. A recital of one or more synonyms does not exclude the use of other synonyms. The use of examples anywhere in this specification including examples of any terms discussed herein is illustrative only, and is not intended to further limit the scope and meaning of the disclosure or of any exemplified term. Likewise, the disclosure is not limited to various embodiments given in this specification.

Without intent to further limit the scope of the disclosure, examples of instruments, apparatus, methods and their related results according to the embodiments of the present disclosure are given below. Note that titles or subtitles may be used in the examples for convenience of a reader, which in no way should limit the scope of the disclosure. Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by

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one of ordinary skill in the art to which this disclosure pertains. In the case of conflict, the present document, including definitions, will control.

It will be appreciated that terms such as "front," "back," "top," "bottom," "side," "short," "long," "up," "down," and "below" used herein are merely for ease of description and refer to the orientation of the components as shown in the figures. It should be understood that any orientation of the components described herein is within the scope of the present invention.

While many embodiments are described herein, at least some of the described embodiments provide an apparatus, system, and method for a reciprocating treatment device.

FIG. 1 shows an embodiment of a percussive massage device 400 that includes a rechargeable battery (and replaceable or removable battery) 114 (FIG. 19). As shown in FIG. 1, in a preferred embodiment, the percussive massage device 400 includes three handle portions (referred to herein as first handle portion 143, second handle portion 145 and third handle portion 147) that cooperate to define a central or handle opening 149. All of the handle portions are generally straight and long enough that they are configured such that a person can grasp that particular handle portion to utilize the device. The ability to grasp the different handle portions allows a person (when using the device on their own body) to use the device on different body parts and from different angles, thus providing the ability to reach body parts, such as the back, that might not be possible without the three handle portions. Generally straight means that the majority of the handle portion is straight, but can include rounded edges or corners where the different handle portions meet or where the handle portions meet the bulge portion or the finger protrusion, etc.

As shown in FIG. 1, the first handle portion 143 defines a first handle portion axis A1, the second handle portion 145 defines a second handle portion axis A2 and the third handle portion 147 defines a third handle portion axis A3 that cooperate to form a triangle. In a preferred embodiment, the battery 114 is housed in the second handle portion 145 and the motor 406 (FIG. 19) is housed in the third handle portion 147.

In a preferred embodiment, the first handle portion 143 has an interior edge 143a, the second handle portion 145 has an interior edge 145a and the third handle portion 147 has an interior edge 147a, which all cooperate to at least partially define the handle opening 149. As shown in FIG. 1, in a preferred embodiment, the first handle portion 143 includes a finger protrusion 151 that includes a finger surface 151a or fourth interior surface that extends between the interior edge 143a of the first handle portion and the interior edge 147a of the third handle portion 147 and at least partially defines the handle opening 149. In use, a user can place their index finger against the finger surface 151a. The finger protrusion and surface provide a feedback point or support surface such that when a user places their index finger against the surface it helps the user with control and comfort of using the device. In a preferred embodiment, at least a portion of the finger surface 151a is straight, as shown in FIG. 1 (as opposed to the other "corners" of the handle opening 149 being rounded).

As shown in FIG. 1, with the finger surface 151a being straight, the first handle portion interior surface, second handle portion interior surface, third handle portion interior surface and finger surface cooperate to define a quadrilateral with radii or rounded edges between each of the straight surfaces.

FIGS. 2-20 show embodiments in accordance with a percussion massage device with a force meter. FIG. 2 is a block diagram showing interconnected components of a percussive therapy device with a force meter 400. In an embodiment, the percussive therapy device with force meter 400 includes a microcontroller unit 701, a battery pack management unit 702, an NTC sensor 703, a power charging management unit 704, a wireless charging management unit 705, a wireless charging receiving system 706, a voltage management unit 707 (5V 3.3V Voltage Management in drawings), battery charging inputs 708 (20V 2.25 A Charging Inputs in drawings), a display 709 (Force/Battery/Speed Display in drawings), a wireless control unit 710 (Bluetooth Control in drawings), an OLED screen 711, an OLED screen control system 712, a motor 713, a motor drive system 714, a PWM speed setup unit 715, an over-current protection unit 716, and a power switch unit 717 (Power On/Off OLED Screen SW in drawings). In the embodiment shown in accordance with FIG. 2, each block in the diagram is shown as a separate component. In alternative embodiments, however, certain components may be combined without departing from the scope of the present disclosure.

The microcontroller unit 701, in an embodiment, is a microcontroller unit including a processor, a memory, and input/output peripherals. In other embodiments, however the microcontroller unit 701 is an ST Microelectronics STM32F030K6 series of microcontroller units, STM32F030C8T6 series of microcontrollers, STM32F030CCT6 series of microcontrollers, or an equivalent microcontroller.

One of ordinary skill would understand that the memory of the microcontroller unit 701 is configured to store machine-readable code for processing by the processor of the microcontroller unit 701. Various other configurations may exist depending on whether the designer of the percussive massage device with force meter 400 desires to implement the machine-readable code in software, firmware, or both. In an embodiment, the machine-readable code is stored on the memory and configured to be executed by a processor of the microcontroller 701. In an embodiment, the machine-readable code is stored on computer-readable media.

The battery pack management unit 702, in an embodiment, is implemented in firmware or software and configured to be used in connection with the microcontroller unit 701. In this embodiment, the firmware or software is stored in memory (not shown) and configured to be obtainable by the microcontroller unit 701. The battery pack management unit 702 may also be a combination of firmware, software, and hardware, in another embodiment. The battery pack management unit 702 is coupled with the NTC sensor 703. The NTC sensor 703 is a negative temperature coefficient thermistor used by the battery pack management unit 702 to sense temperature of the battery pack. For example, the NTC sensor 703 is a thermistor with B value of 3950+/-1%, and a resistance of 10 kΩ. In another example, the thermistor has a resistance of 100 kΩ. One of ordinary skill in the art would recognize that a thermistor is a resistor whose resistance is dependent upon temperature. In other embodiments, however, the NTC sensor 703 may be another type of temperature sensing device or component used in connection with the battery pack management unit 702.

The power charging management unit 704, in an embodiment, is implemented in firmware or software and configured to be used in connection with the microcontroller unit 701. Similarly to the battery pack management unit 702, the power charging management unit 704 firmware or software is stored in memory (not shown) and configured to be

obtainable by the microcontroller unit 701. The power charging management unit 704 may also be a combination of firmware, software, and hardware, in another embodiment. In various embodiments, the power charging management unit 704 is configured to charge a battery pack via a direct connection or through an external charger, such as when configured to be operable with rechargeable batteries.

The wireless charging management unit 705, in an embodiment, is coupled to the battery pack management unit 702 and the battery charging inputs 708. In other embodiments, the battery or battery pack is charged using other conventional methodologies, such as, for example, charging the battery or battery pack using a wire or cord coupled to the battery charging inputs 708.

The wireless charging receiving system 706, in an embodiment, is coupled to the power charging management unit 704 and the display 709. The wireless charging receiving system 706 includes one or more of firmware, software, and hardware. In an embodiment, the wireless charging receiving system 706 is configured to receive information pertaining to battery capacity, charging metrics, and other information pertaining to wireless charging, and to pass along the information to the power charging management unit 704. The wireless charging receiving system 706 preferably includes a wireless charging pad used to charge the percussive massage device with force meter 400. One of ordinary skill in the art would understand that a variety of wireless charging devices may be utilized to wirelessly charge the percussive massage device with force meter 400. As one example, the Qi wireless charging standard and related devices may be utilized to wirelessly charge the percussive massage device with force meter 400.

The voltage management unit 707, in an embodiment, is a DC voltage regulator that steps down 5 volt to 3.3 volt power for use by the microcontroller unit 701. The voltage management unit 707 may also perform additional functions for management of 3.3 volt power for use by the microcontroller unit 701. In an embodiment, the voltage management unit 707 is implemented using a series of electronic components such as, for example, implementing a resistive divider using electronic components. In another embodiment, the voltage management unit 707 is a stand-alone voltage regulator module and/or device designed to step down voltage from 5 volts to 3.3 volts. One of ordinary skill in the art would understand the various methodologies and devices available to step down 5 volts to 3.3 volts.

The battery charging inputs 708, in an embodiment, are interfaces by which a wire or cord may be inserted for charging the percussive massage device with force meter 400. For example, a standardized barrel connector is the battery charging inputs 708. In another example, the battery charging inputs 708 is a USB connector. Other more specialized charging methodologies may require a particular battery charging input not described above.

The display 709, in an embodiment, displays a series of LEDs depicting an amount of force applied by the percussive massage device with force meter 400. In an alternative embodiment, the display 709 displays a series of LEDs depicting the current battery or battery pack charge of the percussive massage device with force meter 400. In yet another embodiment, the display 709 displays a series of LEDs depicting the current speed of the percussive massage device with force meter 400. One of ordinary skill in the art would recognize that while LEDs have been specified in the above-referenced embodiments, other embodiments not using LEDs are within the scope of this disclosure, such as, for example, liquid crystal displays, OLEDs, CRT displays,

or plasma displays. One of ordinary skill in the art would also understand that it may be advantageous in an embodiment utilizing a battery or battery pack to use low-power options to ensure battery power longevity. In an embodiment, the display **709** is a 128×64 pixel OLED display.

The wireless control unit **710** is a wireless connectivity device that may be implemented in a wireless microcontroller unit. In an embodiment, the wireless control unit **710** is a Bluetooth transceiver module configured to couple, via Bluetooth, to a remote device. In an embodiment, the Bluetooth module is a Bluetooth Low-Energy (BLE) module configured to be run in broadcast mode. The wireless control unit **710** is coupled to the microcontroller unit **701**. In an embodiment, the remote device is a smartphone having an embedded Bluetooth module. In an alternative embodiment, the remote device is a personal computer having Bluetooth connectivity. In other embodiments, other wireless connectivity standards besides the Bluetooth wireless standard may be utilized. It will be appreciated that the Bluetooth connectivity or other wireless connectivity may be described herein as being implemented in a wireless connection device. The wireless connection device can be a separate module, can be included in the MCU or other component of the device, or can be a separate chip. In summary, the percussive therapy device including a wireless connection device means that the percussive massage device can connect to another electronic device wirelessly (e.g., a phone, tablet, computer, computer, voice controlled speaker, regular speaker, etc.). One of ordinary skill in the art would recognize that low-power wireless control modules may be advantageous when the percussive massage device with force meter **400** is utilizing a battery or battery pack.

The OLED screen **711** and the OLED screen control system **712**, in an embodiment, are configured to display substantially the same information as the display **709** referenced above. The OLED screen **711** is coupled to the OLED screen control system **511**. The OLED screen control system **712** is coupled to the microcontroller unit **701**, the OLED screen **711**, and the power switch unit **717**. In an embodiment, the display **709** and the OLED screen **711** may be redundant and it may only be necessary to utilize one or the other.

The motor **713**, in an embodiment, is a brushless direct current (BLDC) motor. The motor **713** and the motor drive system **714**, in an embodiment, are configured to vary the speed (i.e., rotational motion) that may be converted to reciprocal motion. In other embodiments, the motor **713** is a brushed DC motor, a brushed AC motor, or a brushless AC motor. One of ordinary skill in the art would understand that choosing a brushless or brushed motor, or direct current or alternating current, may vary depending on the application and intended size, battery power, and use.

The PWM speed setup unit **715**, in an embodiment, is used to control pulse width modulation utilized to drive the motor **713**. The PWM speed setup unit **715** is coupled to the microcontroller unit **701** and the over-current protection unit **716**. One of ordinary skill in the art would understand that pulse width modulation is one way to vary the average power applied to the motor **713**, resulting in varying speed as desired. In alternative embodiments, one of ordinary skill in the art would understand that there are a variety of methods to vary the speed of a brushless DC motor. For example, voltage to the motor **713** may be controlled in other non-PWM methods.

The over-current protection unit **716**, in an embodiment, may be a feature of an integrated system-in-package to prevent damage caused by high currents to the motor. In

other embodiments, the over-current protection unit **716** is implemented using a series of electronic components configured to protect the motor from excessively high current.

The power switch unit **717**, in an embodiment, is configured to turn on and turn off the percussive massage device with force meter **400**. The power switch unit **717** is coupled to the OLED screen control system **712** and the microcontroller unit **701**. In an embodiment, the power switch unit **717** is the switch **405**.

FIG. 3 shows a circuit diagram of the microcontroller unit **701** with pin outputs. In this embodiment, the STM32F030K6 series of microcontroller units is utilized. The circuit diagram depicts +3.3 volt power being provided to the VDD inputs of the microcontroller unit **701**. Input PA3 is labeled "Motor_VOL", the voltage of the motor **713**. Input PA2 is "bt_v", the battery or battery pack voltage. The microcontroller unit is configured to receive analog voltage on inputs PA2 and PA3 and to convert it to digital voltage using the microcontroller's analog-to-digital converter. In this embodiment, the analog-to-digital converter is a 12-bit ADC. One of ordinary skill in the art would understand that other microcontrollers may utilize voltage sensing and analog-to-digital converters to perform similar functions. In yet other embodiments, an analog-to-digital converter module separate from a microcontroller may be utilized.

FIG. 4 shows a circuit diagram used for battery voltage detection. In this embodiment, +BT, the positive battery terminal **602**, is coupled to a circuit consisting of a P-channel MOSFET **604**, an N-Channel MOSFET **608**, 0.1 μ F capacitor **610**, 100 k Ω resistors **612**, **614**, 68 k Ω resistor **616**, 1 k Ω resistors **618**, **620**, and 10 k Ω resistors **622**, **624**. The circuit is configured to provide an input analog voltage of the battery or battery pack, or bt_v, to the microcontroller unit **701** of FIG. 2. In other embodiments, voltage of the battery or battery pack may be achieved using a voltage reader coupled to the terminals of the battery or battery pack.

FIG. 5 shows a circuit diagram for detection and measurement of voltage of the motor **713** of the percussive massage device. In this embodiment, voltage sensing resistor **626** is coupled in parallel with the microcontroller unit **701**, and coupled to the motor **713**. In an embodiment, the voltage sensing resistor has a value of 0.0025 Ω . The circuit depicted in FIG. 5 is configured to provide the Motor_VOL input into the microcontroller unit **701** of FIG. 2. In an embodiment, the input analog voltage is amplified. In another embodiment, the voltage of the motor **713** is measured or sensed using a separate series of electronic components or a standalone device and input into a microprocessor for use with the method of displaying a force on the percussive massage device.

FIG. 6 is a flow diagram showing a method **800** of detecting force applied by the percussive massage device in accordance with a preferred embodiment. At Step **802**, a voltage magnitude V is obtained. In an embodiment, voltage magnitude V is an analog voltage obtained by using the circuit disclosed in FIG. 2. In that circuit, a block curve signal from the motor **713** (i.e., a Hall effect sensor) is simulated in the circuit as current using the resistor R, which is placed in parallel with the microcontroller unit **701**. In other embodiments, voltage that corresponds to the current operating speed of the motor **713** may be generated in a variety of other ways. The voltage magnitude V may be input to a microcontroller unit **701** that converts analog voltage to digital voltage using an analog-to-digital converter, such as that implemented in the STM32F030K6 microcontroller unit. The STM32F030K6 microcontroller unit converts analog voltage magnitude to a digital code

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corresponding to the 12-bit ADC (i.e., 0 to 4096). The digital code represents a voltage magnitude corresponding to the original voltage magnitude V obtained.

At Step **804**, a lookup table is generated that correlates voltage V to force magnitude F. In an embodiment, the lookup table is generated using a method **900** of generating a lookup table correlating voltage to force. For example, the force magnitude F may be expressed in pounds of force. In an alternative embodiment, the force magnitude F may be expressed in Newtons of force.

At Step **806**, the force magnitude F corresponding to voltage magnitude V is displayed on the percussive massage device with force meter **400**. In an embodiment, a series of LED lights may be utilized to depict varying amounts of force as the force is being applied by the percussive massage device with force meter **400**. Thus, as the amount of force magnitude F increases, more LEDs on the series of LED lights will be lit. Preferably, the series of LED lights consists of 12 LED lights.

FIG. **7** is a flow diagram showing a method **900** of generating a lookup table correlating voltage to force. At Step **902**, a maximum magnitude of force, F_{MAX} , is determined. The magnitude of F_{MAX} may be determined by assessing the maximum desired force to apply using the percussive massage device with force meter **400**. As an example, F_{MAX} is 60 pounds of force.

At Step **904**, a maximum magnitude of voltage, V_{MAX} , is determined. The magnitude of V_{MAX} may be determined by assessing the maximum theoretical voltage change possible by the percussive massage device with force meter **400**. As an example, V_{MAX} is 1.8 volts.

At Step **906**, F_{MAX} is divided into equal increments. Using the above example from Step **902**, 60 pounds of force is divided into 60 one-pound increments.

At Step **908**, V_{MAX} is divided into the same amount of increments as determined in Step **906** above. Thus, using the above example from Step **904**, 1.8 volts is divided into 60 0.03-volt increments.

At Step **910**, a lookup table (LUT) is generated that correlates the increments of pounds of force with the increments of voltage. This necessarily creates a linear relationship between force and voltage. FIG. **8** is a graph plotting the LUT for use by the method of detecting force of FIG. **6** that was generated using the specific example identified in FIG. **7**. The graph depicts calculated force that was calculated using the method **900**.

A problem may arise in that the theoretical maximum voltage assumption at Step **904** in the method **900** is inaccurate. It may also be the case that as the percussive massage device with force meter **400** is used, the maximum available voltage degrades over time. In other words, the battery or battery pack voltage may decrease.

Accordingly, a method **1000** of calibrating the LUT generated by method **900** may be advantageous. FIG. **9** is a flow diagram showing a method **1000** of calibrating a LUT. At Step **1002**, battery pack voltage BV is obtained. In an embodiment, battery pack voltage magnitude BV is an analog voltage obtained by using the circuit disclosed in FIG. **4**. In that circuit, the battery pack voltage magnitude BV may be input to a microcontroller unit **701** that converts analog voltage to digital voltage using an analog-to-digital converter, such as that implemented in the STM32F030K6 microcontroller unit. The STM32F030K6 microcontroller unit converts analog voltage magnitude to a digital code corresponding to the 12-bit ADC (i.e., 0 to 4096). The digital code represents a voltage magnitude corresponding to the original battery pack voltage magnitude BV obtained.

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At Step **1004**, V_{MAX} is set to the actual battery voltage magnitude BV output. As an example, may decrease from 1.8 volts to 1.74 volts, a 0.6 volt decrease. At Step **1006**, the LUT linear correlation is adjusted to reflect the lower V_{MAX} . FIG. **10** is a graph plotting the LUT calculated by the method **900** against the LUT calibrated by using the method **1000**. The LUT resulting from method **1000** depicts a calibrated force rather than a calculated force.

FIG. **11** is a flow diagram showing a method **1100** of calibrating a LUT. The method **1100** may be performed after the method **900**, or entirely separately from the method **900**. At Step **1102**, battery pack voltage BV is measured. In an embodiment, the measurement is done without applying any force from the percussive massage device with force meter **400**. In an embodiment, the battery pack voltage BV is measured using an external voltage meter. In another embodiment, the battery pack and/or microcontroller unit **701** have embedded solutions for directly measuring battery pack voltage BV.

At Step **1104**, the display on the percussive massage device with force meter **400** that displays the force magnitude F is read to determine the force magnitude F corresponding to the measured battery pack voltage BV.

At Step **1106**, a force meter is used to measure actual force being applied. In an embodiment, the force meter is a push/pull force meter. The direct measurement of force allows calibration of the LUT by comparing the displayed force magnitude F with the measured actual force. At Step **1108**, the LUT is updated with a corrected force corresponding with the measured battery pack voltage BV. After Step **1108**, Steps **1102-1106** are repeated for each successive voltage increment. In the embodiment depicted in accordance with the method **900**, Steps **1102-1106** are repeated for every 0.03-volt increment. FIG. **12** is a graph plotting the LUT calculated by the method **1100** after all 3-volt increments had been updated.

FIG. **13** is a flow diagram showing a method **1200** of detecting force applied by a percussive massage device in accordance with a preferred embodiment. At Step **1202**, current magnitude C of a battery pack is obtained. In an embodiment, current magnitude C is input into the microcontroller unit **701**. At Step **1204**, voltage magnitude BV of a battery pack is obtained. In an embodiment, voltage magnitude BV is input into the microcontroller unit **701**. At Step **1206**, power is calculated using the product of C and BV. In an embodiment, the microcontroller unit **701** is configured to calculate power by multiplying C and BV. At Step **1208**, a lookup table is generated that correlates power magnitude P to force magnitude F. In an embodiment, the lookup table is generated using a method **1300** of generating a lookup table correlating power to force. For example, the power magnitude P may be expressed in watts. In an alternative embodiment, force magnitude F may be expressed in pounds of force or Newtons of force.

At Step **1210**, the force magnitude F corresponding to power magnitude P is displayed on the percussive massage device with force meter **400**. In an embodiment, a series of LED lights may be utilized to depict varying amounts of force as the force is being applied by the percussive massage device with force meter **400**. Thus, as the amount of force magnitude F increases, more LEDs on the series of LED lights will be lit. Preferably, the series of LED lights consists of 12 LED lights.

FIG. **14** is a flow diagram showing a method **1300** of generating a lookup table correlating power to force. At Step **1302**, a maximum magnitude of power, F_{MAX} , is determined. A theoretical maximum magnitude of power, however, is not

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a reasonable assumption if the total effective power may be calculated. Equation 1 may be utilized to determine Total Maximum Effective Power (EP_{MAX}).

$$\text{Total } EP_{MAX} = P_{MAX} \times \text{Total } EP \quad \text{Equation 1}$$

Equation 2 may be utilized to calculate Total EP, which is then input into Equation 1 above.

$$\text{Total } EP = EP_{BATTERY} \times EP_{PCBA} \times EP_{MOTOR} \quad \text{Equation 2}$$

where Total EP, $EP_{BATTERY}$, EP_{PCBA} , and EP_{MOTOR} are all expressed in percentages, and where PCBA is a printed circuit board assembly.

In an embodiment, EP (Battery) is 85%, EP (PCBA) is 95%, and EP (Motor) is 75%. Thus, using Equation 2, Total EP is $85\% \times 95\% \times 75\% = 60.5625\%$.

In this embodiment, P_{MAX} is calculated by multiplying the maximum voltage V_{MAX} and the maximum amperage C_{MAX} of the battery pack such as in Equation 3. P_{MAX} is then input into Equation 1.

$$P_{MAX} = V_{MAX} \times C_{MAX}$$

In this embodiment, V_{MAX} is 16.8 volts and C_{MAX} is 20 amperes. Thus, P_{MAX} is 336 watts.

Turning back now to Equation 1, if P_{MAX} is 336 watts and Total EP is 60.5625%, then Total EP_{MAX} is 203 watts.

At Step 1304, a minimum amount of power P_{MIN} , is determined. It will be recognized by one of ordinary skill in the art that the power without any force being applied (i.e., no load) will be non-zero. Thus, P_{MIN} of 12 watts is assumed. One of ordinary skill will also understand that the value of is equivalent to the rated power without load, which may be derived from V_{MAX} and C_{MIN} .

At Step 1306, a maximum magnitude of force, F_{MAX} , is determined. The magnitude of F_{MAX} may be determined by assessing the maximum desired force to apply using the percussive massage device with force meter 400. As an example, F_{MAX} is 60 pounds of force.

At Step 1308, Total EP_{MAX} is divided into equal increments. In an embodiment, Total EP_{MAX} is divided in 3 watt increments per one pound of force, starting at P_{MIN} (12 watts). It will be recognized by one of ordinary skill in the art that if F_{MAX} is 60 pounds of force, the total desired force output of the percussive massage device with force meter 400, then 60 pounds of force correlates to 189 watts, within the calculated Total EP_{MAX} .

At Step 1310, a LUT is generated that correlates the increments of pounds of force with the increments of power in watts. This necessarily creates a linear relationship between force and voltage. FIG. 15 is a graph plotting the LUT for use by the method of detecting force of FIG. 13 that was generated using the specific example identified in FIG. 10. The graph depicts calculated force that was calculated using the method 1200.

Similarly to the method 900, a problem may arise in that the measured voltage of the battery pack at Step 1204 in the method 1200 is inaccurate. It may also be the case that as the percussive massage device with force meter 400 is used, the

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maximum available voltage degrades over time. In other words, the battery or battery pack voltage may decrease.

FIG. 16 is a flow diagram showing a method 1400 of calibrating a LUT. The method 1400 may be performed after the method 900 or the method 1200, or entirely separately from the method 900 or the method 1200. At Step 1402, current magnitude C of a battery pack is obtained. In an embodiment, current magnitude C is input into the microcontroller unit 701.

At Step 1404, battery pack voltage BV is measured. In an embodiment, the measurement is done without applying any force from the percussive massage device with force meter 400. In an embodiment, the battery pack voltage BV is measured using an external voltage meter. In another embodiment, the battery pack and/or microcontroller unit 701 have embedded solutions for directly measuring battery pack voltage BV. At Step 1406, power is calculated using the product of C and BV. In an embodiment, the microcontroller unit 701 is configured to calculate power by multiplying C and BV.

At Step 1408, the display on the percussive massage device with force meter 400 that displays the force magnitude F is read to determine the force magnitude F corresponding to the calculated power. At Step 1410, a force meter is used to measure actual force being applied. In an embodiment, the force meter is a push/pull force meter. The direct measurement of force allows calibration of the LUT by comparing the displayed force magnitude F with the measured actual force. At Step 1412, the LUT is updated with a corrected force corresponding with the measured power. After Step 1412, Steps 1402-1410 are repeated for each power or force increment. In the embodiment depicted in accordance with the method 900, Steps 1402-1410 are repeated for every 3-watt increment FIG. 17 is a graph plotting the LUT calculated by the method 1400 after all 3-watt increments had been updated.

FIGS. 18-19 show an exemplary percussive massage device 400 that embodies the features disclosed herein. Generally, the percussive massage device 400 includes a housing 101, an electrical source or battery pack 114, a motor 406 positioned in the housing 101, and a switch 405 for activating the motor 406. The electronics (see printed circuit board 408 in FIG. 19) includes the controller that is configured to obtain a voltage of the motor, generate a lookup table correlating voltage to force applied by the percussive massage device, and display a force magnitude corresponding to the obtained voltage using the lookup table. FIG. 20 is a perspective view of the motor 406.

As shown in FIGS. 21-23, in a preferred embodiment, the motor 406 is located in the head portion 12. The percussive massage device 400 can include a rotatable arm that is part of rotation housing 44. The motor 406 is located in the rotation housing 44, which is housed with the head portion 12 of the housing 101. In another embodiment, the rotation capability can be omitted.

In a preferred embodiment, the device includes a push rod or shaft 14 that is connected directly to a shaft 16 that is rotated by the motor 406 and the motor shaft 21 extending therefrom. The shaft 16 can be part of a counterweight assembly 17 that includes a counterweight 19. In a preferred embodiment, the push rod 14 is L-shaped or includes an arc shape, as shown in FIGS. 22A-22B. Preferably, the point where the push rod 14 is connected to the shaft 16 is offset from the reciprocating path that the distal end 18 of the push rod 14 (and the massage attachment 628) travel. This capability is provided by the arc or L-shape. It should be appreciated that the push rod 14 is designed such that it can

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transmit the force at least partially diagonally or in an arc along its shape instead of vertically so the motor can be located at or near the middle of the device, otherwise a large protrusion would be necessary to keep the shaft in the center with the motor offset therefrom (and positioned in the protrusion). The arc also allows the push rod **14** to have a close clearance with the motor, as shown in FIGS. **22A** and **22B** and allows the outer housing to be smaller than similar prior art devices, therefore making the device **400** lower profile. FIG. **22A** shows the push rod **14** at the bottom dead center of its travel and FIG. **22B** shows the push rod **14** at the top dead center of its travel. Preferably one or more bearings **20** are included at the proximal end of the push rod **14** where it connects to the motor to counteract the diagonal forces and preventing the push rod **14** from moving and touching the motor **406**. The bearing **20** is received on shaft **16** and a threaded fastener **26** is received in a co-axial opening **16a** in shaft **16**. The proximal end of the push rod **14** is received on bearing **20**. These components are all shown in FIG. **23**.

In a preferred embodiment, device **400** includes a number of dampening components that are made of an elastomer or the like and damp vibrations to keep the device relatively quiet. For example, as shown in FIG. **23**, device **400** includes dampening rings **426** (similar to inner suspension rings **219**) that surround the rotation housing **44** (with first and second rotation housing halves **44a** and **44b**) and help dampen the sound of vibration between the rotation housing and outer housing **101**.

As shown in FIGS. **23** and **23A**, the device **400** preferably also includes a motor mount **24** that secures the motor **406** in place and is secured to the housing **101**. Motor **406** includes a receiving member **28** with three protrusions **30** (and number between one and ten can be included) that is received in a protrusion opening **32** defined in the motor mount **24** (in first wall **38**). Flanges **34** extending from the motor mount **24** help keep the protrusions **30** in place. The motor **406** is preferably secured via threaded fasteners or the like to the motor mount **24**. Motor shaft **21** extends into the motor mount interior **36**, which is defined between first and second walls **38** and a side **40** that extends part of the way around the circumference. The counterweight assembly **17**, proximal end of the push rod **14** and related components for converting the rotation of the motor shaft **21** to reciprocating motion are position in the motor mount interior **36**. The push rod **14** extends downwardly out of the motor mount interior and through a push rod opening **42** in the side **40**. In a preferred embodiment, the motor mount **24** is connected directly to the housing **101** via fasteners **46** that are secured to mounting members **48** in the housing (see FIG. **23A**). It will be appreciated that the term push rod assembly used herein includes any of the components discussed herein or combinations thereof, e.g., push rod **14**, output shaft **108**, reciprocator **310**, second rod portion **236**, that extend from the rotating motor shaft **21**, shaft **246** or the like that provide reciprocating motion and include the attachment on the distal end thereof. The push rod assembly also includes the male connector **110** (and any related components) or any other connector at the end of the reciprocating components that allows connection of an attachment to be used for massage or therapy. The housing comprises a first portion that extends away from the handle portion and encloses a portion of the push rod assembly, and the housing comprises a second portion that extends away from the handle portion and the first portion.

In a preferred embodiment, the device **400** is associated with and can be operated by an app or software that runs on

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a mobile device such as a phone, watch or tablet (or any computer). The app can connect to the device **400** via bluetooth or other wireless connection protocol. The app can have any or all of the following functions. Furthermore, any of the functions discussed herein can be added to the touch screen/scroll wheel or button(s) capability directly on the device. If the user walks or is located too far away from the device, the device will not work or activate. The device can be turned on an off using the app as well as the touch screen or button on the device. The app can control the variable speeds (e.g., anywhere between 1750-3000 RPM). A timer can be implemented so the device stops after a predetermined period of time.

In a preferred embodiment the device, via the app or the touch screen and other functional buttons, etc. includes different treatment protocols or routines associated therewith. During the routine, the device can vary different aspects or outputs of the device or make changes based on time, speed (frequency), amplitude (stroke), arm position, force, temperature, grip (i.e., which handle portion to grip), attachment (e.g., cone, ball, dampener, etc.) and body part. The device (via the app, touch screen, haptic feedback or audibly via a speaker) can also prompt the user to make some of these changes at certain points throughout the routine, e.g., arm position, grip, attachment changes and body part changes. One of ordinary skill in the art will understand that, depending upon the particular design of the device, one or more of these outputs are applicable, while in other devices, all options described are applicable.

When the start of the protocol is selected, the device runs through a preprogrammed routine. For example, the device may operate at a first RPM for a first period of time and then operate at a second RPM for a second period of time and/or at a first amplitude for a first period of time and then operate at a second amplitude for a second period of time. The routines can also include prompts (e.g., haptic feedback) for letting the user to know to move to a new body part. These routines or treatments can be related to recovery, blood flow increase, performance, etc. and can each include a preprogrammed routine or protocol. These routines can also help facilitate certain activities, such as sleep, interval training, stairs, post-run, post-workout, recovery, wellness, post-core exercise, high intensity (plyometric) workouts, among others. The routines can also assist in providing relief and recovery from ailments such as plantar fasciitis, "tech neck," muscle cramps, jet lag, sciatica, carpal tunnel, knots, and shin splints, among others. The routines can also prompt or instruct the user to switch attachments (e.g., attachment **628** shown in FIG. **21**) or positions of the arm or rotation housing. The prompts can include sounds, haptic feedback (e.g., vibration of the device or mobile device), textual instructions or visual representation such as a graphic or picture on the app or touch screen, etc. For example, the app may instruct the user to start with the ball attachment with the arm in position two. Then the user hits start and the device runs at a first frequency for a predetermined amount of time. The app or device then prompts the user to begin the next step in the routine and instructs the user to change to the cone attachment and to place the arm in position 1 (e.g., see the arm position in FIG. **18**). The arm can include any number of positions, e.g., 1-10 positions or 1-3 positions or 1-2 positions. The user hits start again and the device runs at a second frequency for a predetermined amount of time. The protocol can be divided into steps where, at each step, varied outputs are predetermined or specified.

Referring again to FIGS. **18-19**, in a preferred embodiment, the device **400** includes a housing **101**, an electrical

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source 114, a motor 406 positioned in the housing 101, a switch 405 (which can be any of the touch screen 409, rocker button 447, button 403 or any other switch or button) for activating the motor 406, and a routine controller 630. The device 400 is configured to mate with an attachment 628. The attachment can be, for example, the attachment 628 shown in FIG. 21. The attachment is affixed to the male connector 110 so that the shaft or push rod assembly 108 moves the attachment reciprocally in accordance with a specified amplitude. For example, the amplitude is depicted in FIGS. 22A and 22B, where FIG. 22A shows the attachment at a maximum extended position and FIG. 22B shows the attachment at a minimum extended position. The distance between maximum and minimum extended positions can, in an embodiment, define the amplitude.

The routine controller 630 is configured to perform a routine in connection with one or more specified protocols. The routine controller 630 can be, for example, the microcontroller unit 701 depicted in FIG. 2. The routine controller 630 can also be a standalone microcontroller separate from the microcontroller 701. The routine controller can step through different steps of a specified protocol designed to target specified muscle groups and to provide certain therapeutic effects, as described herein.

FIG. 24 is a table showing an example of a protocol in accordance with a preferred embodiment. Protocol 1 is divided into four steps, each depicting a specified time, speed, amplitude, attachment, force, temperature, and grip. At Step 1, the device 400 is activated for 30 seconds at a speed of 1550 RPM. A routine controller 630 may be utilized to turn on the percussive massage device and implement a speed of the attachment 628 of 1550 RPM. One of ordinary skill in the art would understand that the speed of the attachment 628 is directly proportional to the speed of the motor 406. The amplitude of the percussive massage device is set to be 2 in accordance with Protocol 1. This may translate to a specified distance that an attachment 628 moves while in use, as described above. Step 1 also specifies a dampener attachment affixed to the device 400, a force of "1" be applied by the device 400, and a temperature of 21° C. be applied to the attachment.

One of ordinary skill in the art would understand that the force to be applied by the device 400 may depend upon the pressure exerted by the user in pressing the attachment onto a person's body part. As described more fully herein, the force to be applied by the device 400 may be the target force. In an embodiment where the user provides pressure to exert a particular force upon a person's body part, the routine controller 630 may adjust the output of the device 400 to ensure that the force actually applied by the attachment is the target force. The routine controller 630 may also be configured to provide feedback to the user to increase or decrease pressure on a person's body part to meet the target force. Each of these embodiments is applicable to each of the steps of a given protocol, including in Steps 2-4 below, as well as Steps 1-4 of the protocol shown in FIG. 25.

Step 1 also specifies that the device 400 is to be operated using grip 1. Grip 1, for example, may be a grip on the first handle portion 143, otherwise referred to as a "regular" or "standard" grip. Grip 2, for example, may be a grip on the third handle portion 147, otherwise referred to as a "reverse" grip. An "inverse" grip can also be used on third handle portion 147. Grip 3, for example, may be a grip shown on the second handle portion 145, otherwise referred to as a "base" grip.

At Step 2, Protocol 1 specifies that the device 400 be activated for 15 seconds at 2100 RPM, with an amplitude of

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"3", a force of "3", and a temperature of 26° C. Step 2 specifies that the small ball attachment 628 be used, and that the device 400 is to be operated using grip 1. Step 2 therefore requires that the dampener attachment in Step 1 be replaced by the small ball attachment, but specifies that the same grip is to be used.

At Step 3, Protocol 1 specifies that the device 400 be activated for 30 seconds, at 2200 RPM, with an amplitude of "1", a force of "3", and a temperature of 29° C. Step 3 specifies that the dampener attachment 628 be used, and that the device 400 is to be operated using grip 1. Step 3 therefore requires that the small ball attachment in Step 2 be replaced by the dampener attachment, but specifies that the same grip is to be used.

At Step 4, Protocol 1 specifies that the device 400 be activated for 45 seconds, at 2400 RPM, with an amplitude of "4", a force of "2", and a temperature of 32° C. Step 3 specifies that the large ball attachment be used, and that the device 400 is to be operated using grip 1. Step 3 therefore requires that the dampener attachment in Step 2 be replaced by the large ball attachment, but specifies that the same grip is to be used. It will be appreciated that Protocol 1 is provided as an example to the reader of many of the different outputs that can be changed during a myriad of treatment protocols that can be provided or developed. It will be further appreciated that any one or more of the outputs can be a part of a protocol or routine and any of the outputs discussed herein can be omitted. For example, a protocol may only include time and speed or only time speed and force, or only time, speed and grip or any other combination of the outputs described herein.

FIG. 25 is a table showing an example of a "Shin Splints" protocol in accordance with a preferred embodiment. Like Protocol 1, the Shin Splints protocol is divided into four steps, each depicting a specified time, speed, amplitude, attachment, force, temperature, and grip, but also specifying a particular arm position and body part to which to apply the attachment. At Step 1, the device 400 is activated for 1 minute at a speed of 1500 RPM, with an amplitude of "1", a force of "2", and a temperature of 21° C. Step 1 specifies that the dampener attachment be used, and that the device 400 is to be operated using grip 2 ("Reverse"), to the right shin.

Step 1 also specifies the arm position to be used is arm position 1. One of ordinary skill in the art would understand that the numbers of arm position (e.g., 1, 2, 3, 4, etc.) are predetermined arm positions intended to be used during a particular protocol. The part of the body to which the attachment 628 is to be applied is one of the factors in determining an optimal arm position. The arm position, however, may be determined by the user and is not required to otherwise implement a protocol. As discussed above, a "standard" grip may be utilized with arm position to apply to specific parts of the body, a "reverse" grip may be utilized with arm position to apply to specific parts of the body, and a "base" grip may be utilized with arm position to apply to specific parts of the body. One of ordinary skill in the art would recognize that the any arm position in combination with the particular grip 143, 145, 147 may vary depending on the application. One of ordinary skill in the art will understand that setting the arm position of a device 400 depends upon the specific device. For example, certain devices may allow a user to adjust arm position while others do not. For those that do not, this step does not apply. In other embodiments, this step may be performed during execution of the steps of the particular protocol.

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At Step 2, the Shin Splints protocol specifies that the device 400 be activated for 1 minute at 1500 RPM, with an amplitude of “1”, a force of “2”, and a temperature of 21° C. Step 2 specifies that the dampener attachment be used, and that the device 400 is to be operated using grip 2 (“Reverse”), at an arm position 1, to the left shin. Step 2 therefore uses the same attachment, grip, and arm position as Step 1, but is applied to the other shin.

At Step 3, the Shin Splints protocol specifies that the device 400 be activated for 1 minute at 2000 RPM, with an amplitude of “3”, a force of “3”, and a temperature of 24° C. Step 2 specifies that the dampener attachment be used, and that the device 400 is to be operated using grip 3 (“Base”), at an arm position 1, to the right calf. Step 3 therefore requires that the user change grips from “reverse” to “base” grips, but specifies that the same attachment and arm position be used.

At Step 4, the Shin Splints protocol specifies that the device 400 be activated for 1 minute at 2000 RPM, with an amplitude of “3”, a force of “3”, and a temperature of 24° C. Step 2 specifies that the dampener attachment be used, and that the device 400 is to be operated using grip 3 (“Base”), at an arm position 1, to the left calf. Step 2 therefore uses the same attachment, grip, and arm position as Step 1, but is applied to the other calf.

FIGS. 26A-C are a series of flow diagrams showing a method 1500 of executing a routine for a percussive massage device.

FIG. 26A is a flow diagram showing an exemplary protocol initiation. At Step 1502, Protocol 1 is initiated. Protocol 1, for example, is the Protocol 1 depicted in FIG. 24 or the “Shin Splints” Protocol depicted in FIG. 25. One of ordinary skill in the art would understand that Protocol 1 depicted in FIG. 24 does not include all of the outputs that are specified in the Shin Splints Protocol depicted in FIG. 25, and thus, not all steps of the method 1500 apply to the Protocol 1 depicted in FIG. 24.

At Step 1504, a user is prompted to set the arm position to the specified arm position. The user may be the person using the device 400 on their own body or on the body of another person. The arm position specified in the Shin Splints Protocol is arm position 1, for example.

At Step 1506, the user is prompted to use a specified grip or handle portion 143, 145, 147 on the device 400. The grip specified in the Shin Splints Protocol is the third handle portion 147, for example. As described herein, the grip may vary depending on the particular protocol or step.

At Step 1508, the user is prompted to affix a specified attachment to the device 400. As described herein, the attachment may vary depending on the particular protocol or step.

At Step 1510, the method determines whether the arm position and the grip position 143, 145, 147 are configured appropriately and whether the attachment 628 is affixed. Step 1510 may involve a prompt to the user by haptic feedback, application interface, or touch screen (among other types of prompts) in which the user is asked to proceed when the appropriate arm position, grip, and attachment are ready. In other embodiments, the device 400 may sense that the arm position and grip are appropriate and that an attachment is affixed before proceeding automatically. In an embodiment, Step 1510 is repeated until the arm position, grip, and attachment are ready.

FIG. 26B is a flow diagram showing an exemplary Step 1 of the protocol, continuing the method 1500 where FIG. 26A left off.

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At Step 1512, Step 1 of the protocol is initiated. Step 1, for example, is Step 1 depicted in FIGS. 24 and 25, for example.

At Step 1514, the method 1500 applies a specified time period (T_1) in which the device 400 is activated, a speed of the attachment, an amplitude of the attachment, a force of the attachment, and a temperature of the attachment. In an embodiment, one or more of these outputs of the device 400 are applied. These outputs may be applied by the routine controller 630. One of ordinary skill in the art would understand that a user’s implementation of the device 400 on a body part is not required to apply certain of these outputs. For example, the time period, speed, amplitude, and temperature are not necessarily dependent upon a user applying pressure to a body part. On the other hand, the force applied by the attachment 628 may require a user to exert pressure on a body part for a target force (or a target force range) to be reached. Further, the temperature may vary depending on whether the attachment 628 is applied to a body part, or not, and to which body part it is applied. Thus, the temperature may need to be adjusted during application of the attachment 628 to reach a desired temperature predetermined by the protocol. In another embodiment, the temperature may be adjusted by a user.

After time period T_1 , the user may be prompted to change the attachment 628, arm position, and/or grip position 143, 145, 147. These outputs may need to be implemented prior to the start of Step 2 of a protocol. In the Shin Splints Protocol depicted in FIG. 25, the attachment 628, arm position and grip position 143, 145, 147 remain the same. At Step 1516, after time period T_1 , the user is prompted to set the arm position to the specified arm position. The user may be the person using the device 400 on their own body or on the body of another person.

At Step 1518, the user is prompted to use a specified grip 143, 145, 147 on the device 400. As described herein, the grip may vary depending on the particular protocol or step.

At Step 1520, the user is prompted to affix a specified attachment 628 to the device 400. As described herein, the attachment 628 may vary depending on the particular protocol or step.

At Step 1522, the method determines whether the arm position and the grip position 143, 145, 147 are configured appropriately and whether the attachment 628 is affixed. This step and all other like steps are optional. Step 1510 may involve a prompt to the user by haptic feedback, application interface, or touch screen (among other types of prompts) in which the user is prompted to move to the next step in the routine and/or requested to proceed when the appropriate arm position, grip, and attachment are ready. In other embodiments, the device 400 may sense that the arm position and grip are appropriate and that an attachment is affixed before proceeding automatically. In an embodiment, Step 1522 is repeated until the arm position, grip, and attachment are ready.

FIG. 26C is a flow diagram showing an exemplary Step 2 of the protocol, continuing the method 1500 where FIG. 26B left off.

At Step 1524, Step 2 of the protocol is initiated. Step 2, for example, is Step 2 depicted in FIGS. 44 and 45, for example.

At Step 1526, the method 1500 applies a specified time period (T_2) in which the device 400 is activated, a speed of the attachment, an amplitude of the attachment, a force of the attachment, and a temperature of the attachment. In an embodiment, one or more of these outputs of the device 400 are applied. These outputs may be applied by the routine

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controller 630. One of ordinary skill in the art would understand that a user's implementation of the device 400 on a body part is not required to apply certain of these outputs. For example, the time period, speed, amplitude, and temperature are not necessarily dependent upon a user applying pressure to a body part. On the other hand, the force applied by the attachment 628 may require a user to exert pressure on a body part for a target force to be reached. Further, the temperature may vary depending on whether the attachment 628 is applied to a body part, or not, and to which body part it is applied. Thus, the temperature may need to be adjusted during application of the attachment 628 to reach a desired temperature predetermined by the protocol. In another embodiment, the temperature may be adjusted by a user.

After time period T_2 , the user may be prompted to change the attachment 628, arm position and/or grip position 143, 145, 147. These outputs may need to be implemented prior to the start of Step 3 of a protocol. In the Shin Splints Protocol depicted in FIG. 25, the attachment 628 and arm position remain the same, but the grip 143, 145, 147 is adjusted to the base grip. At Step 1528, after time period T_2 , the user is prompted to set the arm position to the specified arm position. The user may be the person using the device 400 on their own body or on the body of another person.

At Steps 1528-1534, therefore, steps substantially the same as Steps 1516-1522 are performed. After Step 1534, Steps 3-4 are initiated in substantially the same manner as Steps 1-2. For example, Steps 3 and 4 may be Steps 3 and 4 of the Protocol 1 depicted in FIG. 24 or the Shin Splints Protocol depicted in FIG. 25. Furthermore, Step 1534 can be omitted in a device where none of the grip, arm position or attachment can be sensed by the device. In this embodiment, the given protocol simply moves from step 1 to step 2 prompting the user to make a change (but regardless of whether the user has actually made a change).

As an alternative to FIG. 26C, FIG. 26D is a flow diagram depicting an alternative Step 2 of a protocol. In the alternative Step 2, a force meter adjustment is implemented.

Steps 1536-1538 are performed substantially the same as Steps 1524-1526 in previous Step 2 above.

At Step 1540, the force being applied by the attachment 628 is monitored. In the embodiment shown in FIG. 26D, the method 1500 utilizes the force meter 400 to monitor the force actually being applied by the user.

At Step 1542, the force is displayed to the user. In an embodiment, the force is displayed on an application interface 1584 such as a graphical user interface. In other embodiments, individual use or combined use of the application interface 1584, touch screen 1582, the OLED screen 711, or the like, may be used to display the force.

At Step 1546, the user is prompted to increase or decrease the force being applied to a body part according to the specified protocol during T_2 . FIG. 28 is a diagram showing a touch screen 1582 in accordance with an exemplary embodiment of the display of the force. A force display 1590 shows an exemplary embodiment of Step 1546. The force display 1590 shows a series of force measurements over the course of the "Right Bicep" step of a protocol. A force display prompt 1592 is used to display a message to the user such as "PERFECT PRESSURE: WELL DONE" when the force applied by the attachment 628 matches or corresponds to a target force predetermined by the protocol. In this embodiment, the force display prompt 1592 may recite "INCREASE PRESSURE" or the like if the measured force applied by the attachment 628 is lower than the target force predetermined by the protocol. Consequently, if the measured force applied by the attachment 628 is higher than the

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target force predetermined by the protocol, then the force display prompt 1592 may recite "DECREASE PRESSURE" or the like. The user may then adjust the pressure the user is exerting on the body part to either increase pressure or decrease pressure according to the force display prompt 1592 so that the measured force is equivalent or substantially equivalent to the target force.

After time period T_2 , the user may be prompted to change the attachment 628, arm position and/or grip position 143, 145, 147. These outputs may need to be implemented prior to the start of Step 3 of a protocol. In the Shin Splints Protocol depicted in FIG. 25, the attachment 628 and arm position remain the same, but the grip 143, 145, 147 is adjusted to the base grip. At Step 1528, after time period T_2 , the user is prompted to set the arm position to the specified arm position. The user may be the person using the device 400 on their own body or on the body of another person.

At Steps 1548-1552, therefore, steps substantially the same as Steps 1516-1522 are performed. After Step 1534, Steps 3-4 are initiated in substantially the same manner as Steps 1-2. For example, Steps 3 and 4 may be Steps 3 and 4 of the Protocol 1 depicted in FIG. 24 or the Shin Splints Protocol depicted in FIG. 25.

FIG. 27 is a diagram in accordance with an exemplary embodiment of an application interface 1584. At the top of the interface 1584, a protocol field 1556 is displayed to the user. In this embodiment, the protocol field 1556 is "TECH NECK." The protocol title 1556 also shows the overall time period of the protocol.

The next portion of the interface 1584 shows step fields 1558-1568 of the protocol that are displayed to the user. In this embodiment, the step fields identify the title of the step and time period of the step. For example, step field 1558 is titled "RIGHT BICEP" (where the treatment will be provided) and the time period of activation is "0:30 MIN."

The interface 1584 also includes a current step field 1570 that identifies the current step title 1570, a grip title display 1572, and an attachment title display 1574.

The interface 1584 also includes a time display 1576 and a time remaining display 1578 to show the user how much time has occurred during that step and the time remaining in that step. Finally, the interface 1584 includes a control field 1580 to play, skip back, and skip forward from step to step.

As described above, FIG. 28 shows a touch screen 1582 on a mobile device. The touch screen 1582 displays a graphic depicting a starting point 1586 "A" and an end point 1588 "B" (thereby defining a treatment path) showing the user where to apply the attachment 628 to the specified body part. In FIG. 27, the display instructs the user to move the attachment from the lower portion of the right bicep to the upper portion of the right bicep (the treatment path) during the current step. In some embodiments, during a single step, the user may be prompted or shown on the graphical user interface more than one treatment path (or a first treatment path and a second treatment path) on the same body part/muscle or on different body parts/muscles. For example, during the right bicep step, the user may be prompted to first move the device along the path shown in FIG. 28, but, during the same thirty second step may also be prompted or shown a path that is parallel to the path shown in FIG. 28.

FIGS. 29-33 show a device 457 similar to device 400 described above. However, the motor 402 is oriented differently (the motor shaft axis A4 extends perpendicular to the motor shaft axis in device 400), as shown in FIG. 29. It will be appreciated that all embodiments discussed herein or shown in different drawings are interchangeable and the components or inventive concepts in one embodiment can

be substituted with or into components or inventive concepts in other embodiments. All parts in all embodiments are optional and are interchangeable or usable with parts from or with other embodiments. As shown in FIG. 30, the motor mount 401 includes a mounting wall 427 with first and second mounting flanges 429 extending therefrom and a shaft opening 430 defined therein. The boss members 432 include a threaded opening 433 defined therein. The boss members 432 receive cylindrical dampening feet 461 with annular slots 425 defined therein on the outside thereof and threaded fasteners 46 in the threaded openings 433. As shown in FIGS. 31-33, the motor mount 401 attaches to both housing halves 103 of the housing 101. The mounting members 48, which are essentially an inwardly extending ring are received in annular slots 425 of the cylindrical dampening members 461. In other words, the cylindrical dampening members 461 are received in the opening 435 of mounting members 48 and the ring portion 434 of the mounting members 48 is received in the annular slots 425. The threaded fasteners 46 extend through the central openings of the cylindrical dampening members 461 (and the openings in the mounting members 48) and are threaded into the threaded openings 433 in the boss members 432. This secures the motor mount 401 to the housing halves 103 and the housing 101. The cylindrical dampening members are made of rubber or the like and help reduce vibrations.

Furthermore, the motor mount 401 mounts the motor 402 so that the motor shaft axis A4 (the rotation axis), extends forwardly and backwardly with respect to the orientation of the device 457 in use. This direction is also considered longitudinally. The motor shaft axis A4 (or a plane defined by the motor shaft axis) bisects the housing 101.

FIGS. 34-36 show another embodiment where the percussive massage device 436 includes a heart rate sensor 437 that is located on the top handle or first handle portion 143 of the device. Any type of heart rate sensor is within the scope of the invention. Heart rate sensor 437 is a heart rate sensor that uses infrared to measure and record heart rate and can also measure and record heart rate variability, if desired. In an exemplary use, heart rate is measured using a process called photoplethysmography or PPG. This involves shining a specific wavelength of light, which usually appears green, from a pulse oximeter sensor on the underside or upper side (e.g., top of the first handle portion) of the device where it touches the skin. As the light illuminates the tissue, the pulse oximeter measures changes in light absorption and the device then uses this data to generate a heart rate measurement. The electronics associated with heart rate sensor 437 are included in the housing 101 and can be separate or on the main PCB. The screen 409 displays the heart rate data. A heart rate monitor opening 438 is defined in the housing and the heart rate sensor 437 is mounted therein, as shown in FIG. 34.

FIG. 35 shows another type of heart rate monitor or sensor 439 that can be utilized and includes first and second pulse sensors or contacts 440. A first pulse sensor is positioned so that it contacts the user's palm in use and the second pulse sensor is positioned so that it contacts the user's fingers in use. The first handle portion 143 can also include an indent where the contact is located so the user knows where to place their index finger. It will be appreciated that the any of the heart rate sensors can be positioned on the second and third handle portions or on all three handle portions.

FIGS. 36 and 36A show device 457 including a thermal sensor 462. Any type of thermal sensor is within the scope of the invention. In the embodiment of FIG. 34, the thermal sensor 462 is an infrared thermometer module installed in

the housing 101 of the device (shown in a non-limiting position in FIG. 36 on the third handle portion 147) that allows the user to measure the temperature of the user's muscles or other body part. FIG. 36A shows the temperature readout on the screen 409. The thermal sensor 462 is preferably in data and/or electrical communication with the PCB. The temperature data can also be communicated to the app. In an infrared thermometer, infrared light is focused on the body part to be measured or to be treated or while being treated and the infrared thermometer module measures energy or radiation coming from the surface. The detector then translates the amount of electricity generated into a temperature reading of the particular muscle, body part, etc. The infrared beam (see FIG. 36) is emitted through an opening in the third handle portion 147 of the housing 101 and the module is mounted within the housing.

In a preferred embodiment, the temperature reading capability is integrated with and a part of the treatment routines or protocols described herein. For example, instead of a routine or a step within a routine running or extending for a predetermined period of time, the routine or step (i.e., the amount of time a particular muscle or body part is treated or targeted) can extend until the muscle or body part (referred to generally herein as a body part) reaches a predetermined temperature. Accordingly, reaching a predetermined temperature can be substituted for predetermined period of time for any of the routines discussed herein. For example, step 1526 in FIG. 26C can be substituted with the method 1500 applies the device 400 is activated until a specified temperature is reached. This can be used to be sure that a body part has been warmed up properly prior to exercise. Therefore, in use, the temperature will rise from a starting temperature to a predetermined finishing temperature and the routine can then go to the next step or end. There also may be a number of "temperature steps" that are each part of the a routine. For example, in the first step, the muscle may go from the starting temperature and move to a second temperature. The next step may treatment and temperature reading from the second temperature to a higher third temperature. The temperature range between the starting and the finish temperature within the routine may also be different for each user. Furthermore, haptic feedback or other notification or instructions can be provided to let the user know when the finish temperature or predetermined temperature has been reached and they can move to the next step in the routine.

As shown in FIG. 34, in a preferred embodiment, the device 400 includes screen 409, which may or may not be a touch screen, as well as button(s) for operating the device. In the embodiment shown in FIG. 34, the device also includes a center button 403 for turning the device on and off and a ring/rocker button 447 that provides the ability to scroll left and right (e.g., to the preset treatments discussed herein) and up and down (e.g., to control the speed or frequency).

As shown in FIG. 35, in a preferred embodiment, the arm cover 449 includes a rounded edge or surface to prevent a user's fingers from getting caught therein, and the upper portion of the male connector 110 each include rounded edges. As shown in FIG. 29, in a preferred embodiment, the male connector 110 includes an alignment tab 497 above each ball that mates with a slot in the female opening. These tabs 497 help with proper alignment with the treatment structure.

In another preferred embodiment, any of the devices taught herein can include a mechanism for heating or changing the temperature of the attachment (massage element, treatment structure, Ampbit) on the end of the recip-

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rotating shaft. The attachment can include an electrical resistance element therein that provides heat to the muscles. In a preferred embodiment, the electrical resistance element is connected to the PCB via a hollow shaft. The two outwardly biased metal spring balls on the male connector act as the electrical connector to the attachment.

FIGS. 37-40 show embodiments of a percussive massage device that includes a heated massage attachment or massage member. In the embodiment shown in FIG. 37, the male attachment member 110 includes a heating pad or heating element 502 therein. The heating element 502 is preferably electrically connected via electrical wiring 506 or the like to the PCB 504 of the device. Any type of heating is within the scope of the present invention. In a preferred embodiment, the heating element is an electrical resistance member that is located in the end of the male connector 110. In this embodiment, a wire connects the electrical resistance member to the PCB and the battery. The wiring 506 may extend through a hollow shaft or other conduit and is guided through the housing, down the shaft and into the male connector 110. The heating element 502 may be internal within the male connector 110 or may be part of the exterior surface, as shown in FIG. 37. In an embodiment with a female connector on the device (at the end of the shaft), the heating element can be in the female connector. In use, the heated male attachment member transfers heat to the massage member, which heats the outer surface of the massage member, which can then be applied to the user's body part. The PCB can include a controller for controlling the temperature. More than one temperature setting can be provided (e.g., 2-10 settings) so that different temperatures can be utilized by the user as desired. Cooler temperatures can also be provided. The attachment member and the massage member can be made of or partially made of a material that is a good conductor of heat.

FIGS. 38-40 show another preferred embodiment with a heated or temperature controlled massage member 508. All disclosure related to the FIG. 37 embodiment is repeated for this embodiment. In this embodiment, the female or male attachment member 110 is electrically connected to the complementary male or female attachment member in the massage member to provide power to heat or cool the massage member 508. FIG. 38 shows the device with power running from the PCB 504 to the male attachment member 110. As shown in FIG. 39, the male attachment member 110 includes positive and negative electrical contacts 510 that mate with opposing positive and negative electrical contacts 512 in the female attachment member in the massage member 508, as shown in FIG. 40. FIG. 39 shows a male attachment member with metal balls 514 that are received in indentations in the female attachment member. The metal balls 514 can be the electrical contacts 510 and the electrical contacts 512 can be positioned in the indentations in the female attachment member. The heating element 502 may be internal within the massage member 508 or may be part of the exterior surface.

In use, an electrical connection is made when the massage member 508 is secured to the device and to the male attachment member 110. When heating or cooling is turned on, the heating element 502 in the massage member 508 is heated, which can then be applied to the user's body part. The heating element or electrical resistance member (e.g., heated pad) can be located in or on the massage member (e.g., ball, cone, etc.) and the metal connection between the male connector and the massage member is used to electrically connect to the battery.

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The electrical connection between the male or female attachment member 110 permits a variety of uses beyond heating with the heating element 502. In a preferred embodiment, a heating element 502 radiates wavelengths to produce heat on a user's body part. The male or female attachment member 110, for example, may be utilized for a variety of other uses, such as vibration, percussion, cooling, and exfoliating. The male or female attachment member 110 may be configured as an actuator designed to provide these uses. For example, percussion is already achieved using the attachment 628. However, the attachment 628 or 508 may be modified to add or replace the heating element 502 with a cooling, vibration, or exfoliating element. Other uses and actuators may be utilized without departing from the scope of the present invention.

As shown in FIGS. 41-42C, in a preferred embodiment, the percussive therapy device 100 includes an angular position sensor 516 and a linear position sensor 518. See FIG. 37. For example, the angular position sensor 516 is a gyroscope 516 and the linear position sensor 518 is an accelerometer 518. One or more gyroscopes, accelerometers, sensors or the like can be included on or in the device for detecting and gathering data. The system including the device 100 and the angular position sensor 516 and the linear position sensor 518 allows data to be gathered regarding the angular and linear positioning of the device 100. Data can include angular positioning (α, β, γ) (i.e., angular position data) and linear movement in three axes (x, y, z) (i.e., linear position data), for example. In a preferred embodiment, a sensor chipboard 504 is included in the device 100 to measure variations in its angular position in three axes, α , β and γ via a gyroscope 516 and to track linear movement of the device in three axes x , y and z via an accelerometer 518. See FIG. 37. The angular position sensor 516 and the linear position sensor 518 may be implemented on the sensor chipboard 504, or they may constitute separate electronic devices operably connected to the sensor chipboard 504. Other suitable configurations of the angular position sensor 516 and the linear position sensor 518 exist without departing from the scope of this invention.

In an embodiment, the printed circuit board 408 of the device 100 powers the angular position sensor 516 and a linear position sensor 518 and stores the data the sensors generate. For example, the sensor data may be stored in a memory (not shown). In another embodiment, the PCB 408 integrally incorporates the sensor chipboard 504. Preferably, the PCB 408 broadcasts and/or transmits data generated by the sensors through a wireless connectivity standard, such as Bluetooth. For example, the wireless connectivity standard is implemented via the wireless control unit 710 (FIG. 2). The sensors are configured to accurately map how the device 100 moves with respect to the user's muscle during the treatment. In an embodiment, the sensors may also include an oxygen saturation sensor to monitor an amount of oxygen content in the user's blood (e.g., a pulse oximeter or the like), and a blood flow sensor to monitor magnitude and/or velocity of the user's blood flow.

FIGS. 42A-42C show exemplary angular positioning using the angular position sensor 516. As the device 100 is rotated left and right (see FIGS. 42A and 42B) in x and y axes, and tilted upwardly (see FIG. 42C) in the z axis, the angles and direction of the device 100 are shown on a computer monitor or display. The depictions shown in FIGS. 42A-42C illustrate a graphical representation of the device 100 as the device 100 is moved. While FIGS. 42A-42C illustrate angular movement of the device 100, the linear movement of the device 100 is also graphically represented

on a computer monitor or display in like manner. It will be appreciated that the movement is shown on the computer monitor in the drawings to provide an example of how the angular position sensor **516** senses the movement.

In a preferred embodiment, the angular and linear position sensors **516**, **518**, coupled with the force meter of the percussive therapy device **400** discussed above, can be used to map the treatment of a muscle or body part as the device **400** is being used in a three-dimensional display. This “map” or data can be displayed through or on an application or on the touch screen **1582**. For example, angular and linear position data obtained from the angular and linear position sensors **516**, **518** can be graphically represented via the application or on the touch screen **1582**. The angular and linear position data can assist the user in applying a particular protocol or routine, for example, such as those depicted in FIGS. **24-28** and accompanying descriptions, or the like. In addition to angular and linear movement, the force meter of device **400** (or device **457**) can obtain force magnitude data to assist the user in administering a routine or protocol constituting a therapeutic treatment to the user (or to another person to whom the user is administering the treatment). For example, the map of angular and linear position and force magnitude can be compared against the routine or protocol. The routine or protocol, in this example, will specify a muscle group, a linear and/or angular path (see FIG. **28**, for example, with the starting point **1586** and the ending point **1588**, in two dimensions), and a force magnitude that the user is intended to exert on the muscle group (see FIG. **28**, for example, with the force display **1590** and force display prompt **1592**). In a preferred embodiment, the muscle group, linear and angular position, and force magnitude (i.e., depression on the muscle group) is graphically presented in a three dimensional display. Preferably, the display also graphically illustrates when the user’s linear movement, angular movement, or force magnitude exerted on the muscle group is following the protocol or routine. If the user is not following the routine or protocol, the user will receive a prompt to take corrective action to follow the routine or protocol correctly. For example, the prompt may alert the user that the user is applying the attachment **628** to a different muscle group than that specified by the protocol. The prompt may be haptic feedback, application interface, or touch screen (among other types of prompts). The prompt may also be presented in a two-dimensional or three-dimensional graphical representation. As a result, the device can track over time what regions of a user’s muscles or body parts are being worked the most and whether the user is positioning the device correctly. The prompt may also let the user know they are positioning the device incorrectly or they are working on the wrong body part (e.g., during the treatment protocols).

Referring again to FIG. **36**, the device **457** is shown depressing the attachment **628** onto a user’s body part. In accordance with the description above, the depression may be graphically represented in two or three dimensions on a display. In practice, the attachment **628** shown in FIG. **36** is configured to provide percussive effect to the user’s body part, and thus, exerts a force onto the user’s body part. The force meter measures the force magnitude of the attachment **628** when depressed onto the user’s body part. The force magnitude data is then transmitted to a monitor/display, application, or touch screen **1582**, or the like, to show a user (or other person) the amount of force exerted on the user’s body part during a protocol or routine. Gathering multi-sensory data allows for augmented reality features that can

be used to train users and recovery professionals virtually on how to use the device **400**, **457**.

As an example, while a user’s quad muscle is not a uniform shape, it is possible to simplify the user’s quad muscle to the shape of a cylinder. The angular and linear position can be ascertained, and thus, a determination can be made concerning how the device **400**, **457** is positioned relative to the cylinder. Further, a determination can be made concerning the direction the percussive arm (e.g., push rod assembly **14**, shaft **16**, and/or attachment **628**) is directed of the device **400**, **457**. The determination can also be made concerning how the device is moving relative to the cylinder in linear coordinates. The force magnitude from the force meter of the device **400**, **457** allows confirmation that the device **400**, **457** is in contact with the muscle, as well as the intensity and duration of that interaction.

Similarly, the device **400**, **457** can also include a thermal sensor **462** or thermometer **462** that can determine the temperature of the user’s muscle and to provide feedback to the device and/or application. See FIG. **36**, thermal sensor **462**. For example, an electronic thermometer **462** that reads the temperature of the user’s skin or muscle before, during and/or after treatment can be included. In an embodiment, the thermal sensor **462** is located in the housing **12** of the device **400**, **457** where infrared radiation or wavelengths can be used to measure temperature. In another embodiment, the thermometer **462** can be positioned to require direct contact to measure the temperature and/or it may utilize wireless technology, like an infrared sensor, to make the temperature readings. For example, FIG. **40** illustrates how the attachment **508** may function as (or include) a thermal sensor **462**, a heating element **502**, or both. Similarly to the heating element **502** as shown in FIG. **37**, for example, the thermal sensor **462** may be connected to the PCB **504** via the electrical wiring **506** and may be located in the attachment **628**. The electrical contacts **510**, **512** (or metal balls **514**) as shown in the embodiments of FIGS. **38-39** provide electrical connectivity between the PCB **504**, the male or female connector **110**, and thus, the thermal sensor **462**. As with the heating element **502**, a thermal sensor **462** may be utilized as part of a protocol or routine.

In an embodiment, a three-dimensional rendering of thermal readings from the thermal sensor **462** is provided to a user to show incremental increases in temperature over time. For example, a three-dimensional rendering may show varying colors from blue (e.g., cool) to yellow/orange (e.g., medium temperature) to red (e.g., hot) to illustrate to the user the increase in temperature over time.

An accessory, module or attachment module **520** can be used with and attached or secured to a percussive massage or percussive therapy device **100**, **400**, **457** as part of a percussive therapy system **500**. In a preferred embodiment, the attachment module **520** includes a thermal sensor or thermometer **462** that can determine the temperature of the user’s muscle and to provide feedback to a device and/or application. In a preferred embodiment, the thermal sensor **462** allows the application to determine or customize the timing of each step within a protocol. The temperature can be used to determine blood flow and therefore muscle readiness for a specific goal (e.g., relaxation, performance, focus).

As shown in FIGS. **43-45**, in a preferred embodiment, the attachment module **520** includes a housing **522**, a thermal sensor **524**, a battery **526**, a printed circuit board (PCB) **528** (that includes a gyroscope **516** or other angular/positional device, e.g., the angular position sensor **516**, and/or an accelerometer **518** or other linear/positional device, e.g., the

linear position sensor **518**), a button **530** and a wireless communication module **532** (e.g., a Bluetooth module). In a preferred embodiment, the housing **522** includes a securement portion **534** defined therein so that the attachment module **520** can be secured to a percussive therapy device **400**, **457**. The securement portion **534** or recess **534** can include rubber on the inside thereof to provide grip on the percussive therapy device. Protrusions **536** are preferably included on both sides of the housing **522** to provide grip when securing and removing the attachment module **520** from the percussive therapy device **400**, **457**. In another embodiment, the wireless connection module can be omitted and the attachment module can include a display or screen for displaying information, such as temperature, angular and linear position, or any other information obtained or sensed by the attachment module.

As described above with respect to FIG. **36**, any type of thermal sensor **524** is within the scope of the invention. In the embodiment shown in FIGS. **43-45**, the thermal sensor **524** is an infrared thermometer module installed in the housing **522** and directed downwardly when installed on a percussive therapy device **100** as shown in FIGS. **46-47** (shown in a non-limiting position on the front arm of the percussive therapy device **100**). In another embodiment, the thermal sensor **524** is the thermal sensor **462** and can be secured to the third handle portion **147** or bottom of a percussive therapy device **400**, **457** or on any handle portion **143**, **145**, **147** or part of a percussive therapy device **400**, **457** where it can be positioned and allow the user to measure the temperature of the user's muscles or other body part. See FIG. **36**. The attachment module **520** can be used with any type of percussive therapy device **500**, massage device or other device where temperature and/or positioning measurements are desired. It will be appreciated that all embodiments and components thereof are interchangeable with all other embodiments and components thereof.

In a preferred embodiment, the attachment module **520** communicates wirelessly with the percussive therapy device **400** and/or the application on the user's mobile device. See FIG. **2**, the wireless control unit **710**, and accompanying discussion. In another embodiment, the attachment module **520** is physically and electrically connected to the device **400** and no wireless module is needed as communication is achieved through conventional electrical wires or the like.

Referring again to FIG. **36A**, a temperature readout on the screen **409** of the percussive therapy device **100** is shown. The thermal sensor **524** is preferably in data and/or electrical communication with the PCB **528** and the data is communicated to one or both of the device **400** or application.

In a preferred embodiment, the temperature reading capability is integrated with and a part of the treatment routines or protocols described herein or by reference. For example, instead of a routine or a step within a routine running or extending for a predetermined period of time, the routine or step (i.e., the amount of time a particular muscle or body part is treated or targeted) can extend until the muscle or body part (referred to generally herein as a body part) reaches a predetermined temperature. Accordingly, reaching a predetermined temperature can be substituted for predetermined period of time for any of the routines. For example, step **1526** in FIG. **26C** can be substituted for the step of "apply attachment to specified body part until a specified temperature is reached." This can be used to be sure that a body part has been warmed up properly prior to exercise. Therefore, in use, the temperature will rise from a starting temperature to a predetermined finishing temperature and the routine can then go to the next step or end. There also may be a number

of "temperature steps" that are each part of the a routine. For example, during the first step, the muscle may increase in temperature from the starting temperature to a second temperature. The next step may involve additional treatment until the temperature reading increases from the second temperature to a higher third temperature. The temperature range between the starting and the finish temperature within the routine may also be different for each user. Furthermore, haptic feedback or other notification or instructions can be provided to let the user know when the finish temperature or predetermined temperature has been reached and that they can move to the next step in the routine.

In a preferred embodiment, the attachment module **520** includes an angular position sensor **516** (e.g., gyroscope **516**) and/or a linear position sensor **518** (e.g., accelerometer **518**). Each or both can be implemented as part of the PCB **18**. One or more gyroscopes **516**, accelerometers **518**, sensors or the like can be included on or in the device **400** for detecting and gathering data. One or more actuators may also be included on or in the device **400** for providing at least one therapeutic effect. Thus, the description above referencing gyroscopes, **516**, accelerometers **518**, attachments **628**, **508**, male or female attachment members **110**, or sensors or actuators within or without the housing **101** is instructive and within the scope of the attachment module **520**. See FIGS. **36-42C**. For example, a heating element **502** may be implemented in the attachment module **520** to utilize radiation to penetrate skin and muscle to a certain depth. This treatment can result in muscle recovery.

In an embodiment, the percussive therapy system **500** is configured to determine at least one characteristic of the attachment **628**, **508**. For example, a percussive therapy device **100**, **400** itself may include circuitry and wired or wireless communication to sense the type of attachment the user intends to use in connection with the device **100**, **400**. For example, the device **100**, **400** may sense that the attachment **628** is a dampener. Other characteristics of the attachment **628**, **508** may be sensed. For example, the existence of one or more sensors included in the attachment **628**, **508** may be sensed. In addition, the existence of one or more actuators included in the attachment **628**, **508** may be sensed. In an embodiment, the device **100**, **400** senses when the attachment **628**, **508** is attached to a distal end of the push rod assembly **14**. Once the attachment **628**, **508** is attached, then the device may, through wired connections (e.g., positive/negative contacts **510**, **512** or the like, or other wired electrical connections), sense the various characteristics of the attachment **628**, **508**. In this embodiment, the wired connections may communicate with the PCB **408**, **504** so that the device **100**, **400** determines the characteristics. In another embodiment, the attachment **628**, **508** may include wireless communication capabilities and communicate the characteristics wirelessly. One of ordinary skill in the art would understand that there are a variety of methodologies to employ to communicate the characteristics to the device **100**, **400** and/or the user, preferably through communication on a remote device or touch screen **1582**.

FIG. **48** is a flow diagram of a method **1600** of providing at least one therapeutic effect to a user in accordance with an embodiment of the present invention. At Step **1602**, a percussive therapy device **400**, **457** is operated on a user's body part. For example, the user initiates a protocol such as that shown in FIGS. **24-28** and accompanying descriptions, or the like. In accordance with the specified protocol initiated, the user typically is instructed to operate the percussive therapy device (or other suitable therapeutic treatment or effect) in accordance with steps of the protocol in a specified

fashion. For example, the user may be instructed to orient the device 400, 457 at a specified angle relative to a muscle group, along a linear path relative to the specified muscle group, and/or with a certain amount of force exerted on the specified muscle group. At Step 1604, angular position data is obtained from a gyroscope 516 in three rotational axes (α, β, γ). The gyroscope may also be an angular position sensor 516 or suitable replacement. At Step 1606, adjustment of an angular position of the percussive massage device 400, 457 is recommended in response to the angular position data. As illustrated in FIGS. 42A-C, the angular position data may show that the angular position of the device 400, 457 is correctly oriented relative to a body part. It may also reveal that the angular position of the device 400, 457 is incorrectly oriented. Thus, the recommendation preferably instructs the user to orient the device 400, 457 properly relative to the body part.

At Step 1608, linear position data is obtained from an accelerometer 518 in three linear axes (x, y, z). The accelerometer may also be a linear position sensor 518 or suitable replacement. At Step 1610, adjustment of a linear position of the percussive massage device 400, 457 is recommended in response to the linear position data. For example, in FIG. 28, a right bicep routine is shown that instructs the user to move the device 400, 457 from the starting point 1586 (A) to the ending point 1588 (B). If the user correctly follows the linear path from (A) to (B), then the recommendation may indicate so to the user. If the user is not correctly following the linear path from (A) to (B), then the recommendation preferably instructs the user to adjust the linear position of the device 400, 457 and/or attachment 628 to correctly follow the linear path and the predetermined routine.

At Step 1612, force magnitude data is obtained from a force meter included in the percussive therapy device 400, 457. At Step 1614, application of the attachment 628 of device 400, 457 to the user's body part is recommended if the attachment 628 is not in contact with the user's body part in response to the force magnitude data. For example, the force magnitude is approximately zero (or a de minimus threshold amount) that may be predetermined if the attachment is not in contact with the user's body part.

At Step 1616, adjustment of a force magnitude exerted on the user by the attachment 628 of the device 400, 457 is recommended in response to the force magnitude data. For example, in FIG. 28, a force magnitude exerted on a right bicep is illustrated in accordance with the force display 1590. In that embodiment, the force display prompt 1592 reads "PERFECT PRESSURE: WELL DONE", indicating that the pressure the user is exerting on the right bicep is in accordance with the pressure specified by the predetermined right bicep routine. In the event that the force magnitude is lower or higher than the pressure specified by the routine, the recommendation will read "INCREASE PRESSURE" or "DECREASE PRESSURE" as needed.

At Step 1618, a three-dimensional representation of the device 400, 457 and its angular and/or linear position and/or force magnitude is displayed on a display. The angular position of the device 400, 457, in an embodiment, is displayed similarly to the graphic shown in FIG. 42A-C. The display may be situated on a touch screen 1582, a mobile device, or other remote device. The display of the three-dimensional device is utilized to assist the user in adjustment of the angular and/or linear position of the device and/or the pressure (e.g., force magnitude) exerted on the user's body part. See FIGS. 42A-C and accompanying description concerning "mapping" of device 400, 457 relative to the user's body part.

FIG. 49 is a flow diagram of a method 1620 of preparing a user's body part for exercise in accordance with an embodiment of the present invention. At Step 1622, a therapeutic effect is provided to the user's body part using the percussive therapy device 400, 457. The therapeutic effect may include a variety of massage or other treatments, including vibration, concussion, heat, or exfoliation. A heating element 502 or other heating actuator may be implemented to increase the temperature during the time that the therapeutic effect is provided to the user.

At Step 1624, a temperature of the user's body part is monitored. At Step 1626, it is determined whether the temperature reading is greater than or equal to a predetermined threshold temperature. Once the temperature reaches the predetermined threshold temperature, for example, the user's body part is ready for exercise. This may vary depending on the user and the user's body part. If the temperature is less than the predetermined threshold temperature, Steps 1622 and 1624 are repeated. If the temperature is greater than or equal to the predetermined threshold temperature, then Step 1628 is implemented. At Step 1628, user instructions are provided to cease providing the therapeutic effect to the user's body part. The user's body part is warm enough to exercise safely and effectively with lower risk for exercise-related injury, and can also improve performance of the user during the exercise.

Although the operations of the method(s) herein are shown and described in a particular order, the order of the operations of each method may be altered so that certain operations may be performed in an inverse order or so that certain operations may be performed, at least in part, concurrently with other operations. In another embodiment, instructions or sub-operations of distinct operations may be implemented in an intermittent and/or alternating manner.

Unless the context clearly requires otherwise, throughout the description and the claims, the words "comprise," "comprising," and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of "including, but not limited to." As used herein, the terms "connected," "coupled," or any variant thereof, means any connection or coupling, either direct or indirect, between two or more elements; the coupling of connection between the elements can be physical, logical, or a combination thereof. Additionally, the words "herein," "above," "below," and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description of the Preferred Embodiments using the singular or plural number may also include the plural or singular number respectively. The word "or" in reference to a list of two or more items, covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

Embodiments are envisioned where any of the aspects, features, component or steps herein may be omitted and/or are option. Furthermore, where appropriate any of these optional aspects, features, component or steps discussed herein in relation to one aspect of the invention may be applied to another aspect of the invention.

The above-detailed description of embodiments of the disclosure is not intended to be exhaustive or to limit the teachings to the precise form disclosed above. While specific embodiments of and examples for the disclosure are described above for illustrative purposes, various equivalent modifications are possible within the scope of the disclosure, as those skilled in the relevant art will recognize. For

example, while processes or blocks are presented in a given order, alternative embodiments may perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified to provide alternative or subcombinations. Each of these processes or blocks may be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks may instead be performed in parallel, or may be performed, at different times. Further any specific numbers noted herein are only examples: alternative implementations may employ differing values or ranges.

The above-detailed description of embodiments of the disclosure is not intended to be exhaustive or to limit the teachings to the precise form disclosed above. While specific embodiments of and examples for the disclosure are described above for illustrative purposes, various equivalent modifications are possible within the scope of the disclosure, as those skilled in the relevant art will recognize. Further, any specific numbers noted herein are only examples: alternative implementations may employ differing values, measurements or ranges. It will be appreciated that any dimensions given herein are only exemplary and that none of the dimensions or descriptions are limiting on the present invention.

The teachings of the disclosure provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various embodiments described above can be combined to provide further embodiments.

Any patents and applications and other references noted above, including any that may be listed in accompanying filing papers, are incorporated herein by reference in their entirety. Aspects of the disclosure can be modified, if necessary, to employ the systems, functions, and concepts of the various references described above to provide yet further embodiments of the disclosure.

These and other changes can be made to the disclosure in light of the above Detailed Description of the Preferred Embodiments. While the above description describes certain embodiments of the disclosure, and describes the best mode contemplated, no matter how detailed the above appears in text, the teachings can be practiced in many ways. Details of the system may vary considerably in its implementation details, while still being encompassed by the subject matter disclosed herein. As noted above, particular terminology used when describing certain features or aspects of the disclosure should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features or aspects of the disclosure with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the disclosures to the specific embodiments disclosed in the specification unless the above Detailed Description of the Preferred Embodiments section explicitly defines such terms. Accordingly, the actual scope of the disclosure encompasses not only the disclosed embodiments, but also all equivalent ways of practicing or implementing the disclosure under the claims.

While certain aspects of the disclosure are presented below in certain claim forms, the inventors contemplate the various aspects of the disclosure in any number of claim forms. For example, while only one aspect of the disclosure is recited as a means-plus-function claim under 35 U.S.C. § 112, ¶6, other aspects may likewise be embodied as a means-plus-function claim, or in other forms, such as being

embodied in a computer-readable medium. (Any claims intended to be treated under 35 U.S.C. § 112, ¶6 will begin with the words “means for”). Accordingly, the applicant reserves the right to add additional claims after filing the application to pursue such additional claim forms for other aspects of the disclosure.

Accordingly, although exemplary embodiments of the invention have been shown and described, it is to be understood that all the terms used herein are descriptive rather than limiting, and that many changes, modifications, and substitutions may be made by one having ordinary skill in the art without departing from the spirit and scope of the invention.

What is claimed is:

1. A percussive therapy system comprising:

a percussive therapy device comprising:

a housing;

a handle portion;

a motor positioned in the housing and configured to output torque about a motor axis;

an eccentric shaft configured to travel in a circle about the motor axis in response to activation of the motor; and

a push rod assembly operatively connected to the motor and configured to reciprocate in response to activation of the motor, wherein the push rod assembly comprises an output shaft configured to travel along a reciprocation axis when the push rod assembly reciprocates, the reciprocation axis extends transverse to the handle portion on a dividing plane normal to the motor axis, a proximal end of the push rod assembly is connected to the eccentric shaft at a location offset from the dividing plane in a first direction, the motor extends transverse to the handle portion and away from the dividing plane in a second direction opposite the first direction, and an interior angle defined between the handle portion and the output shaft is obtuse; and

an attachment configured to be operatively connected to the output shaft of the percussive therapy device.

2. A percussive therapy device comprising:

a motor configured to output torque about a motor axis;

a handle portion extending along a handle portion axis;

a housing; and

a push rod assembly comprising:

an output shaft configured to reciprocate along a reciprocation axis in response to activation of the motor and having a distal end that comprises a connector for an attachment, wherein the output shaft is centered on the reciprocation axis and the reciprocation axis extends on a dividing plane; and

a push rod having a proximal end operatively connected to the motor and a distal end connected to the output shaft; wherein:

the dividing plane is normal to the motor axis,

the handle portion axis extends at an oblique angle to the reciprocation axis and intersects the reciprocation axis and the motor axis,

the motor is located on a first side of the dividing plane, the device includes an eccentric shaft that is configured to travel in a circle about the motor axis when the motor is active,

the proximal end of the push rod is operatively connected to the motor by the eccentric shaft, wherein the proximal end of the push rod is coupled to the

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eccentric shaft on a second side of the dividing plane that is opposite from the first side of the dividing plane, and

the proximal end of the push rod is farther from the reciprocation axis than the distal end is.

3. The percussive therapy system of claim 1, wherein the handle portion extends longitudinally away from the reciprocation axis.

4. The percussive therapy system of claim 1, wherein the percussive therapy device comprises a bearing between the eccentric shaft and the push rod assembly, the output shaft is centered on the reciprocation axis, and the bearing and the motor are positioned on opposite sides of the reciprocation axis.

5. The percussive therapy system of claim 3, wherein the push rod assembly comprises a push rod, the push rod comprises a proximal end coupled to the bearing, the push rod comprises a distal end coupled to the output shaft, and the proximal end of the push rod is farther from the reciprocation axis than the distal end of the push rod is.

6. The percussive therapy system of claim 3, wherein the push rod assembly comprises a push rod, the push rod comprises a proximal end, the push rod comprises a distal end, and the proximal end of the push rod is farther from the reciprocation axis than the distal end of the push rod is.

7. The percussive therapy system of claim 1, wherein the device comprises a switch for activating the motor, and the switch is located on a portion of the housing that faces an opposite direction from a distal end of the push rod assembly.

8. The percussive therapy system of claim 1, wherein the device comprises a switch and an electronic display that faces a same direction as the switch, wherein the switch is configured for activating the motor.

9. The percussive therapy system of claim 1, wherein the handle portion extends along a handle portion axis, the reciprocation axis and the handle portion axis both lie on a common plane, and the common plane bisects the housing.

10. The percussive therapy system of claim 1, comprising a battery housed in the handle portion.

11. The percussive therapy system of claim 1, wherein the housing comprises a first portion that extends away from the handle portion and encloses a portion of the push rod assembly, and the housing comprises a second portion that extends away from the handle portion and the first portion.

12. The percussive therapy device of claim 2, comprising a backward facing electronic display and a backward facing switch configured to activate the motor.

13. The percussive therapy device of claim 11, comprising a switch configured to activate the motor, wherein the switch is located on a portion of the housing that faces an opposite direction from the distal end of the output shaft.

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14. The percussive therapy device of claim 2, wherein the handle portion extends longitudinally away from the reciprocation axis.

15. The percussive therapy device of claim 2, wherein an interior angle defined between the handle portion and the output shaft is obtuse.

16. The percussive therapy device of claim 2, comprising a battery and a handle, wherein the handle comprises the handle portion and the battery is housed in the handle.

17. The percussive therapy device of claim 2, wherein the housing comprises a first portion that extends away from the handle portion and encloses the push rod assembly, and the housing comprises a second portion that extends away from the handle portion and the first portion.

18. A percussive therapy device comprising:
a motor configured to output torque about a motor axis;
a handle portion extending along a handle portion axis;
a housing; and
a push rod assembly comprising:

an output shaft configured to reciprocate along a reciprocation axis in response to activation of the motor and having a distal end that comprises a connector for an attachment, wherein the output shaft is centered on the reciprocation axis and the reciprocation axis extends on a dividing plane; and

a push rod having a proximal end operatively connected to the motor and a distal end connected to the output shaft; wherein:

the dividing plane is normal to the motor axis,
the handle portion axis extends at an oblique angle to the reciprocation axis and intersects the reciprocation axis and the motor axis,

the motor is located on a first side of the dividing plane, the device includes an eccentric shaft that is configured to travel in a circle about the motor axis when the motor is active,

the proximal end of the push rod is operatively connected to the motor by the eccentric shaft, wherein the proximal end of the push rod is coupled to the eccentric shaft on a second side of the dividing plane that is opposite from the first side of the dividing plane, and

the proximal end of the push rod is coupled to the eccentric shaft at a first location, the distal end of the push rod is connected to the output shaft at a second location, and the first location is farther from the reciprocation axis than the second location is.

19. The percussive therapy device of claim 18, wherein the handle portion extends longitudinally away from the reciprocation axis.

20. The percussive therapy device of claim 18, wherein an interior angle defined between the handle portion and the output shaft is obtuse.

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